"The purpose of learning mathematics is not merely to find answers, but to understand the ideas behind them."

Abstract Algebra

(MMT522J1)

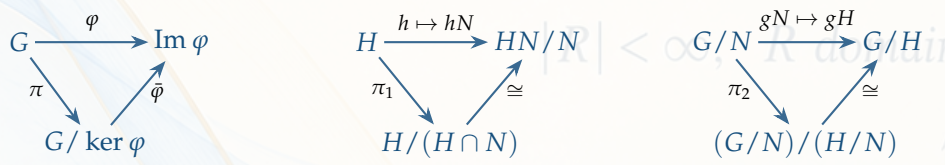
Lecture Notes for Undergraduate Students

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CHAPTER
1

GROUPS AND SUBGROUPS

1.1 Binary Operations and Equivalence Relations

The study of algebraic structures begins with sets equipped with operations that combine their elements according to prescribed rules. Familiar examples include addition and multiplication of numbers, multiplication of matrices, and composition of functions. Although these operations arise in different settings, they often satisfy common algebraic properties. The systematic study of these properties leads to important algebraic structures such as groups, rings, and fields.

Before introducing groups, we examine two fundamental concepts: binary operations and equivalence relations. Binary operations provide the algebraic framework for combining elements, whereas equivalence relations allow us to classify elements into disjoint classes. Both concepts play a central role in the subsequent development of abstract algebra.

1.1.1 Binary Operations

Definition 1.1.1. Let S be a nonempty set. A *binary operation* on S is a function

$$* : S \times S \longrightarrow S.$$

For each ordered pair $(a, b) \in S \times S$, the image under $*$ is denoted by $a * b$.

Thus, a binary operation assigns to every ordered pair of elements of S a unique element of S. In particular, the result of combining any two elements must again belong to S. This requirement is called the *closure property*.

It is important to observe that an operation is defined relative to a particular set. A rule may define a binary operation on one set but fail to do so on another.

Example 1.1.2. The following illustrate binary operations and the importance of closure.

(i) Addition, subtraction, and multiplication are binary operations on $\mathbb{Z}$, since the sum, difference, and product of any two integers are again integers.

(ii) Subtraction is not a binary operation on $\mathbb{N} = \{1, 2, 3, \ldots\}$. Indeed, $2, 5 \in \mathbb{N}$, but $2 - 5 = -3 \notin \mathbb{N}$.

(iii) Multiplication is a binary operation on $M_2(\mathbb{R})$, the set of all 2×2 matrices with real

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entries, because the product of two such matrices is again a 2×2 real matrix.

(iv) The rule $a * b = a + b + ab$ defines a binary operation on $\mathbb{Z}$. For example, $2 * 3 = 2 + 3 + 6 = 11$.

(v) Division does not define a binary operation on $\mathbb{R}$, since division by zero is undefined. Moreover, division does not define a binary operation on $\mathbb{Z} \setminus \{0\}$, since the quotient of two nonzero integers need not be an integer.

Properties of Binary Operations

Certain properties of binary operations occur frequently in algebra. The two most important are associativity and commutativity.

Definition 1.1.3. A binary operation $*$ on a set S is said to be

(i) *associative* if $(a * b) * c = a * (b * c)$ for all $a, b, c \in S$;

(ii) *commutative* if $a * b = b * a$ for all $a, b \in S$.

Associativity permits us to change the grouping of elements without changing their order. Thus, for an associative operation, the expression $a * b * c$ is unambiguous. Commutativity, on the other hand, permits us to interchange the order of the elements.

These properties are independent: an operation may satisfy either one, both, or neither.

Example 1.1.4. (i) Addition and multiplication on $\mathbb{Z}$ are both associative and commutative.

(ii) Subtraction on $\mathbb{Z}$ is neither associative nor commutative. For example,

$$(8 - 5) - 2 = 1, \qquad 8 - (5 - 2) = 5,$$

while $8 - 5 = 3$ and $5 - 8 = -3$.

(iii) Matrix multiplication is associative but is not commutative in general. Consider

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Then

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Hence $AB \neq BA$.

(iv) On $\mathbb{Z}$, define $a * b = 2a + 2b$. This operation is commutative, but it is not associative. Indeed,

$$(1 * 2) * 3 = 18, \qquad 1 * (2 * 3) = 22.$$

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Identity and Inverse Elements

Besides associativity and commutativity, the existence of identity and inverse elements is fundamental in the study of groups.

Definition 1.1.5. Let $*$ be a binary operation on a nonempty set S. An element $e \in S$ is called an *identity element* if

$$a * e = e * a = a$$

for every $a \in S$.

The identity element leaves every element unchanged when combined with it, regardless of the order in which the operation is performed.

For example, 0 is the identity for addition on $\mathbb{Z}$, while 1 is the identity for multiplication on $\mathbb{R}$.

Proposition 1.1.6. *If a binary operation on a nonempty set has an identity element, then that identity element is unique.*

Proof. Suppose e and f are identity elements of S. Since e is an identity, $e * f = f$. Similarly, since f is an identity, $e * f = e$. Therefore $e = f$, proving uniqueness. ■

Definition 1.1.7. Suppose a binary operation $*$ on S has an identity element e. An element $b \in S$ is called an *inverse* of $a \in S$ if

$$a * b = b * a = e.$$

The inverse of an element therefore reverses its effect under the given operation. For addition on $\mathbb{Z}$, the inverse of a is $-a$, whereas for multiplication on $\mathbb{R} \setminus \{0\}$, the inverse of a is $1/a$.

Example 1.1.8. On $\mathbb{Z}$, define

$$a * b = a + b + 1.$$

We examine its algebraic properties.

First, $*$ is a binary operation, since $a + b + 1 \in \mathbb{Z}$ whenever $a, b \in \mathbb{Z}$.

To verify associativity, let $a, b, c \in \mathbb{Z}$. Then

$$(a * b) * c = (a + b + 1) + c + 1 = a + b + c + 2,$$

and

$$a * (b * c) = a + (b + c + 1) + 1 = a + b + c + 2.$$

Therefore $(a * b) * c = a * (b * c)$, so the operation is associative.

Also, $a * b = a + b + 1 = b + a + 1 = b * a$, showing that $*$ is commutative.

To determine the identity, suppose $e \in \mathbb{Z}$ satisfies $a * e = a$. Then $a + e + 1 = a$, which gives $e = -1$. Since $(-1) * a = a * (-1) = a$, the element -1 is the identity.

Finally, the inverse of a is an element b satisfying $a * b = -1$. Thus,

$$a + b + 1 = -1,$$

which gives $b = -a - 2$. Hence every element $a \in \mathbb{Z}$ has inverse $-a - 2$.

Consequently, $(\mathbb{Z}, *)$ satisfies the group axioms. In fact, it is an abelian group.

1.1.2 Equivalence Relations

In mathematics, two objects may be regarded as equivalent even when they are not literally equal. For example, two integers may leave the same remainder upon division by a fixed positive integer. Equivalence relations provide a precise mathematical framework for expressing such situations.

Definition 1.1.9. Let S be a nonempty set. A relation $\sim$ on S is called an *equivalence relation* if the following conditions hold for all $a, b, c \in S$:

(i) *Reflexivity:* $a \sim a$.

(ii) *Symmetry:* If $a \sim b$, then $b \sim a$.

(iii) *Transitivity:* If $a \sim b$ and $b \sim c$, then $a \sim c$.

Reflexivity means that every element is equivalent to itself. Symmetry means that equivalence works in both directions, while transitivity ensures that equivalence is preserved through intermediate elements.

Example 1.1.10. Let n be a fixed positive integer. Define a relation on $\mathbb{Z}$ by

$$a \sim b \quad \Longleftrightarrow \quad n \mid (a - b).$$

This relation is called *congruence modulo* n and is written $a \equiv b \pmod{n}$.

We verify that it is an equivalence relation.

Reflexivity. Since $a - a = 0$ and $n \mid 0$, we have $a \equiv a \pmod{n}$.

Symmetry. If $a \equiv b \pmod{n}$, then $n \mid (a - b)$. Since $b - a = -(a - b)$, it follows that $n \mid (b - a)$, so $b \equiv a \pmod{n}$.

Transitivity. Suppose $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$. Then $n \mid (a - b)$ and $n \mid (b - c)$. Consequently,

$$n \mid \big((a - b) + (b - c)\big) = a - c,$$

and hence $a \equiv c \pmod{n}$.

Therefore congruence modulo n is an equivalence relation on $\mathbb{Z}$.

Example 1.1.11. (i) Equality on any set is an equivalence relation.

(ii) On $\mathbb{R}$, define $a \sim b$ if $|a| = |b|$. This is an equivalence relation. For instance, $3 \sim -3$.

(iii) On the set of all people, the relation "has the same month of birth as" is an equivalence relation.

(iv) The usual relation $\leq$ on $\mathbb{R}$ is not an equivalence relation. Although it is reflexive

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and transitive, it is not symmetric. For example, $2 \leq 5$, but $5 \not\leq 2$.

(v) On $\mathbb{Z}$, define $a \sim b$ if $a - b$ is odd. This relation is symmetric, but it is neither reflexive nor transitive. Indeed, $a - a = 0$ is not odd, and although $1 \sim 2$ and $2 \sim 3$, we have $1 \not\sim 3$.

1.1.3 Equivalence Classes

An equivalence relation groups together elements that are equivalent to one another. Each such collection is called an equivalence class.

Definition 1.1.12. Let $\sim$ be an equivalence relation on a set S. For $a \in S$, the *equivalence class* of a is the subset

$$[a] = \{x \in S : x \sim a\}.$$

The element a is called a *representative* of the class $[a]$.

Since $a \sim a$, every equivalence class contains its representative and is therefore nonempty. Different representatives may determine the same equivalence class.

Example 1.1.13. Consider congruence modulo 3 on $\mathbb{Z}$. Its equivalence classes are

$$[0] = \{3k : k \in \mathbb{Z}\},$$
$$[1] = \{3k + 1 : k \in \mathbb{Z}\},$$
$$[2] = \{3k + 2 : k \in \mathbb{Z}\}.$$

By the division algorithm, every integer belongs to exactly one of these three classes.

For example, $8 \equiv 2 \pmod 3$ and $-4 \equiv 2 \pmod 3$. Hence $[8] = [-4] = [2]$.

Similarly, $[6] = [0]$ and $[10] = [1]$. Thus there are exactly three distinct equivalence classes, even though $\mathbb{Z}$ contains infinitely many elements.

Example 1.1.14. On $\mathbb{R}$, define $a \sim b$ if $|a| = |b|$. The equivalence class of a is

$$[a] = \{a, -a\}.$$

In particular, $[0] = \{0\}$, while $[5] = \{5, -5\}$. This shows that equivalence classes need not contain the same number of elements.

The following result provides a useful criterion for determining when two elements represent the same equivalence class.

Proposition 1.1.15. *Let $\sim$ be an equivalence relation on S. For any $a, b \in S$,*

$$[a] = [b] \quad \Longleftrightarrow \quad a \sim b.$$

Proof. Suppose $[a] = [b]$. Since $a \in [a]$, we also have $a \in [b]$. By the definition of $[b]$, this gives $a \sim b$.

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Conversely, suppose $a \sim b$. Let $x \in [a]$. Then $x \sim a$, and transitivity gives $x \sim b$. Thus $x \in [b]$, proving $[a] \subseteq [b]$.

By symmetry, $b \sim a$, and the same argument gives $[b] \subseteq [a]$. Hence $[a] = [b]$. ■

1.1.4 Partitions and Equivalence Relations

Definition 1.1.16. A *partition* of a nonempty set S is a collection $\{S_i : i \in I\}$ of nonempty subsets of S such that

(i) $S = \bigcup_{i \in I} S_i$;

(ii) $S_i \cap S_j = \varnothing$ whenever $i \neq j$.

A partition divides a set into nonempty, pairwise disjoint subsets whose union is the entire set. Every element therefore belongs to exactly one part of the partition.

Example 1.1.17. Let $S = \{1, 2, 3, 4, 5, 6\}$. The collection

$$\{\{1, 3, 5\}, \{2, 4, 6\}\}$$

is a partition of S into odd and even elements.

However, $\{\{1, 2, 3\}, \{3, 4, 5, 6\}\}$ is not a partition because the two subsets overlap.

Theorem 1.1.18. *Let $\sim$ be an equivalence relation on a nonempty set S. Then the collection of all distinct equivalence classes of S forms a partition of S.*

Proof. We verify the defining properties of a partition.

First, every equivalence class is nonempty. Indeed, for each $a \in S$, reflexivity gives $a \sim a$, so $a \in [a]$.

Next, every element of S belongs to an equivalence class, namely its own. Consequently, the union of all equivalence classes is S.

It remains to show that two distinct equivalence classes are disjoint. Suppose $[a] \cap [b] \neq \varnothing$. Choose $c \in [a] \cap [b]$. Then $c \sim a$ and $c \sim b$. By symmetry, $a \sim c$, and hence transitivity gives $a \sim b$.

By the preceding proposition, $[a] = [b]$. Therefore any two equivalence classes are either equal or disjoint.

Thus the collection of all distinct equivalence classes forms a partition of S. ■

The converse is equally important: every partition determines an equivalence relation.

Theorem 1.1.19. *Let $\{S_i : i \in I\}$ be a partition of a nonempty set S. Define a relation $\sim$ on S by declaring $a \sim b$ if a and b belong to the same member of the partition. Then $\sim$ is an equivalence relation.*

Proof. Since the subsets S_i form a partition, every element $a \in S$ belongs to some S_i. Thus a and itself belong to the same subset, giving $a \sim a$. Hence the relation is reflexive.

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If $a \sim b$, then a and b belong to the same subset S_i. Consequently, b and a belong to that same subset, so $b \sim a$. Thus the relation is symmetric.

Finally, suppose $a \sim b$ and $b \sim c$. Then $a, b \in S_i$ and $b, c \in S_j$ for some $i, j \in I$. Since $b \in S_i \cap S_j$ and the members of a partition are pairwise disjoint, we must have $S_i = S_j$. Therefore a and c belong to the same subset, so $a \sim c$.

Hence $\sim$ is an equivalence relation. ■

1.2 Groups and Their Basic Properties

A group is one of the fundamental structures of abstract algebra. It consists of a set equipped with a binary operation satisfying certain conditions that make it possible to combine elements, identify an element that leaves others unchanged, and reverse the effect of an operation.

Many familiar mathematical systems possess a group structure. The integers under addition, the nonzero real numbers under multiplication, invertible matrices under matrix multiplication, and permutations under composition are important examples. Although these systems differ considerably in their nature, they satisfy the same fundamental algebraic laws.

The concept of a group provides a common framework for studying these systems and the properties they share.

Definition 1.2.1. A *group* is a nonempty set G equipped with a binary operation $*$ satisfying the following conditions:

(G1) *Associativity:* $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$.

(G2) *Identity element:* There exists an element $e \in G$ such that $e * a = a * e = a$ for every $a \in G$.

(G3) *Inverse element:* For every $a \in G$, there exists $b \in G$ such that $a * b = b * a = e$.

The element e is called an *identity element*, and b is called an *inverse* of a.

Since $*$ is a binary operation on G, the closure condition $a * b \in G$ is already included in the definition. Nevertheless, closure is often stated explicitly when verifying whether a particular set and operation form a group.

A group is generally denoted by $(G, *)$. When the operation is understood, we simply write G. Two common notational conventions are used:

(i) In *additive notation*, the operation is written $+$, the identity is denoted by 0, and the inverse of a is written $-a$.

(ii) In *multiplicative notation*, the operation is written by juxtaposition, the identity is denoted by e or 1, and the inverse of a is written a^{-1}.

These are not different definitions of a group; they are simply different ways of writing the same algebraic structure.

Example 1.2.2. The set $\mathbb{Z}$ forms a group under ordinary addition. Indeed, the sum of two integers is an integer, addition is associative, 0 is the identity element, and every integer a has additive inverse $-a$. Thus $(\mathbb{Z}, +)$ is a group.

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The same argument shows that $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, and $(\mathbb{C}, +)$ are groups.

Example 1.2.3. Let $G = \mathbb{Q} \setminus \{0\}$. Then $(G, \cdot)$ is a group under ordinary multiplication.

The product of two nonzero rational numbers is again a nonzero rational number, and multiplication is associative. The identity element is 1, while the inverse of $a \in G$ is $1/a$, which also belongs to G.

Similarly, $\mathbb{R} \setminus \{0\}$ and $\mathbb{C} \setminus \{0\}$ are groups under multiplication.

Example 1.2.4. Let $n \geq 2$ and consider the set of residue classes

$$\mathbb{Z}_n = \{[0], [1], \ldots, [n-1]\}.$$

Define addition by $[a] + [b] = [a + b]$, where the class on the right is taken modulo n.

This operation is well-defined. Indeed, if $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$, then n divides both $a - a'$ and $b - b'$. Consequently,

$$n \mid \big((a + b) - (a' + b')\big),$$

so $[a + b] = [a' + b']$.

Associativity follows from integer addition. The identity element is $[0]$, and the additive inverse of $[a]$ is $[-a]$. Thus $(\mathbb{Z}_n, +)$ is a group.

For example, in $\mathbb{Z}_5$, we have $[3] + [4] = [2]$ and $[3] + [2] = [0]$. Hence $[2]$ is the additive inverse of $[3]$.

Example 1.2.5. Let $M_n(\mathbb{R})$ denote the set of all $n \times n$ matrices with real entries, and define

$$GL_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det A \neq 0\}.$$

Then $GL_n(\mathbb{R})$ is a group under matrix multiplication.

If $A, B \in GL_n(\mathbb{R})$, then $\det(AB) = \det(A)\det(B) \neq 0$, so $AB \in GL_n(\mathbb{R})$. Matrix multiplication is associative, and the identity matrix I_n is the identity element.

Furthermore, every matrix $A \in GL_n(\mathbb{R})$ is invertible, and its inverse A^{-1} also has nonzero determinant. Therefore $GL_n(\mathbb{R})$ satisfies all the group axioms.

Not every set with a binary operation forms a group. The failure of even one group axiom is sufficient to show that a given structure is not a group.

Example 1.2.6. (i) $(\mathbb{N}, +)$, where $\mathbb{N} = \{1, 2, 3, \ldots\}$, is not a group because it does not contain the additive identity 0.

(ii) $(\mathbb{Z}, \cdot)$ is not a group because most integers do not have multiplicative inverses in $\mathbb{Z}$. In particular, there is no integer b satisfying $2b = 1$.

(iii) $(\mathbb{Z}, -)$ is not a group because subtraction is not associative.

(iv) $M_n(\mathbb{R})$ under matrix multiplication is not a group, since singular matrices do not

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possess multiplicative inverses.

(v) $(\mathbb{N}_0, +)$, where $\mathbb{N}_0 = \{0, 1, 2, \dots\}$, has an identity but is not a group, since positive integers have no additive inverses in $\mathbb{N}_0$.

1.2.1 Finite, Infinite, Abelian and Nonabelian Groups

Groups may be classified according to the number of elements they contain and whether their operation is commutative.

Definition 1.2.7. The *order* of a group G, denoted by $|G|$, is the number of elements in G. If G has finitely many elements, it is called a *finite group*; otherwise, it is called an *infinite group*.

For example, the additive group $\mathbb{Z}_n$ has order n, whereas the additive groups $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ are infinite.

Definition 1.2.8. A group G is called *abelian* or *commutative* if

$$ab = ba$$

for every $a, b \in G$. A group that is not abelian is called *nonabelian*.

To establish that a group is abelian, one must show that every pair of its elements commutes. To establish that it is nonabelian, it is sufficient to find a single pair of elements that does not commute.

Example 1.2.9. The groups $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, and $(\mathbb{Z}_n, +)$ are abelian because addition is commutative.

The multiplicative groups $\mathbb{Q} \setminus \{0\}$ and $\mathbb{R} \setminus \{0\}$ are also abelian.

Example 1.2.10. Consider the matrices

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Both belong to $GL_2(\mathbb{R})$, since their determinants are nonzero. However,

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Thus $AB \neq BA$, proving that $GL_2(\mathbb{R})$ is nonabelian.

1.2.2 Permutation Groups and Symmetries

An important source of groups arises from permutations and geometric symmetries.

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Definition 1.2.11. A *permutation* of a nonempty set S is a bijective mapping from S onto itself.

The composition of two permutations is again a permutation. Since function composition is associative, the identity mapping is an identity element, and every permutation has an inverse, the set of all permutations of S forms a group under composition.

When $S = \{1, 2, \ldots, n\}$, this group is called the *symmetric group of degree* n, denoted by S_n.

Proposition 1.2.12. *The symmetric group S_n has order $n!$.*

Proof. A permutation is completely determined by the images of $1, 2, \ldots, n$. There are n possible images for 1, followed by $n - 1$ choices for the image of 2, and so on. Since distinct elements must have distinct images, the number of permutations is

$$n(n-1)\cdots 2 \cdot 1 = n!.$$

Therefore $|S_n| = n!$. ■

Example 1.2.13. The symmetric group on $\{1, 2, 3\}$ consists of six permutations:

$$S_3 = \{e, (12), (13), (23), (123), (132)\}.$$

Here (12) denotes the permutation interchanging 1 and 2 while fixing 3, and (123) sends 1 to 2, 2 to 3, and 3 to 1.

Let $\sigma = (12)$ and $\tau = (23)$. Using the convention that the rightmost permutation is applied first, we obtain

$$\sigma\tau = (123), \qquad \tau\sigma = (132).$$

Since $\sigma\tau \neq \tau\sigma$, the group S_3 is nonabelian.

Example 1.2.14. The symmetries of an equilateral triangle consist of three rotations, including the identity, and three reflections.

Composition of symmetries is associative, the identity symmetry leaves the triangle unchanged, and every symmetry has an inverse. Consequently, these six symmetries form a group.

If the three vertices are labelled $1, 2, 3$, every symmetry determines a permutation of the vertices. In fact, every permutation of the three vertices is induced by exactly one symmetry. Thus the symmetry group of an equilateral triangle is isomorphic to S_3.

1.2.3 Elementary Properties of Groups

The group axioms have several important consequences. We now establish the elementary results that will be used repeatedly in the study of subgroups, cosets, and homomorphisms.

Theorem 1.2.15. *Every group has a unique identity element.*

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Proof. Suppose e and f are both identity elements of a group G. Since e is an identity, $ef = f$. Since f is also an identity, $ef = e$. Hence $e = f$, proving that the identity element is unique. ■

Theorem 1.2.16. *Every element of a group has a unique inverse.*

Proof. Let $a \in G$, and suppose b and c are both inverses of a. Then $ab = ba = e$ and $ac = ca = e$. By associativity,

$$b = be = b(ac) = (ba)c = ec = c.$$

Therefore $b = c$, proving uniqueness. ■

Since the inverse of each element is unique, it may be denoted unambiguously by a^{-1}.

Theorem 1.2.17. *Let G be a group and let $a, b, c \in G$. Then:*

(i) $ab = ac$ *implies* $b = c$;

(ii) $ba = ca$ *implies* $b = c$.

Proof. Suppose $ab = ac$. Multiplying on the left by a^{-1} and using associativity, we obtain

$$a^{-1}(ab) = a^{-1}(ac),$$

so $(a^{-1}a)b = (a^{-1}a)c$, giving $eb = ec$. Hence $b = c$.

Similarly, if $ba = ca$, multiplication on the right by a^{-1} yields

$$(ba)a^{-1} = (ca)a^{-1},$$

which simplifies to $be = ce$. Thus $b = c$. ■

Remark 1.2.18. In a nonabelian group, cancellation is valid only when the common factor occurs on the same side. In particular, the equation $ab = ca$ does not generally imply $b = c$.

Theorem 1.2.19. *For every $a, b \in G$,*

(i) $(a^{-1})^{-1} = a$;

(ii) $(ab)^{-1} = b^{-1}a^{-1}$;

(iii) $e^{-1} = e$.

Proof. Since $aa^{-1} = a^{-1}a = e$, the element a is an inverse of a^{-1}. Uniqueness of inverses gives $(a^{-1})^{-1} = a$.

Next, consider $b^{-1}a^{-1}$. By associativity,

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aa^{-1} = e.$$

Similarly,

$$(b^{-1}a^{-1})(ab) = b^{-1}(a^{-1}a)b = b^{-1}b = e.$$

Thus $b^{-1}a^{-1}$ is the inverse of ab, proving the second assertion.

Finally, since $ee = e$, the identity element is its own inverse. ■

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Corollary 1.2.20. *For any $a_1, a_2, \ldots, a_n \in G$,*

$$(a_1 a_2 \cdots a_n)^{-1} = a_n^{-1} \cdots a_2^{-1} a_1^{-1}.$$

Proof. The result follows by repeated application of the identity $(ab)^{-1} = b^{-1}a^{-1}$. ■

Proposition 1.2.21. *Let $a, b \in G$. Then:*

(i) the equation $ax = b$ has the unique solution $x = a^{-1}b$;

(ii) the equation $ya = b$ has the unique solution $y = ba^{-1}$.

Proof. If $ax = b$, multiplication on the left by a^{-1} gives

$$x = (a^{-1}a)x = a^{-1}b.$$

Conversely, $a(a^{-1}b) = (aa^{-1})b = b$, so this element is indeed a solution. Uniqueness follows from left cancellation.

Similarly, from $ya = b$, multiplication on the right by a^{-1} gives $y = ba^{-1}$. Substitution verifies the solution, and right cancellation establishes uniqueness. ■

Example 1.2.22. Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

We solve $AX = B$ in $GL_2(\mathbb{R})$.

Since

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix},$$

the unique solution is

$$X = A^{-1}B = \begin{pmatrix} 2 & -3 \\ 0 & 3 \end{pmatrix}.$$

Indeed, direct multiplication gives $AX = B$.

Notice that the order of multiplication is essential: the solution is $A^{-1}B$, not BA^{-1}.

Proposition 1.2.23. *A group G is abelian if and only if*

$$(ab)^{-1} = a^{-1}b^{-1}$$

for all $a, b \in G$.

Proof. If G is abelian, then $b^{-1}a^{-1} = a^{-1}b^{-1}$. Hence the usual inverse law gives $(ab)^{-1} = a^{-1}b^{-1}$.

Conversely, suppose $(ab)^{-1} = a^{-1}b^{-1}$ for all $a, b \in G$. Since $(ab)^{-1} = b^{-1}a^{-1}$ in every group, we obtain

$$a^{-1}b^{-1} = b^{-1}a^{-1}.$$

Taking inverses of both sides gives $ba = ab$. Since a, b were arbitrary, G is abelian. ■

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1.2.4 Powers of Group Elements

Let G be a group, and let $a \in G$. For a positive integer n, the power a^n denotes the product of n copies of a. We also define $a^0 = e$ and $a^{-n} = (a^{-1})^n$.

The familiar laws of exponents remain valid for powers of a fixed group element.

Proposition 1.2.24. *For every $a \in G$ and $r, s \in \mathbb{Z}$,*

(i) $a^r a^s = a^{r+s}$;

(ii) $(a^r)^s = a^{rs}$;

(iii) $(a^r)^{-1} = a^{-r}$.

Proof. For positive integers r, s, the first identity follows from associativity by combining r copies of a with s additional copies. The identity remains valid when either exponent is zero.

For negative exponents, the relation $a^{-m} = (a^{-1})^m$, together with $aa^{-1} = a^{-1}a = e$, allows opposite powers to cancel. This establishes $a^r a^s = a^{r+s}$ for arbitrary integers r, s.

The second identity follows by repeated application of the first when $s > 0$, is immediate when $s = 0$, and follows from inverses when $s < 0$.

Finally, since $a^r a^{-r} = a^0 = e = a^{-r}a^r$, the uniqueness of inverses gives $(a^r)^{-1} = a^{-r}$. ■

Remark 1.2.25. The exponent laws above concern powers of the same element. For arbitrary $a, b \in G$, the identity $(ab)^n = a^n b^n$ need not hold.

However, if $ab = ba$, then $(ab)^n = a^n b^n$ for every integer n.

1.2.5 Order of an Element

The order of a group is the number of elements it contains. The order of an individual element is a different, though closely related, concept.

Definition 1.2.26. Let a be an element of a group G. If there exists a positive integer n such that $a^n = e$, the least such integer is called the *order of* a, denoted by $\operatorname{ord}(a)$.

If no positive integer n satisfies $a^n = e$, the element a is said to have *infinite order*.

For additive groups, the condition $a^n = e$ is replaced by $na = 0$, where na denotes the sum of n copies of a.

Example 1.2.27. (i) In $(\mathbb{Z}, +)$, the identity 0 has order 1, while every nonzero integer has infinite order.

(ii) In $(\mathbb{Z}_8, +)$, the element $[2]$ has order 4, since

$$[2], \quad [4], \quad [6], \quad [0]$$

are its successive positive multiples.

(iii) In $(\mathbb{Z}_6, +)$, the element $[2]$ has order 3, while $[3]$ has order 2.

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(iv) In $GL_2(\mathbb{R})$, the matrix

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

has order 4. Indeed, $A^2 = -I_2$, $A^3 = -A$, and $A^4 = I_2$, with no smaller positive power equal to I_2.

Theorem 1.2.28. *Let $a \in G$ have finite order n. Then, for every integer k,*

$$a^k = e \iff n \mid k.$$

Proof. Suppose $a^k = e$. By the division algorithm, there exist integers q, r such that $k = qn + r$, where $0 \leq r < n$. Therefore,

$$e = a^k = a^{qn+r} = (a^n)^q a^r = a^r.$$

Since n is the least positive integer for which $a^n = e$, the only possibility is $r = 0$. Hence $k = qn$, showing that $n \mid k$.

Conversely, if $n \mid k$, then $k = qn$ for some integer q. Consequently, $a^k = (a^n)^q = e$, as required. ■

Corollary 1.2.29. *If $\operatorname{ord}(a) = n$, then*

$$a^r = a^s \iff r \equiv s \pmod{n}$$

for all integers r, s.

Proof. We have $a^r = a^s$ if and only if $a^{r-s} = e$. By the preceding theorem, this holds precisely when $n \mid (r - s)$, which is equivalent to $r \equiv s \pmod{n}$. ■

Proposition 1.2.30. *An element a has infinite order if and only if all the powers a^k, $k \in \mathbb{Z}$, are distinct.*

Proof. Suppose a has infinite order. If $a^r = a^s$ for integers $r > s$, then $a^{r-s} = e$, contradicting the infinite order of a. Hence distinct integers give distinct powers.

Conversely, if a has finite order n, then $a^n = e = a^0$, so its powers are not all distinct. ■

Theorem 1.2.31. *Suppose $a \in G$ has finite order n. Then, for every integer k,*

$$\operatorname{ord}(a^k) = \frac{n}{\gcd(n,k)}.$$

Here $\gcd(n,k)$ denotes the positive greatest common divisor.

Proof. Put $d = \gcd(n,k)$, and write $n = dn_1$ and $k = dk_1$, where $\gcd(n_1, k_1) = 1$.

Since

$$(a^k)^{n_1} = a^{kn_1} = a^{k_1 n} = e,$$

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the element a^k has finite order at most n_1.

Now suppose $(a^k)^m = e$ for some positive integer m. Then $a^{km} = e$, so $n \mid km$. Substituting $n = dn_1$ and $k = dk_1$, we obtain $n_1 \mid k_1 m$.

Since $\gcd(n_1, k_1) = 1$, Euclid's lemma implies $n_1 \mid m$. Thus every positive exponent giving $(a^k)^m = e$ is a multiple of n_1.

Therefore the least such exponent is $n_1 = n/d$, proving the result. ■

Example 1.2.32. In $(\mathbb{Z}_{12}, +)$, the element $[1]$ has order 12. Consequently, the order of $[8] = 8[1]$ is

$$\operatorname{ord}([8]) = \frac{12}{\gcd(12, 8)} = \frac{12}{4} = 3.$$

Indeed, $[8] + [8] = [4]$ and $[8] + [8] + [8] = [0]$.

Similarly, $\operatorname{ord}([9]) = 12/\gcd(12, 9) = 4$.

1.3 Subgroups

A group may contain subsets that themselves form groups under the same operation. Such subsets are called *subgroups*. The study of subgroups provides a means of understanding the structure of a group through its smaller algebraic systems.

For example, the even integers form a group under addition and are contained in the additive group of all integers. Similarly, certain collections of invertible matrices form groups under matrix multiplication. These are instances of subgroups.

Definition 1.3.1. Let $(G, *)$ be a group. A nonempty subset H of G is called a *subgroup* of G if $(H, *)$ is itself a group under the operation inherited from G. We write $H \leq G$.

Every group G contains at least two subgroups, namely $\{e\}$ and G itself, where e is the identity element of G. These are called the *trivial subgroups*. A subgroup H is called *proper* if $H \neq G$, and we write $H < G$.

Since the operation on H is inherited from G, associativity holds automatically in H. Thus, to determine whether a nonempty subset is a subgroup, it is sufficient to verify closure, the presence of the identity, and the existence of inverses.

The following theorem combines these requirements into a single condition.

Theorem 1.3.2 (Subgroup Test). *A nonempty subset H of a group G is a subgroup of G if and only if*

$$ab^{-1} \in H$$

for all $a, b \in H$.

Proof. Suppose first that H is a subgroup of G. Since H is a group, it contains the inverse b^{-1} of every $b \in H$. Closure under the group operation then gives $ab^{-1} \in H$ for all $a, b \in H$.

Conversely, suppose H is nonempty and satisfies the stated condition. We show that H is a group under the operation inherited from G.

Since H is nonempty, choose $a \in H$. Taking $b = a$ gives $aa^{-1} = e \in H$. Thus H contains the identity of G.

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Next, let $b \in H$. Since $e \in H$, the assumed condition gives $eb^{-1} = b^{-1} \in H$. Hence every element of H has its inverse in H.

Finally, let $a, b \in H$. Since $b^{-1} \in H$, applying the condition to a and b^{-1} yields

$$a(b^{-1})^{-1} = ab \in H.$$

Therefore H is closed under multiplication.

Associativity is inherited from G, so H satisfies all the group axioms and is a subgroup of G. ■

Example 1.3.3. For any integer m, define $m\mathbb{Z} = \{mk : k \in \mathbb{Z}\}$. We show that $m\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$.

Since $0 = m \cdot 0$, the set $m\mathbb{Z}$ is nonempty. If $ma, mb \in m\mathbb{Z}$, then

$$ma - mb = m(a - b) \in m\mathbb{Z}.$$

The subgroup test therefore gives $m\mathbb{Z} \leq \mathbb{Z}$.

In particular, $2\mathbb{Z}$ is the subgroup of even integers, while $3\mathbb{Z}$ consists of all multiples of 3. For $m = 0$, we obtain the trivial subgroup $\{0\}$.

Example 1.3.4. Consider the additive group $\mathbb{Z}_{12}$ and the subset

$$H = \{[0], [3], [6], [9]\}.$$

The sum or difference modulo 12 of any two elements of H again belongs to H. Hence H is a subgroup of $\mathbb{Z}_{12}$.

On the other hand, $K = \{[0], [4], [6]\}$ is not a subgroup, since $[4] + [6] = [10] \notin K$. Thus K is not closed under addition.

Example 1.3.5. Let $GL_n(\mathbb{R})$ be the group of all invertible $n \times n$ real matrices, and define

$$SL_n(\mathbb{R}) = \{A \in GL_n(\mathbb{R}) : \det A = 1\}.$$

Then $SL_n(\mathbb{R})$ is a subgroup of $GL_n(\mathbb{R})$.

Indeed, the identity matrix belongs to $SL_n(\mathbb{R})$. Moreover, if $A, B \in SL_n(\mathbb{R})$, then

$$\det(AB^{-1}) = \det(A)\det(B^{-1}) = \frac{\det A}{\det B} = 1.$$

Consequently $AB^{-1} \in SL_n(\mathbb{R})$, and the subgroup test proves the assertion.

1.3.1 Subgroup Test for Finite Subsets

For a finite subset of a group, the subgroup test can be simplified considerably.

Theorem 1.3.6 (Finite Subgroup Test). *A nonempty finite subset H of a group G is a subgroup of G if and only if H is closed under the group operation.*

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Proof. If H is a subgroup, closure follows immediately.

Conversely, suppose H is a nonempty finite subset of G that is closed under multiplication. Let $a \in H$. By closure, every positive power a^k belongs to H.

Since H is finite, two positive powers must be equal. Thus $a^r = a^s$ for some integers $r > s \geq 1$. Multiplying by a^{-s} in G, we obtain $a^{r-s} = e$. Since $r - s \geq 1$, closure implies $e \in H$.

Moreover, $a^{r-s-1} \in H$, where $a^0 = e$, and

$$aa^{r-s-1} = a^{r-s} = e.$$

Thus $a^{r-s-1} = a^{-1} \in H$.

Therefore H contains the identity and the inverse of each of its elements. Together with closure and associativity inherited from G, this shows that H is a subgroup. ■

Remark 1.3.7. The finiteness assumption is essential. For example, $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$ is closed under addition but is not a subgroup of $(\mathbb{Z}, +)$, since it does not contain the additive inverses of positive integers.

1.3.2 Intersections and Unions of Subgroups

The intersection of subgroups always retains the group structure. This observation is useful in constructing new subgroups from known ones.

Theorem 1.3.8. *Let $\{H_i : i \in I\}$ be a nonempty family of subgroups of a group G. Then*

$$H = \bigcap_{i \in I} H_i$$

is a subgroup of G.

Proof. Since every subgroup contains the identity e of G, we have $e \in H_i$ for all $i \in I$. Hence $e \in H$, so H is nonempty.

Let $a, b \in H$. Then $a, b \in H_i$ for every $i \in I$. Since each H_i is a subgroup, $ab^{-1} \in H_i$ for every $i \in I$. Therefore $ab^{-1} \in H$, and the subgroup test shows that H is a subgroup of G. ■

Unlike intersections, unions of subgroups need not be subgroups.

Example 1.3.9. Consider the subgroups $2\mathbb{Z}$ and $3\mathbb{Z}$ of $(\mathbb{Z}, +)$. Their union contains 2 and 3, but it does not contain $2 + 3 = 5$. Thus $2\mathbb{Z} \cup 3\mathbb{Z}$ is not a subgroup of $\mathbb{Z}$.

The precise condition under which the union of two subgroups is again a subgroup is given by the following theorem.

Theorem 1.3.10. *Let H and K be subgroups of a group G. Then $H \cup K$ is a subgroup of G if and only if $H \subseteq K$ or $K \subseteq H$.*

Proof. If $H \subseteq K$, then $H \cup K = K$, which is a subgroup. Similarly, if $K \subseteq H$, then $H \cup K = H$ is a subgroup.

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Conversely, suppose $H \cup K$ is a subgroup, but neither subgroup is contained in the other. Choose $h \in H \setminus K$ and $k \in K \setminus H$.

Since $h, k \in H \cup K$ and this union is a subgroup, we have $hk \in H \cup K$.

If $hk \in H$, then $k = h^{-1}(hk) \in H$, a contradiction.

If $hk \in K$, then $h = (hk)k^{-1} \in K$, again a contradiction.

Therefore one of the inclusions $H \subseteq K$ or $K \subseteq H$ must hold. ■

1.3.3 Cyclic Subgroups

An especially important subgroup is obtained by taking all integer powers of a single group element.

Definition 1.3.11. Let $a \in G$. The subset

$$\langle a \rangle = \{a^k : k \in \mathbb{Z}\}$$

is called the *cyclic subgroup generated by* a. The element a is called a *generator* of $\langle a \rangle$.

Theorem 1.3.12. *For every $a \in G$, the set $\langle a \rangle$ is a subgroup of G. Moreover, it is the smallest subgroup of G containing a.*

Proof. Since $a^0 = e$, the set $\langle a \rangle$ is nonempty. If $a^r, a^s \in \langle a \rangle$, then

$$a^r (a^s)^{-1} = a^{r-s} \in \langle a \rangle.$$

Thus $\langle a \rangle$ is a subgroup by the subgroup test.

Now let H be any subgroup of G containing a. Since H is closed under multiplication and inverses, it contains every integer power of a. Hence $\langle a \rangle \subseteq H$.

Therefore $\langle a \rangle$ is the smallest subgroup of G containing a. ■

Proposition 1.3.13. *For every $a \in G$,*

$$|\langle a \rangle| = \operatorname{ord}(a),$$

where both sides are understood to be infinite when a has infinite order.

Proof. Suppose $\operatorname{ord}(a) = n < \infty$. Then $a^n = e$, and every integer power of a is equal to one of $e, a, a^2, \ldots, a^{n-1}$.

These n elements are distinct. Indeed, if $a^r = a^s$ for $0 \leq s < r < n$, then $a^{r-s} = e$, contradicting the minimality of n.

Therefore $\langle a \rangle = \{e, a, \ldots, a^{n-1}\}$ and $|\langle a \rangle| = n$.

If a has infinite order, then all its integer powers are distinct, so $\langle a \rangle$ is infinite. ■

Example 1.3.14. In the additive group $\mathbb{Z}_{12}$, the cyclic subgroup generated by $[4]$ is

$$\langle [4] \rangle = \{[0], [4], [8]\}.$$

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Thus $\operatorname{ord}([4]) = 3$. Similarly,

$$\langle [3] \rangle = \{[0], [3], [6], [9]\},$$

so $\operatorname{ord}([3]) = 4$.

1.4 Cosets and Lagrange's Theorem

Subgroups allow us to organize the elements of a group into collections called *cosets*. These collections partition the group into disjoint subsets of equal size. This observation leads to Lagrange's theorem, which establishes a fundamental relationship between the orders of a finite group and its subgroups.

1.4.1 Left and Right Cosets

Definition 1.4.1. Let H be a subgroup of a group G, and let $a \in G$. The set

$$aH = \{ah : h \in H\}$$

is called the *left coset* of H determined by a. Similarly,

$$Ha = \{ha : h \in H\}$$

is called the *right coset* of H determined by a.

A coset is obtained by multiplying every element of a subgroup by a fixed element of the group. The distinction between left and right cosets is important in nonabelian groups, since the two sets need not be equal.

Example 1.4.2. Consider the additive group $(\mathbb{Z}, +)$ and its subgroup $H = 4\mathbb{Z}$. Since the operation is addition, the cosets are written as $a + H$.

The distinct cosets are

$$\begin{aligned} H &= \{\ldots, -8, -4, 0, 4, 8, \ldots\}, \\ 1 + H &= \{\ldots, -7, -3, 1, 5, 9, \ldots\}, \\ 2 + H &= \{\ldots, -6, -2, 2, 6, 10, \ldots\}, \\ 3 + H &= \{\ldots, -5, -1, 3, 7, 11, \ldots\}. \end{aligned}$$

These are precisely the four residue classes modulo 4. Every integer belongs to exactly one of them.

Example 1.4.3. Let $G = \mathbb{Z}_6$ and $H = \{[0], [3]\}$. The distinct cosets of H are

$$H = \{[0], [3]\}, \qquad [1] + H = \{[1], [4]\}, \qquad [2] + H = \{[2], [5]\}.$$

Thus the six elements of $\mathbb{Z}_6$ are partitioned into three cosets, each containing two elements.

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1.4.2 Equality and Partition of Cosets

We first establish a criterion for determining when two elements produce the same left coset.

Theorem 1.4.4. *Let $H \leq G$ and let $a, b \in G$. Then*

$$aH = bH \iff a^{-1}b \in H.$$

Proof. Suppose $aH = bH$. Since $b = be \in bH$, we have $b \in aH$. Thus $b = ah$ for some $h \in H$, giving $a^{-1}b = h \in H$.

Conversely, suppose $a^{-1}b \in H$. Then $b = ah$ for some $h \in H$. Consequently,

$$bH = (ah)H = a(hH).$$

Since $h \in H$ and H is a subgroup, $hH = H$. Hence $bH = aH$. ■

Corollary 1.4.5. *For $H \leq G$ and $a, b \in G$,*

$$Ha = Hb \iff ab^{-1} \in H.$$

Proof. The proof is analogous to that of the left coset criterion, with multiplication performed on the right. ■

Theorem 1.4.6. *Let H be a subgroup of G. Then any two left cosets of H are either equal or disjoint, and their union is G.*

Proof. Suppose $aH \cap bH \neq \varnothing$. Then there exists $x \in G$ such that $x = ah_1 = bh_2$ for some $h_1, h_2 \in H$.

It follows that

$$a^{-1}b = h_1 h_2^{-1} \in H.$$

By the coset equality criterion, $aH = bH$.

Thus any two distinct left cosets are disjoint.

Furthermore, every $a \in G$ belongs to aH, since $a = ae$ and $e \in H$. Hence the union of all left cosets is G.

Therefore the distinct left cosets form a partition of G. ■

1.4.3 Cardinality of Cosets

A remarkable property of cosets is that every coset contains exactly as many elements as the subgroup from which it is formed.

Theorem 1.4.7. *Let $H \leq G$ and $a \in G$. Then H and aH have the same cardinality. In particular, if H is finite, then*

$$|aH| = |H|.$$

Proof. Define $\phi : H \to aH$ by $\phi(h) = ah$.

The map is surjective by the definition of aH. If $\phi(h_1) = \phi(h_2)$, then $ah_1 = ah_2$. By left cancellation, $h_1 = h_2$, so ϕ is injective.

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Thus ϕ is a bijection, proving that H and aH have the same cardinality. ■

Definition 1.4.8. Let H be a subgroup of G. The number of distinct left cosets of H in G is called the *index* of H in G, denoted by $[G : H]$.

Example 1.4.9. In $(\mathbb{Z}, +)$, the subgroup $5\mathbb{Z}$ has exactly five distinct cosets:

$$5\mathbb{Z}, \quad 1 + 5\mathbb{Z}, \quad 2 + 5\mathbb{Z}, \quad 3 + 5\mathbb{Z}, \quad 4 + 5\mathbb{Z}.$$

Therefore $[\mathbb{Z} : 5\mathbb{Z}] = 5$.

1.4.4 Lagrange's Theorem

The partition of a group into cosets, together with their equal cardinality, yields one of the most important results in finite group theory.

Theorem 1.4.10 (Lagrange's Theorem). *Let G be a finite group and let H be a subgroup of G. Then*

$$|G| = [G : H]\,|H|.$$

Consequently, the order of every subgroup of a finite group divides the order of the group.

Proof. Let $[G : H] = k$, and let $a_1H, a_2H, \ldots, a_kH$ be the distinct left cosets of H in G.

Since the cosets form a partition of G,

$$G = a_1H \cup a_2H \cup \cdots \cup a_kH,$$

where the union is disjoint.

By the equal cardinality theorem, every coset contains exactly $|H|$ elements. Therefore,

$$|G| = \sum_{i=1}^{k} |a_iH| = k|H| = [G : H]|H|.$$

Since $[G : H]$ is a positive integer, $|H|$ divides $|G|$. ■

Example 1.4.11. Let $G = \mathbb{Z}_{12}$ and $H = \{[0], [4], [8]\}$. Then $|G| = 12$ and $|H| = 3$.

By Lagrange's theorem,

$$[G : H] = \frac{|G|}{|H|} = \frac{12}{3} = 4.$$

The four distinct cosets are

$$H, \quad [1] + H, \quad [2] + H, \quad [3] + H.$$

Each coset contains three elements, and together they contain all twelve elements of G.

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1.4.5 Consequences of Lagrange's Theorem

Lagrange's theorem has several important consequences concerning the orders of elements and the structure of finite groups.

Corollary 1.4.12. *If G is a finite group and $a \in G$, then*

$$\operatorname{ord}(a) \mid |G|.$$

Proof. The cyclic subgroup $\langle a \rangle$ has order $\operatorname{ord}(a)$. Since $\langle a \rangle \leq G$, Lagrange's theorem gives

$$|\langle a \rangle| \mid |G|.$$

Therefore $\operatorname{ord}(a) \mid |G|$. ■

Corollary 1.4.13. *If G is a finite group, then $a^{|G|} = e$ for every $a \in G$.*

Proof. Let $\operatorname{ord}(a) = n$. By the preceding result, $n \mid |G|$. Hence $|G| = kn$ for some positive integer k, and

$$a^{|G|} = a^{kn} = (a^n)^k = e.$$

■

Corollary 1.4.14. *Every group of prime order is cyclic.*

Proof. Let G be a group of prime order p. Choose $a \in G$ with $a \neq e$.

By Lagrange's theorem, $\operatorname{ord}(a)$ divides p. Since $a \neq e$, its order is not 1. As p is prime, we must have $\operatorname{ord}(a) = p$.

Thus the cyclic subgroup $\langle a \rangle$ has p elements. Since $|G| = p$, it follows that $\langle a \rangle = G$. Therefore G is cyclic. ■

Corollary 1.4.15. *A group of prime order has no proper nontrivial subgroups.*

Proof. Let $|G| = p$, where p is prime, and let $H \leq G$. By Lagrange's theorem, $|H|$ divides p. Therefore $|H| = 1$ or $|H| = p$.

In the first case, $H = \{e\}$; in the second case, $H = G$. Hence there are no proper nontrivial subgroups. ■

Corollary 1.4.16. *Every group of prime order is abelian.*

Proof. Every group of prime order is cyclic. Since every cyclic group is abelian, the result follows. ■

Example 1.4.17. Can a group of order 18 contain an element of order 4?

Suppose such an element a exists. Then $\langle a \rangle$ is a subgroup of order 4. By Lagrange's theorem, 4 must divide 18, which is impossible.

Therefore no group of order 18 can contain an element of order 4.

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Remark 1.4.18. Lagrange's theorem gives a necessary condition for the existence of a subgroup of a given order, but the converse is not true in general.

In other words, if d divides $|G|$, it does not follow that G must contain a subgroup of order d.

Example 1.4.19. The alternating group A_4 has order 12, but it has no subgroup of order 6. Thus, although 6 divides 12, there need not exist a subgroup of the corresponding order.

This example shows that the converse of Lagrange's theorem fails in general.

1.5 Cyclic Groups and Their Structure

Among the various classes of groups, cyclic groups occupy a particularly important position because their entire structure is determined by a single element. They provide the simplest examples of groups and play a fundamental role in understanding more general algebraic structures.

We have already seen that every element a of a group G determines a cyclic subgroup $\langle a \rangle$, consisting of all integer powers of a. When this subgroup is the whole group, G is called cyclic.

Definition 1.5.1. A group G is said to be *cyclic* if there exists an element $a \in G$ such that

$$G = \langle a \rangle = \{a^k : k \in \mathbb{Z}\}.$$

Such an element a is called a *generator* of G.

Thus, a cyclic group is one in which every element can be obtained by repeatedly applying the group operation to a single element and its inverse.

In additive notation, the cyclic subgroup generated by a is written

$$\langle a \rangle = \{ka : k \in \mathbb{Z}\},$$

where ka denotes the sum of k copies of a when $k > 0$, with the usual interpretation for $k = 0$ and negative integers.

A cyclic group may have more than one generator. The problem of determining all its generators is an important part of the study of cyclic groups.

Example 1.5.2. The additive group $(\mathbb{Z}, +)$ is cyclic because every integer is an integral multiple of 1. Thus $\mathbb{Z} = \langle 1 \rangle$. It is also generated by -1, since every integer is an integral multiple of -1.

The group $(\mathbb{Z}_6, +)$ is cyclic, with

$$\mathbb{Z}_6 = \langle [1] \rangle = \langle [5] \rangle.$$

However, $[2]$ is not a generator of $\mathbb{Z}_6$, because

$$\langle [2] \rangle = \{[0], [2], [4]\}.$$

This is a proper subgroup of $\mathbb{Z}_6$.

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Example 1.5.3. Consider the group of units modulo 10,

$$U(10) = \{[1], [3], [7], [9]\},$$

under multiplication modulo 10.

The successive powers of $[3]$ are

$$[3]^1 = [3], \quad [3]^2 = [9], \quad [3]^3 = [7], \quad [3]^4 = [1].$$

Therefore $U(10) = \langle [3] \rangle$, and hence $U(10)$ is cyclic of order 4.

1.5.1 Elementary Properties of Cyclic Groups

The first important property of cyclic groups is that their operation is necessarily commutative.

Theorem 1.5.4. *Every cyclic group is abelian.*

Proof. Let $G = \langle a \rangle$ be a cyclic group. If $x, y \in G$, then there exist integers r, s such that $x = a^r$ and $y = a^s$. By the laws of exponents,

$$xy = a^r a^s = a^{r+s} = a^{s+r} = a^s a^r = yx.$$

Since x and y were arbitrary, G is abelian. ■

The converse of this theorem is not true. An abelian group need not be cyclic.

Example 1.5.5. Consider the direct product

$$G = \mathbb{Z}_2 \times \mathbb{Z}_2 = \{([0], [0]), ([1], [0]), ([0], [1]), ([1], [1])\},$$

under componentwise addition modulo 2.

The group G is abelian because addition in each component is commutative. However, every nonidentity element has order 2. Thus no element generates all four elements of G, and consequently G is not cyclic.

Proposition 1.5.6. *Every group of order at most 3 is cyclic.*

Proof. The group of order 1 is generated by its identity element. If G has order 2 or 3, then its order is prime. By the consequence of Lagrange's theorem established earlier, every group of prime order is cyclic. ■

1.5.2 Generators of Finite Cyclic Groups

A cyclic group may possess several generators. For finite cyclic groups, the generators can be characterized completely in terms of the greatest common divisor.

Recall that if $\text{ord}(a) = n$, then

$$\text{ord}(a^k) = \frac{n}{\gcd(n, k)}$$

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for every integer k.

This formula immediately gives a criterion for determining when a power of a generator is again a generator.

Theorem 1.5.7. *Let $G = \langle a \rangle$ be a finite cyclic group of order n. Then a^k is a generator of G if and only if*

$$\gcd(k, n) = 1.$$

Proof. Since $G = \langle a \rangle$, the order of a is n. The element a^k generates G if and only if its order is n.

By the formula for the order of a power,

$$\operatorname{ord}(a^k) = \frac{n}{\gcd(n, k)}.$$

Consequently, $\operatorname{ord}(a^k) = n$ if and only if $\gcd(n, k) = 1$.

Therefore a^k generates G precisely when k and n are relatively prime. ■

Example 1.5.8. The additive group $\mathbb{Z}_{12}$ is cyclic of order 12, generated by $[1]$.

An element $[k]$ generates $\mathbb{Z}_{12}$ if and only if $\gcd(k, 12) = 1$. The integers between 1 and 11 relatively prime to 12 are $1, 5, 7, 11$.

Therefore the generators of $\mathbb{Z}_{12}$ are

$$[1], \quad [5], \quad [7], \quad [11].$$

In contrast, the element $[8]$ has order

$$\operatorname{ord}([8]) = \frac{12}{\gcd(12, 8)} = 3,$$

and generates only the subgroup $\{[0], [4], [8]\}$.

The number of generators of a finite cyclic group is also determined by the order of the group.

Definition 1.5.9. For a positive integer n, the number of integers k satisfying $1 \leq k \leq n$ and $\gcd(k, n) = 1$ is denoted by $\varphi(n)$ and is called *Euler's totient function*.

Corollary 1.5.10. *A cyclic group of order n has exactly $\varphi(n)$ generators.*

Proof. Let $G = \langle a \rangle$ have order n. Its elements are $e, a, \ldots, a^{n-1}$, and the preceding theorem shows that a^k is a generator precisely when $\gcd(k, n) = 1$.

There are exactly $\varphi(n)$ such residue classes modulo n. Hence G has $\varphi(n)$ generators. ■

Example 1.5.11. The group $\mathbb{Z}_8$ has $\varphi(8) = 4$ generators, namely $[1], [3], [5], [7]$. Similarly, $\mathbb{Z}_{10}$ has $\varphi(10) = 4$ generators, namely $[1], [3], [7], [9]$.

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1.5.3 Generators of Infinite Cyclic Groups

Infinite cyclic groups have a simpler description: they possess exactly two generators.

Theorem 1.5.12. *Let $G = \langle a \rangle$ be an infinite cyclic group. Then the only generators of G are a and a^{-1}.*

Proof. Clearly a generates G. Also, every element a^k can be written as $(a^{-1})^{-k}$, so a^{-1} generates G.

Now suppose b is any generator of G. Since $G = \langle a \rangle$, we have $b = a^k$ for some integer k.

Since b generates G, there exists an integer m such that

$$a = b^m = (a^k)^m = a^{km}.$$

Because a has infinite order, distinct integer powers of a are distinct. Hence $km = 1$.

The only integers k for which this is possible are $k = 1$ and $k = -1$. Thus $b = a$ or $b = a^{-1}$, proving the result. ■

Example 1.5.13. The additive group $\mathbb{Z}$ is infinite cyclic, and its only generators are 1 and -1.

For example, 2 is not a generator of $\mathbb{Z}$, since $\langle 2 \rangle = 2\mathbb{Z}$, the subgroup of even integers.

1.5.4 Subgroups of Cyclic Groups

One of the most useful properties of cyclic groups is that all their subgroups are themselves cyclic.

Theorem 1.5.14. *Every subgroup of a cyclic group is cyclic.*

Proof. Let $G = \langle a \rangle$ be a cyclic group, and let H be a subgroup of G.

If $H = \{e\}$, then $H = \langle e \rangle$ is cyclic.

Suppose $H \neq \{e\}$. Since every element of G is an integer power of a, there exists a nonzero integer k such that $a^k \in H$. If $k < 0$, then $a^{-k} = (a^k)^{-1} \in H$. Hence H contains a positive power of a.

Let d be the least positive integer such that $a^d \in H$. Since H is a subgroup, every integer power of a^d lies in H. Therefore

$$\langle a^d \rangle \subseteq H.$$

Conversely, let $a^k \in H$. By the division algorithm, write $k = qd + r$, where $q, r \in \mathbb{Z}$ and $0 \leq r < d$. Then

$$a^r = a^k (a^d)^{-q}.$$

Since both factors on the right belong to H, we have $a^r \in H$.

By the minimality of d, we must have $r = 0$. Thus $k = qd$, so $a^k = (a^d)^q \in \langle a^d \rangle$.

Therefore $H \subseteq \langle a^d \rangle$, and hence $H = \langle a^d \rangle$. Thus H is cyclic. ■

Example 1.5.15. Every subgroup of $(\mathbb{Z}, +)$ is cyclic. More precisely, every subgroup of $\mathbb{Z}$

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has the form

$$n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$$

for some integer $n \geq 0$.

For example, $4\mathbb{Z}$ is generated by 4, while $7\mathbb{Z}$ is generated by 7. The trivial subgroup $\{0\}$ corresponds to $n = 0$.

For finite cyclic groups, a stronger classification is possible.

Theorem 1.5.16. *Let $G = \langle a \rangle$ be a cyclic group of order n. For every positive divisor d of n, there exists exactly one subgroup of G of order d, namely*

$$\left\langle a^{n/d} \right\rangle.$$

Proof. Let d be a positive divisor of n, and put $m = n/d$. Since m divides n, we have $\gcd(n, m) = m$. Therefore

$$\operatorname{ord}(a^m) = \frac{n}{\gcd(n, m)} = \frac{n}{m} = d.$$

Thus $\langle a^m \rangle = \langle a^{n/d} \rangle$ is a subgroup of order d.

To prove uniqueness, let H be any subgroup of G with $|H| = d$. Since every subgroup of a cyclic group is cyclic, write $H = \langle a^k \rangle$ for some integer k.

Then

$$d = |H| = \operatorname{ord}(a^k) = \frac{n}{\gcd(n, k)}.$$

Hence $\gcd(n, k) = n/d = m$. In particular, m divides k, so a^k belongs to $\langle a^m \rangle$.

Consequently, $H \subseteq \langle a^m \rangle$. Since both subgroups have order d, they must be equal.

Thus $\langle a^{n/d} \rangle$ is the unique subgroup of order d. ■

Corollary 1.5.17. *A finite cyclic group of order n has exactly as many subgroups as n has positive divisors.*

Proof. By the preceding theorem, each positive divisor d of n determines exactly one subgroup of order d. Conversely, by Lagrange's theorem, the order of every subgroup divides n. Hence there is a one-to-one correspondence between the subgroups of G and the positive divisors of n. ■

Example 1.5.18. The positive divisors of 12 are $1, 2, 3, 4, 6, 12$. Therefore $\mathbb{Z}_{12}$ has exactly six subgroups.

They are

$$\begin{aligned}
\langle [0] \rangle &= \{[0]\}, \\
\langle [6] \rangle &= \{[0], [6]\}, \\
\langle [4] \rangle &= \{[0], [4], [8]\}, \\
\langle [3] \rangle &= \{[0], [3], [6], [9]\}, \\
\langle [2] \rangle &= \{[0], [2], [4], [6], [8], [10]\}, \\
\langle [1] \rangle &= \mathbb{Z}_{12}.
\end{aligned}$$

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Their respective orders are $1, 2, 3, 4, 6$, and 12.

Proposition 1.5.19. *Let $G = \langle a \rangle$ be a finite cyclic group of order n. For integers r, s,*

$$\langle a^r \rangle \subseteq \langle a^s \rangle \quad \Longleftrightarrow \quad \gcd(n, s) \mid \gcd(n, r).$$

Proof. Put $d_r = \gcd(n, r)$ and $d_s = \gcd(n, s)$. By the subgroup classification theorem, $\langle a^r \rangle = \langle a^{d_r} \rangle$ and $\langle a^s \rangle = \langle a^{d_s} \rangle$.

If $d_s \mid d_r$, then every power of a^{d_r} is a power of a^{d_s}, so $\langle a^r \rangle \subseteq \langle a^s \rangle$.

Conversely, suppose $\langle a^{d_r} \rangle \subseteq \langle a^{d_s} \rangle$. Then $a^{d_r} = (a^{d_s})^k$ for some integer k, and hence $n \mid (d_r - kd_s)$.

Since $d_s \mid n$, it follows that $d_s \mid d_r$. ■

1.5.5 Classification of Cyclic Groups

Two groups may be constructed from entirely different sets and operations yet possess the same essential algebraic structure. The concept of an isomorphism makes this precise.

Definition 1.5.20. Let G and K be groups. A bijective mapping $\varphi : G \to K$ is called an *isomorphism* if

$$\varphi(xy) = \varphi(x)\varphi(y)$$

for all $x, y \in G$, where the operations on the two sides are those of G and K, respectively.

If such a mapping exists, the groups are said to be *isomorphic*, written $G \cong K$.

An isomorphism preserves the group operation and therefore the fundamental algebraic properties of the group.

The following theorem shows that, up to isomorphism, there is only one infinite cyclic group and exactly one cyclic group of each finite positive order.

Theorem 1.5.21 (Structure Theorem for Cyclic Groups)**.** *Every infinite cyclic group is isomorphic to $(\mathbb{Z}, +)$. Moreover, every finite cyclic group of order n is isomorphic to $(\mathbb{Z}_n, +)$.*

Proof. Let $G = \langle a \rangle$ be a cyclic group. We consider two cases.

Case 1: G is infinite.

Define $\varphi : \mathbb{Z} \to G$ by $\varphi(k) = a^k$. For any $r, s \in \mathbb{Z}$,

$$\varphi(r + s) = a^{r+s} = a^r a^s = \varphi(r)\varphi(s).$$

Thus φ preserves the group operation.

Since $G = \langle a \rangle$, every element of G has the form a^k, so φ is surjective.

If $\varphi(r) = \varphi(s)$, then $a^r = a^s$. As a has infinite order, distinct integer powers are distinct. Hence $r = s$, so φ is injective.

Therefore φ is an isomorphism and $G \cong \mathbb{Z}$.

Case 2: G is finite of order n.

Define $\psi : \mathbb{Z}_n \to G$ by

$$\psi([k]) = a^k.$$

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We first show that ψ is well-defined. If $[k] = [r]$ in $\mathbb{Z}_n$, then $n \mid (k - r)$. Since $\operatorname{ord}(a) = n$, this implies $a^k = a^r$. Thus the value of $\psi([k])$ does not depend on the chosen representative.

For $[r], [s] \in \mathbb{Z}_n$,

$$\psi([r] + [s]) = \psi([r + s]) = a^{r+s} = a^r a^s = \psi([r])\psi([s]).$$

Hence ψ preserves the operation.

Since a generates G, the map ψ is surjective. Furthermore, if $\psi([r]) = \psi([s])$, then $a^r = a^s$, which implies $n \mid (r - s)$ and hence $[r] = [s]$. Thus ψ is injective.

Therefore ψ is an isomorphism and $G \cong \mathbb{Z}_n$. ■

Corollary 1.5.22. *Any two cyclic groups of the same finite order are isomorphic.*

Proof. Let G and K be cyclic groups of order n. By the structure theorem, both are isomorphic to $\mathbb{Z}_n$. Consequently, $G \cong K$. ■

Example 1.5.23. Let G be the group of rotations of a square through multiples of 90°, under composition. If r denotes counterclockwise rotation through 90°, then

$$G = \{e, r, r^2, r^3\}, \qquad r^4 = e.$$

Thus $G = \langle r \rangle$ is cyclic of order 4.

The mapping $\varphi : \mathbb{Z}_4 \to G$ defined by $\varphi([k]) = r^k$ is an isomorphism. Hence $G \cong \mathbb{Z}_4$.

1.6 Exercises

Unless otherwise stated, all operations are the usual ones.

1. Determine whether each operation on $\mathbb{Z}$ is associative and commutative: (a) $a * b = a - b$; (b) $a * b = a + b + 2$; (c) $a * b = ab$.

2. Prove that the relation $a \sim b$ on $\mathbb{Z}$ defined by $5 \mid (a - b)$ is an equivalence relation. Write all its distinct equivalence classes.

3. Show that $a \sim b$ on $\mathbb{R}$ defined by $a^2 = b^2$ is an equivalence relation. Describe the equivalence class of 3 and of 0.

4. Verify directly that $\mathbb{Z}_5$ is a group under addition modulo 5. Construct its addition table.

5. Determine whether the positive rational numbers form a group under multiplication.

6. Give two noncommuting elements of S_3 and calculate their products in both orders.

7. Prove that every group of order 1 or 2 is abelian.

8. Show that in a group G, $(abc)^{-1} = c^{-1}b^{-1}a^{-1}$.

9. If $a^2 = e$ for every $a \in G$, prove that G is abelian.

10. Solve $axb = c$ for x in an arbitrary group, where $a, b, c \in G$.

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11. Prove that $3\mathbb{Z}$ is a subgroup of $\mathbb{Z}$. Is $2\mathbb{Z} \cup 3\mathbb{Z}$ a subgroup?

12. Determine all subgroups of the additive group $\mathbb{Z}_6$.

13. Prove that the center $Z(G) = \{z \in G : zg = gz \text{ for every } g \in G\}$ is a subgroup of G.

14. List the distinct cosets of $\langle [4] \rangle$ in $\mathbb{Z}_{12}$ and determine their number.

15. Can a finite group of order 21 contain an element of order 6? Justify your answer.

16. Prove that a group of prime order is abelian.

17. Find the orders of $[4]$, $[6]$ and $[9]$ in $\mathbb{Z}_{18}$.

18. Find all generators of $\mathbb{Z}_{10}$ and $\mathbb{Z}_{15}$.

19. Show that every subgroup of $\mathbb{Z}$ is of the form $m\mathbb{Z}$ for a unique nonnegative integer m.

20. Describe all subgroups of a cyclic group of order 18, giving their generators and orders.

21. Prove that $\langle a^r \rangle = \langle a^s \rangle$ in a cyclic group of order n if and only if $\gcd(r, n) = \gcd(s, n)$.

22. Explain why the rule $[k] \mapsto a^k$ requires a well-definedness check when its domain is $\mathbb{Z}_n$, but not when its domain is $\mathbb{Z}$.

CHAPTER
2

NORMAL SUBGROUPS AND HOMOMORPHISMS

The study of subgroups provides important information about the internal structure of a group. Among subgroups, those called *normal subgroups* play a particularly significant role because they allow us to construct new groups whose elements are cosets of the original group. These are known as *quotient groups*.

The notion of a normal subgroup is closely connected with that of a group homomorphism, which is a mapping between groups that preserves their operations. Together, these concepts provide a systematic method for studying relationships between groups and understanding their structure.

In this chapter, we develop the theory of normal subgroups, quotient groups, and homomorphisms. We also study permutation groups, Cayley's theorem, and the symmetry groups of regular polygons. Unless otherwise stated, group operations are written multiplicatively, and permutations are composed from right to left.

2.1 Normal Subgroups

Recall that if H is a subgroup of a group G and $g \in G$, then the *left coset* and *right coset* of H determined by g are, respectively,

$$gH = \{gh : h \in H\}, \qquad Hg = \{hg : h \in H\}.$$

In an abelian group, these cosets are equal because $gh = hg$ for all $g \in G$ and $h \in H$. In a nonabelian group, however, left and right cosets need not coincide.

Subgroups for which these cosets are always equal are called normal subgroups.

Definition 2.1.1. A subgroup N of a group G is called a *normal subgroup* of G if

$$gN = Ng$$

for every $g \in G$. We denote this by $N \trianglelefteq G$.

Normality therefore means that multiplying the elements of a subgroup by any fixed element of G produces the same subset, whether multiplication is performed on the left or on the right.

Normality can also be expressed in terms of *conjugation*. For $g, x \in G$, the element gxg^{-1} is called the conjugate of x by g.

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For a subset N of G, we write

$$gNg^{-1} = \{gng^{-1} : n \in N\}.$$

The following theorem gives several equivalent characterizations of normal subgroups.

Theorem 2.1.2. *Let N be a subgroup of G. The following conditions are equivalent:*

(i) N is normal in G;

(ii) $gNg^{-1} = N$ for every $g \in G$;

(iii) $gng^{-1} \in N$ for all $g \in G$ and $n \in N$.

Proof. Suppose first that $N \trianglelefteq G$. Then $gN = Ng$ for every $g \in G$. Multiplying both sides on the right by g^{-1} gives

$$gNg^{-1} = Ngg^{-1} = N.$$

Thus (i) implies (ii).

Conversely, if $gNg^{-1} = N$, multiplying on the right by g gives $gN = Ng$. Hence (ii) implies (i).

Clearly, (ii) implies (iii), since every element gng^{-1} belongs to the set $gNg^{-1} = N$.

Finally, suppose (iii) holds. Then $gNg^{-1} \subseteq N$ for every $g \in G$.

Replacing g by g^{-1} gives $g^{-1}Ng \subseteq N$. Multiplying this inclusion on the left by g and on the right by g^{-1} yields

$$N \subseteq gNg^{-1}.$$

Combining the two inclusions gives $gNg^{-1} = N$. Therefore (iii) implies (ii), completing the proof. ■

2.1.1 Examples of Normal Subgroups

Example 2.1.3. For every group G, both $\{e\}$ and G are normal subgroups.

Indeed, for every $g \in G$,

$$g\{e\}g^{-1} = \{e\}, \qquad gGg^{-1} = G.$$

Thus $\{e\} \trianglelefteq G$ and $G \trianglelefteq G$.

Example 2.1.4. Every subgroup of an abelian group is normal.

To see this, let $N \leq G$, where G is abelian. For $g \in G$ and $n \in N$, commutativity gives

$$gng^{-1} = ngg^{-1} = n \in N.$$

Hence $N \trianglelefteq G$.

In particular, $m\mathbb{Z} \trianglelefteq \mathbb{Z}$ for every nonnegative integer m.

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Example 2.1.5. Consider the symmetric group

$$S_3 = \{e, (12), (13), (23), (123), (132)\}$$

and its subgroup $H = \{e, (12)\}$.

Choose $g = (13)$. Since $g^{-1} = g$, we obtain

$$g(12)g^{-1} = (13)(12)(13) = (23).$$

But $(23) \notin H$. Therefore H is not normal in S_3.

This example shows that a subgroup of a nonabelian group need not be normal.

Example 2.1.6. Consider the subgroup

$$A_3 = \{e, (123), (132)\}$$

of S_3.

Since $|S_3| = 6$ and $|A_3| = 3$, Lagrange's theorem gives

$$[S_3 : A_3] = \frac{6}{3} = 2.$$

As every subgroup of index two is normal, a result proved below, we conclude that $A_3 \trianglelefteq S_3$.

2.1.2 Further Properties of Normal Subgroups

The following theorem provides an important sufficient condition for normality.

Theorem 2.1.7. *Every subgroup of index two in a group is normal.*

Proof. Let $H \leq G$ and suppose $[G : H] = 2$. Thus there are exactly two distinct left cosets of H in G, and exactly two distinct right cosets.

Let $g \in G$. If $g \in H$, then $gH = H = Hg$.

If $g \notin H$, the left coset gH is distinct from H. Since the left cosets partition G and there are exactly two of them, we must have

$$gH = G \setminus H.$$

Similarly, the right coset Hg is distinct from H, and hence

$$Hg = G \setminus H.$$

Therefore $gH = Hg$ for every $g \in G$, proving that $H \trianglelefteq G$. ■

Example 2.1.8. The subgroup A_n of even permutations in S_n has index two for $n \geq 2$, since $|S_n| = n!$ and $|A_n| = n!/2$.

Consequently, $A_n \trianglelefteq S_n$.

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Theorem 2.1.9. *Let $\{N_i : i \in I\}$ be a nonempty family of normal subgroups of G. Then*

$$N = \bigcap_{i \in I} N_i$$

is a normal subgroup of G.

Proof. Since every N_i is a subgroup, their intersection N is also a subgroup of G.

Let $g \in G$ and $n \in N$. Then $n \in N_i$ for every $i \in I$. Since each N_i is normal,

$$gng^{-1} \in N_i$$

for every $i \in I$. Hence $gng^{-1} \in N$.

By the normal subgroup criterion, $N \trianglelefteq G$. ■

Proposition 2.1.10. *If $N \trianglelefteq G$ and $H \leq G$, then $N \cap H$ is a normal subgroup of H.*

Proof. The intersection $N \cap H$ is a subgroup of H. Let $h \in H$ and $x \in N \cap H$.

Since N is normal in G, we have $hxh^{-1} \in N$. Also, since $h, x \in H$ and H is a subgroup, $hxh^{-1} \in H$.

Therefore $hxh^{-1} \in N \cap H$, and hence $N \cap H \trianglelefteq H$. ■

Proposition 2.1.11. *Let H and K be normal subgroups of G. Then*

$$HK = \{hk : h \in H,\ k \in K\}$$

is a normal subgroup of G.

Proof. We first show that HK is a subgroup. Clearly $e \in HK$. Let $h_1k_1, h_2k_2 \in HK$. Then

$$(h_1k_1)(h_2k_2)^{-1} = h_1(k_1k_2^{-1})h_2^{-1}.$$

Since H is normal, there exists $h_3 \in H$ such that

$$(k_1k_2^{-1})h_2^{-1} = h_3(k_1k_2^{-1}).$$

Hence

$$(h_1k_1)(h_2k_2)^{-1} = (h_1h_3)(k_1k_2^{-1}) \in HK.$$

Therefore HK is a subgroup.

Now let $g \in G$ and $hk \in HK$. Since H and K are normal,

$$g(hk)g^{-1} = (ghg^{-1})(gkg^{-1}) \in HK.$$

Thus $HK \trianglelefteq G$. ■

2.2 Quotient Groups

We have seen that the left cosets of any subgroup partition a group into disjoint subsets. It is natural to ask whether these cosets can themselves be regarded as the elements of a new group.

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For an arbitrary subgroup, multiplication of cosets may depend on the representatives chosen. However, when the subgroup is normal, multiplication of cosets is well-defined and produces a group.

This construction is called a *quotient group*.

Definition 2.2.1. Let N be a normal subgroup of a group G. The set of all left cosets of N in G is denoted by

$$G/N = \{gN : g \in G\}.$$

Define multiplication on G/N by

$$(aN)(bN) = (ab)N$$

for all $a, b \in G$.

The resulting structure is called the *quotient group* or *factor group* of G by N.

The most important point in this definition is that the product must not depend on the particular representatives of the cosets. We establish this in the following theorem.

Theorem 2.2.2. *Let N be a normal subgroup of G. Then the operation*

$$(aN)(bN) = (ab)N$$

is well-defined and makes G/N a group.

The identity element is N, the inverse of aN is $a^{-1}N$, and if G is finite, then

$$|G/N| = [G : N] = \frac{|G|}{|N|}.$$

Proof. We first show that multiplication of cosets is well-defined.

Suppose $aN = a'N$ and $bN = b'N$. Then there exist $n_1, n_2 \in N$ such that

$$a' = an_1, \qquad b' = bn_2.$$

Consequently,

$$a'b' = an_1bn_2 = ab(b^{-1}n_1b)n_2.$$

Since N is normal, $b^{-1}n_1b \in N$. Therefore $(b^{-1}n_1b)n_2 \in N$, giving

$$a'b'N = abN.$$

Hence the product of two cosets is independent of the chosen representatives.

We now verify the group axioms.

Closure. For $aN, bN \in G/N$, their product is $(ab)N$, which is again a coset of N and hence belongs to G/N.

Associativity. For $aN, bN, cN \in G/N$,

$$\begin{aligned}((aN)(bN))(cN) &= ((ab)N)(cN)\\ &= ((ab)c)N\\ &= (a(bc))N\\ &= (aN)((bc)N)\\ &= (aN)((bN)(cN)).\end{aligned}$$

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Thus multiplication is associative.

Identity. Since $eN = N$, we have

$$(aN)N = (ae)N = aN, \qquad N(aN) = (ea)N = aN.$$

Therefore N is the identity element of G/N.

Inverses. For every $aN \in G/N$,

$$(aN)(a^{-1}N) = (aa^{-1})N = N,$$

and similarly $(a^{-1}N)(aN) = N$. Thus the inverse of aN is $a^{-1}N$.

Therefore G/N is a group.

Finally, if G is finite, the elements of G/N are precisely the distinct cosets of N. Hence $|G/N| = [G : N]$. By Lagrange's theorem,

$$|G/N| = \frac{|G|}{|N|}.$$

■

Remark 2.2.3. The identity element of G/N is the coset N, not the identity element e of G. Although $e \in N$, the elements of G/N are cosets rather than individual elements of G.

2.2.1 Normality and Well-defined Coset Multiplication

The requirement that N be normal is not merely sufficient for defining multiplication of cosets; it is also necessary.

Theorem 2.2.4. *Let H be a subgroup of G. The rule*

$$(aH)(bH) = (ab)H$$

defines a well-defined binary operation on the set of left cosets of H if and only if H is normal in G.

Proof. If H is normal, the quotient group theorem shows that the operation is well-defined.

Conversely, suppose the rule is well-defined. Let $g \in G$ and $h \in H$. Since $hH = H = eH$, well-definedness gives

$$(gH)(hH) = (gH)(eH).$$

Therefore $(gh)H = gH$, which implies $g^{-1}(gh) \in H$ but yields no new information. To obtain the required condition, use the equality $hH = eH$ in the first factor:

$$(hH)(gH) = (eH)(gH).$$

Thus $(hg)H = gH$, so $g^{-1}hg \in H$.

Since $g \in G$ and $h \in H$ were arbitrary, the normal subgroup criterion gives $H \trianglelefteq G$. ■

2.2.2 Examples of Quotient Groups

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Example 2.2.5. Let $G = (\mathbb{Z}, +)$ and $N = n\mathbb{Z}$, where $n \geq 2$. Since $\mathbb{Z}$ is abelian, $n\mathbb{Z}$ is normal.

The quotient group $\mathbb{Z}/n\mathbb{Z}$ consists of the n distinct cosets

$$0 + n\mathbb{Z}, \quad 1 + n\mathbb{Z}, \quad \ldots, \quad (n-1) + n\mathbb{Z}.$$

Its operation is

$$(a + n\mathbb{Z}) + (b + n\mathbb{Z}) = (a + b) + n\mathbb{Z}.$$

For example, in $\mathbb{Z}/5\mathbb{Z}$,

$$(3 + 5\mathbb{Z}) + (4 + 5\mathbb{Z}) = 7 + 5\mathbb{Z} = 2 + 5\mathbb{Z}.$$

Thus $\mathbb{Z}/n\mathbb{Z}$ is the familiar additive group of integers modulo n.

Example 2.2.6. Let

$$G = S_3, \qquad N = A_3 = \{e, (123), (132)\}.$$

Since $[S_3 : A_3] = 2$, the subgroup A_3 is normal.

The quotient group has two elements:

$$S_3/A_3 = \{A_3, (12)A_3\}.$$

Since $(12)^2 = e$, we have

$$((12)A_3)^2 = A_3.$$

Therefore S_3/A_3 is a cyclic group of order 2, and hence

$$S_3/A_3 \cong \mathbb{Z}_2.$$

Example 2.2.7. Consider the additive group $G = \mathbb{Z}_{12}$ and the subgroup

$$H = \langle [4] \rangle = \{[0], [4], [8]\}.$$

Since G is abelian, H is normal.

We have $|G| = 12$ and $|H| = 3$, so

$$|G/H| = \frac{12}{3} = 4.$$

The four distinct cosets are

$$H, \quad [1] + H, \quad [2] + H, \quad [3] + H.$$

The coset $[1] + H$ generates G/H, since its successive multiples are

$$[1] + H, \quad [2] + H, \quad [3] + H, \quad H.$$

Thus G/H is cyclic of order 4, and hence

$$\mathbb{Z}_{12}/\langle [4] \rangle \cong \mathbb{Z}_4.$$

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Example 2.2.8. Consider the determinant-one subgroup $SL_2(\mathbb{R})$ of $GL_2(\mathbb{R})$.
For $A \in GL_2(\mathbb{R})$ and $B \in SL_2(\mathbb{R})$,

$$\det(ABA^{-1}) = \det(A)\det(B)\det(A^{-1}) = 1.$$

Therefore $ABA^{-1} \in SL_2(\mathbb{R})$, showing that

$$SL_2(\mathbb{R}) \trianglelefteq GL_2(\mathbb{R}).$$

The quotient group $GL_2(\mathbb{R})/SL_2(\mathbb{R})$ is thus well-defined. In the study of group homomorphisms, we shall see that this quotient is isomorphic to the multiplicative group $\mathbb{R} \setminus \{0\}$.

Proposition 2.2.9. *If G is abelian and $N \trianglelefteq G$, then G/N is abelian.*

Proof. Let $aN, bN \in G/N$. Since G is abelian, $ab = ba$. Therefore

$$(aN)(bN) = (ab)N = (ba)N = (bN)(aN).$$

Hence G/N is abelian. ■

Proposition 2.2.10. *If G is cyclic and N is a subgroup of G, then G/N is cyclic.*

Proof. Let $G = \langle a \rangle$. Since every cyclic group is abelian, N is normal and G/N is defined.
Every coset of N has the form gN for some $g \in G$. Since $G = \langle a \rangle$, we may write $g = a^k$ for some integer k. Hence

$$gN = a^k N = (aN)^k.$$

Therefore every element of G/N is a power of aN, so

$$G/N = \langle aN \rangle.$$

Thus the quotient group is cyclic. ■

2.3 Group Homomorphisms

Having studied subgroups and quotient groups, we now consider mappings between groups that preserve their algebraic structure. Such mappings are called *group homomorphisms*. They provide a means of comparing groups and establishing relationships between algebraic systems that may have entirely different underlying sets.

The essential property of a homomorphism is that combining two elements before applying the mapping gives the same result as applying the mapping to each element and then combining their images.

Definition 2.3.1. Let $(G, *)$ and $(K, \cdot)$ be groups. A mapping $\varphi : G \to K$ is called a *group homomorphism* if

$$\varphi(a * b) = \varphi(a) \cdot \varphi(b)$$

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for all $a, b \in G$.

When the operations are understood, we simply write $\varphi(ab) = \varphi(a)\varphi(b)$.

A homomorphism need not be injective or surjective. A homomorphism that is injective is called a *monomorphism*, while a surjective homomorphism is called an *epimorphism*. A bijective homomorphism is called an *isomorphism*. If an isomorphism exists between G and K, we write $G \cong K$.

Example 2.3.2. Let $n \geq 2$. Define $\varphi : (\mathbb{Z}, +) \to (\mathbb{Z}_n, +)$ by $\varphi(k) = [k]$.
For any $a, b \in \mathbb{Z}$,

$$\varphi(a + b) = [a + b] = [a] + [b] = \varphi(a) + \varphi(b).$$

Thus φ is a homomorphism. It is surjective because every element of $\mathbb{Z}_n$ is represented by an integer.

Example 2.3.3. Consider the mapping

$$\det : GL_n(\mathbb{R}) \longrightarrow \mathbb{R}^\times,$$

where $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$ is a group under multiplication.
Since $\det(AB) = \det(A)\det(B)$ for all invertible matrices A, B, the determinant is a group homomorphism.

Example 2.3.4. For a fixed integer m, define $\varphi : \mathbb{Z} \to \mathbb{Z}$ by $\varphi(k) = mk$. Then

$$\varphi(a + b) = m(a + b) = ma + mb = \varphi(a) + \varphi(b).$$

Hence φ is a homomorphism.
If $m \neq 0$, this homomorphism is injective, while its image is the subgroup $m\mathbb{Z}$. It is surjective precisely when $m = 1$ or $m = -1$.

Example 2.3.5. Consider $f : (\mathbb{Z}, +) \to (\mathbb{Z}, +)$ defined by $f(k) = k + 1$.
For integers a, b, we have

$$f(a + b) = a + b + 1, \qquad f(a) + f(b) = a + b + 2.$$

These are not equal. Therefore f is not a homomorphism.

2.3.1 Elementary Properties of Homomorphisms

The defining property of a homomorphism has several immediate consequences. In particular, a homomorphism preserves identity elements, inverses, and integer powers.

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Theorem 2.3.6. *Let $\varphi : G \to K$ be a group homomorphism, and let e_G and e_K denote the identity elements of G and K, respectively. Then, for every $a \in G$ and $m \in \mathbb{Z}$,*

(i) $\varphi(e_G) = e_K$;

(ii) $\varphi(a^{-1}) = \varphi(a)^{-1}$;

(iii) $\varphi(a^m) = \varphi(a)^m$.

Proof. Since $e_G e_G = e_G$, the homomorphism property gives

$$\varphi(e_G) = \varphi(e_G)\varphi(e_G).$$

Multiplying by $\varphi(e_G)^{-1}$ in K, we obtain $\varphi(e_G) = e_K$.

Next, since $aa^{-1} = e_G$, we have

$$\varphi(a)\varphi(a^{-1}) = \varphi(aa^{-1}) = \varphi(e_G) = e_K.$$

Similarly, $\varphi(a^{-1})\varphi(a) = e_K$. By uniqueness of inverses, $\varphi(a^{-1}) = \varphi(a)^{-1}$.

For positive integers m, repeated application of the homomorphism property gives

$$\varphi(a^m) = \varphi(a)^m.$$

The result is immediate for $m = 0$ and follows for negative integers from the inverse formula. ■

Proposition 2.3.7. *Let $\varphi : G \to K$ be a homomorphism. If $a \in G$ has finite order, then the order of $\varphi(a)$ divides the order of a.*

Proof. Suppose $\operatorname{ord}(a) = n$. Then $a^n = e_G$, so

$$\varphi(a)^n = \varphi(a^n) = \varphi(e_G) = e_K.$$

Therefore $\varphi(a)$ has finite order, say m, and the criterion for the order of an element gives $m \mid n$. ■

2.3.2 Kernel and Image of a Homomorphism

Two subsets associated with a homomorphism are particularly important: its kernel and its image.

Definition 2.3.8. Let $\varphi : G \to K$ be a group homomorphism. The *kernel* of φ is

$$\ker \varphi = \{g \in G : \varphi(g) = e_K\}.$$

The *image* of φ is

$$\operatorname{Im} \varphi = \{\varphi(g) : g \in G\}.$$

The kernel consists of those elements of G that are mapped to the identity of K. The image consists of all elements of K that actually occur as values of φ.

The following theorem explains their algebraic significance.

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Theorem 2.3.9. *Let $\varphi : G \to K$ be a group homomorphism. Then*

(i) $\ker \varphi$ *is a normal subgroup of* G*;*

(ii) $\operatorname{Im} \varphi$ *is a subgroup of* K*.*

Proof. Let $N = \ker \varphi$. Since $\varphi(e_G) = e_K$, we have $e_G \in N$, so N is nonempty.
If $a, b \in N$, then

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e_K e_K^{-1} = e_K.$$

Hence $ab^{-1} \in N$, and the subgroup test shows that $N \leq G$.
To establish normality, let $g \in G$ and $n \in N$. Then

$$\begin{aligned} \varphi(gng^{-1}) &= \varphi(g)\varphi(n)\varphi(g)^{-1} \\ &= \varphi(g)e_K\varphi(g)^{-1} \\ &= e_K. \end{aligned}$$

Thus $gng^{-1} \in N$, proving $N \trianglelefteq G$.
Now let $H = \operatorname{Im} \varphi$. Since $\varphi(e_G) = e_K$, the image is nonempty.
Let $x, y \in H$. Then $x = \varphi(a)$ and $y = \varphi(b)$ for some $a, b \in G$. Consequently,

$$xy^{-1} = \varphi(a)\varphi(b)^{-1} = \varphi(ab^{-1}) \in H.$$

Therefore $H \leq K$ by the subgroup test. ■

The kernel determines whether a homomorphism is injective.

Theorem 2.3.10. *A group homomorphism $\varphi : G \to K$ is injective if and only if*

$$\ker \varphi = \{e_G\}.$$

Proof. Suppose φ is injective. If $a \in \ker \varphi$, then

$$\varphi(a) = e_K = \varphi(e_G).$$

Injectivity gives $a = e_G$. Hence $\ker \varphi = \{e_G\}$.
Conversely, suppose $\ker \varphi = \{e_G\}$. If $\varphi(a) = \varphi(b)$, then

$$\varphi(b^{-1}a) = \varphi(b)^{-1}\varphi(a) = e_K.$$

Thus $b^{-1}a \in \ker \varphi$, so $b^{-1}a = e_G$ and hence $a = b$. Therefore φ is injective. ■

Example 2.3.11. Let $\varphi : \mathbb{Z} \to \mathbb{Z}_6$ be defined by $\varphi(k) = [k]$.
An integer k belongs to the kernel precisely when $[k] = [0]$, that is, when $6 \mid k$. Therefore

$$\ker \varphi = 6\mathbb{Z}.$$

Every residue class modulo 6 has an integer representative, so $\operatorname{Im} \varphi = \mathbb{Z}_6$.
Thus φ is surjective but not injective.

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Example 2.3.12. Define $\operatorname{sgn} : S_n \to \{1, -1\}$ by assigning 1 to every even permutation and -1 to every odd permutation.

The parity law for products of permutations gives

$$\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau).$$

Thus sgn is a homomorphism. Its kernel is the alternating group A_n, and for $n \geq 2$ its image is $\{1, -1\}$.

2.3.3 Images and Inverse Images of Subgroups

Homomorphisms also preserve subgroup structure in the following sense.

Theorem 2.3.13. *Let $\varphi : G \to K$ be a group homomorphism.*

(i) If $H \leq G$, then $\varphi(H) \leq K$.

(ii) If $L \leq K$, then

$$\varphi^{-1}(L) = \{g \in G : \varphi(g) \in L\}$$

is a subgroup of G.

Furthermore, if $L \trianglelefteq K$, then $\varphi^{-1}(L) \trianglelefteq G$.

Proof. Let $H \leq G$. Since $e_G \in H$, the set $\varphi(H)$ contains e_K. If $x = \varphi(a)$ and $y = \varphi(b)$ belong to $\varphi(H)$, where $a, b \in H$, then

$$xy^{-1} = \varphi(ab^{-1}) \in \varphi(H).$$

Thus $\varphi(H)$ is a subgroup.

Next, let $L \leq K$ and put $M = \varphi^{-1}(L)$. Since $\varphi(e_G) = e_K \in L$, the set M is nonempty. If $a, b \in M$, then $\varphi(a), \varphi(b) \in L$, so

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in L.$$

Therefore $ab^{-1} \in M$, showing that M is a subgroup.

Finally, suppose L is normal in K. For $g \in G$ and $m \in M$,

$$\varphi(gmg^{-1}) = \varphi(g)\varphi(m)\varphi(g)^{-1} \in L.$$

Hence $gmg^{-1} \in M$, proving $M \trianglelefteq G$. ∎

2.4 The Isomorphism Theorems

The isomorphism theorems are fundamental results connecting homomorphisms, normal subgroups, and quotient groups. They explain how quotient groups arise naturally from homomorphisms and provide methods for comparing different quotient constructions.

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2.4.1 First Isomorphism Theorem

The kernel of a homomorphism identifies elements that have the same image. The First Isomorphism Theorem states that, after factoring out the kernel, the resulting quotient group is isomorphic to the image.

Theorem 2.4.1 (First Isomorphism Theorem). *Let $\varphi : G \to K$ be a group homomorphism. Then*

$$G/\ker\varphi \cong \operatorname{Im}\varphi.$$

Proof. Put $N = \ker\varphi$. Since $N \trianglelefteq G$, the quotient group G/N is well-defined.

Define $\widetilde{\varphi} : G/N \to \operatorname{Im}\varphi$ by

$$\widetilde{\varphi}(gN) = \varphi(g).$$

We first show that this mapping is well-defined. Suppose $gN = hN$. Then $h^{-1}g \in N$, so

$$\varphi(h^{-1}g) = e_K.$$

Consequently,

$$\varphi(h)^{-1}\varphi(g) = e_K,$$

which implies $\varphi(g) = \varphi(h)$. Hence $\widetilde{\varphi}(gN) = \widetilde{\varphi}(hN)$.

Next, for $gN, hN \in G/N$,

$$\begin{aligned}\widetilde{\varphi}((gN)(hN)) &= \widetilde{\varphi}(ghN)\\ &= \varphi(gh)\\ &= \varphi(g)\varphi(h)\\ &= \widetilde{\varphi}(gN)\widetilde{\varphi}(hN).\end{aligned}$$

Thus $\widetilde{\varphi}$ is a homomorphism.

It is surjective because every element of $\operatorname{Im}\varphi$ has the form $\varphi(g)$ for some $g \in G$.

Finally, suppose $\widetilde{\varphi}(gN) = \widetilde{\varphi}(hN)$. Then $\varphi(g) = \varphi(h)$, so $\varphi(h^{-1}g) = e_K$. Therefore $h^{-1}g \in N$, giving $gN = hN$. Hence $\widetilde{\varphi}$ is injective.

Thus $\widetilde{\varphi}$ is an isomorphism, and

$$G/\ker\varphi \cong \operatorname{Im}\varphi.$$

■

Example 2.4.2. Consider the homomorphism $\varphi : \mathbb{Z} \to \mathbb{Z}_n$ given by $\varphi(k) = [k]$.

Since $\ker\varphi = n\mathbb{Z}$ and $\operatorname{Im}\varphi = \mathbb{Z}_n$, the First Isomorphism Theorem gives

$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n.$$

Example 2.4.3. The determinant map

$$\det : GL_n(\mathbb{R}) \longrightarrow \mathbb{R}^\times$$

is a homomorphism with kernel

$$SL_n(\mathbb{R}) = \{A \in GL_n(\mathbb{R}) : \det A = 1\}.$$

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It is surjective because, for every $t \in \mathbb{R}^\times$, the diagonal matrix $\operatorname{diag}(t, 1, \ldots, 1)$ has determinant t.

Therefore

$$GL_n(\mathbb{R})/SL_n(\mathbb{R}) \cong \mathbb{R}^\times.$$

Corollary 2.4.4. *A normal subgroup N of G is the kernel of a group homomorphism from G.*

Proof. Define the natural projection $\pi : G \to G/N$ by $\pi(g) = gN$.

For $a, b \in G$,

$$\pi(ab) = abN = (aN)(bN) = \pi(a)\pi(b),$$

so π is a homomorphism.

Moreover,

$$\ker \pi = \{g \in G : gN = N\} = N.$$

Thus N is the kernel of a homomorphism. ■

2.4.2 Second Isomorphism Theorem

The Second Isomorphism Theorem describes the relationship between a subgroup, a normal subgroup, and their product.

Theorem 2.4.5 (Second Isomorphism Theorem)**.** *Let $H \leq G$ and $N \trianglelefteq G$. Then HN is a subgroup of G, $N \trianglelefteq HN$, and $H \cap N \trianglelefteq H$. Moreover,*

$$H/(H \cap N) \cong HN/N.$$

Proof. We first show that HN is a subgroup. Clearly $e \in HN$.

Let $h_1n_1, h_2n_2 \in HN$. Then

$$(h_1n_1)(h_2n_2)^{-1} = h_1n_1n_2^{-1}h_2^{-1}.$$

Rearranging, we obtain

$$(h_1n_1)(h_2n_2)^{-1} = (h_1h_2^{-1})(h_2n_1n_2^{-1}h_2^{-1}).$$

Since H is a subgroup, $h_1h_2^{-1} \in H$. Since N is normal in G, the second factor belongs to N. Hence the product belongs to HN, proving that $HN \leq G$.

Since $N \trianglelefteq G$ and $N \leq HN$, we have $N \trianglelefteq HN$.

Also, $H \cap N$ is a subgroup of H. For $h \in H$ and $x \in H \cap N$, the element hxh^{-1} belongs to both H and N. Hence $H \cap N \trianglelefteq H$.

Now define $\varphi : H \to HN/N$ by

$$\varphi(h) = hN.$$

For $h_1, h_2 \in H$,

$$\varphi(h_1h_2) = (h_1h_2)N = (h_1N)(h_2N) = \varphi(h_1)\varphi(h_2).$$

Thus φ is a homomorphism.

Every element of HN/N has the form $(hn)N = hN$ for some $h \in H$ and $n \in N$, so φ is surjective.

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Finally,

$$\ker \varphi = \{h \in H : hN = N\} = H \cap N.$$

By the First Isomorphism Theorem,

$$H/(H \cap N) \cong HN/N.$$

■

Example 2.4.6. Let $G = \mathbb{Z}$, $H = 4\mathbb{Z}$, and $N = 6\mathbb{Z}$. Since $\mathbb{Z}$ is abelian, N is normal.

We have

$$H + N = 4\mathbb{Z} + 6\mathbb{Z} = 2\mathbb{Z}$$

and

$$H \cap N = 4\mathbb{Z} \cap 6\mathbb{Z} = 12\mathbb{Z}.$$

The Second Isomorphism Theorem gives

$$4\mathbb{Z}/12\mathbb{Z} \cong 2\mathbb{Z}/6\mathbb{Z}.$$

Both quotient groups are cyclic of order 3.

2.4.3 Third Isomorphism Theorem

The Third Isomorphism Theorem explains how successive quotient constructions can be simplified.

Theorem 2.4.7 (Third Isomorphism Theorem)**.** *Let N and K be normal subgroups of G such that $N \subseteq K$. Then K/N is a normal subgroup of G/N, and*

$$(G/N)/(K/N) \cong G/K.$$

Proof. Define $\psi : G/N \to G/K$ by

$$\psi(gN) = gK.$$

This mapping is well-defined. Indeed, if $gN = hN$, then $h^{-1}g \in N \subseteq K$, so $gK = hK$.

For $gN, hN \in G/N$,

$$\begin{aligned} \psi((gN)(hN)) &= \psi(ghN) \\ &= ghK \\ &= (gK)(hK) \\ &= \psi(gN)\psi(hN). \end{aligned}$$

Thus ψ is a homomorphism. It is surjective because every coset gK is the image of gN.

Its kernel is

$$\ker \psi = \{gN : gK = K\} = \{gN : g \in K\} = K/N.$$

Thus K/N is normal in G/N, and the First Isomorphism Theorem gives

$$(G/N)/(K/N) \cong G/K.$$

■

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Example 2.4.8. Consider the normal subgroups

$$12\mathbb{Z} \subseteq 4\mathbb{Z} \subseteq \mathbb{Z}.$$

The Third Isomorphism Theorem gives

$$(\mathbb{Z}/12\mathbb{Z})/(4\mathbb{Z}/12\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}.$$

Thus taking the quotient by $12\mathbb{Z}$ and then by $4\mathbb{Z}/12\mathbb{Z}$ has the same effect, up to isomorphism, as taking the quotient by $4\mathbb{Z}$ directly.

2.5 Products of Subgroups, Centralizers and the Centre

The structure of a group can also be investigated through products of subgroups and through elements that commute with other elements. These ideas lead naturally to centralizers, the centre, and normalizers.

2.5.1 Products of Subgroups

Definition 2.5.1. Let H and K be subgroups of G. Their *set product* is

$$HK = \{hk : h \in H,\ k \in K\}.$$

Although H and K are subgroups, their set product need not be a subgroup. The precise condition is given below.

Theorem 2.5.2. *Let $H, K \leq G$. Then HK is a subgroup of G if and only if*

$$HK = KH.$$

Proof. Suppose HK is a subgroup. Then it is closed under inverses, so

$$HK = (HK)^{-1} = K^{-1}H^{-1} = KH,$$

where $H^{-1} = H$ and $K^{-1} = K$ because both are subgroups.

Conversely, suppose $HK = KH$. Clearly $e \in HK$.

Let $h_1k_1, h_2k_2 \in HK$. Since $k_1h_2 \in KH = HK$, there exist $h_3 \in H$ and $k_3 \in K$ such that $k_1h_2 = h_3k_3$. Hence

$$(h_1k_1)(h_2k_2) = h_1h_3k_3k_2 \in HK.$$

Moreover, if $hk \in HK$, then

$$(hk)^{-1} = k^{-1}h^{-1} \in KH = HK.$$

Thus HK contains the identity and is closed under multiplication and inverses. Therefore it is a subgroup. ∎

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Corollary 2.5.3. *If $H, K \leq G$ and either H or K is normal in G, then HK is a subgroup of G.*

Proof. Suppose $H \trianglelefteq G$. Then $gH = Hg$ for every $g \in G$, and in particular $kH = Hk$ for every $k \in K$.

Consequently, $KH = HK$, so the preceding theorem shows that HK is a subgroup. The case in which K is normal is analogous. ■

Proposition 2.5.4. *If H and K are finite subgroups of G, then*

$$|HK| = \frac{|H||K|}{|H \cap K|}.$$

Proof. Consider the surjective mapping $f : H \times K \to HK$ defined by $f(h, k) = hk$.

For $x \in HK$, choose $h_0 \in H$ and $k_0 \in K$ such that $x = h_0 k_0$. The equality $hk = x$ holds precisely when

$$h_0^{-1} h = k_0 k^{-1} \in H \cap K.$$

Thus the pairs mapping to x are exactly

$$(h_0 t, t^{-1} k_0), \qquad t \in H \cap K.$$

Therefore every element of HK has exactly $|H \cap K|$ preimages under f.

Since $|H \times K| = |H||K|$, counting these preimages gives

$$|H||K| = |HK|\,|H \cap K|,$$

which proves the formula. ■

2.5.2 Centralizers

Definition 2.5.5. Let $a \in G$. The *centralizer* of a in G is

$$C_G(a) = \{g \in G : ga = ag\}.$$

Thus $C_G(a)$ consists of all elements of G that commute with a.

Theorem 2.5.6. *For every $a \in G$, the centralizer $C_G(a)$ is a subgroup of G.*

Proof. Since $ea = ae = a$, the identity belongs to $C_G(a)$.

Let $g, h \in C_G(a)$. Then $ga = ag$ and $ha = ah$. The second equality implies $h^{-1}a = ah^{-1}$. Consequently,

$$(gh^{-1})a = g(h^{-1}a) = g(ah^{-1}) = (ga)h^{-1} = a(gh^{-1}).$$

Thus $gh^{-1} \in C_G(a)$, and the subgroup test shows that $C_G(a) \leq G$. ■

Example 2.5.7. In S_3, consider the transposition $a = (12)$. The only permutations com-

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muting with (12) are e and (12). Therefore

$$C_{S_3}((12)) = \{e, (12)\}.$$

2.5.3 The Centre of a Group

Definition 2.5.8. The *centre* of a group G is the subset

$$Z(G) = \{z \in G : zg = gz \text{ for every } g \in G\}.$$

The centre consists of all elements that commute with every element of the group.

Theorem 2.5.9. *The centre $Z(G)$ is a normal subgroup of G.*

Proof. The identity commutes with every element, so $e \in Z(G)$.

Let $a, b \in Z(G)$. For any $g \in G$, both a and b^{-1} commute with g, so

$$(ab^{-1})g = a(b^{-1}g) = a(gb^{-1}) = (ag)b^{-1} = g(ab^{-1}).$$

Thus $ab^{-1} \in Z(G)$. By the subgroup test, $Z(G) \leq G$.

Now let $z \in Z(G)$ and $g \in G$. Since $zg = gz$,

$$gzg^{-1} = zgg^{-1} = z \in Z(G).$$

Hence $Z(G) \trianglelefteq G$. ■

Proposition 2.5.10. *For every group G,*

$$Z(G) = \bigcap_{a \in G} C_G(a).$$

Proof. An element $z \in G$ belongs to the intersection on the right if and only if it belongs to $C_G(a)$ for every $a \in G$. This is equivalent to $za = az$ for every $a \in G$, which is precisely the condition that $z \in Z(G)$. ■

Proposition 2.5.11. *A group G is abelian if and only if $Z(G) = G$.*

Proof. If G is abelian, every element commutes with every other element, so every element belongs to the centre and $Z(G) = G$.

Conversely, if $Z(G) = G$, every element of G commutes with every other element. Hence G is abelian. ■

Example 2.5.12. The centre of S_3 is $\{e\}$. Indeed, no nonidentity permutation in S_3 commutes with every element.

In contrast, if G is abelian, then $Z(G) = G$.

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Example 2.5.13. For $n \geq 2$, the centre of $GL_n(\mathbb{R})$ is

$$Z(GL_n(\mathbb{R})) = \{\lambda I_n : \lambda \in \mathbb{R}^\times\}.$$

Every nonzero scalar matrix commutes with all matrices.

Conversely, let $A \in Z(GL_n(\mathbb{R}))$. Then A commutes with every elementary matrix $I_n + E_{ij}$, where $i \neq j$ and E_{ij} has a single nonzero entry 1 in position (i, j).

Hence $AE_{ij} = E_{ij}A$ for all $i \neq j$. Comparing matrix entries shows that all off-diagonal entries of A vanish and all diagonal entries are equal. Therefore $A = \lambda I_n$. Since A is invertible, $\lambda \neq 0$.

2.5.4 Normalizers

Definition 2.5.14. Let H be a subgroup of G. The *normalizer* of H in G is

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

The normalizer consists of those elements of G whose conjugation preserves H as a set.

Theorem 2.5.15. *Let $H \leq G$. Then $N_G(H)$ is a subgroup of G containing H, and*

$$H \trianglelefteq N_G(H).$$

Moreover, $N_G(H)$ is the largest subgroup of G in which H is normal.

Proof. Since $eHe^{-1} = H$, we have $e \in N_G(H)$.

Let $g, k \in N_G(H)$. Since $gHg^{-1} = H$ and $kHk^{-1} = H$, the latter equality also gives $k^{-1}Hk = H$. Therefore

$$(gk^{-1})H(gk^{-1})^{-1} = g(k^{-1}Hk)g^{-1} = gHg^{-1} = H.$$

Thus $gk^{-1} \in N_G(H)$, so $N_G(H)$ is a subgroup.

If $h \in H$, then $hHh^{-1} = H$, since H is itself a group. Hence $H \subseteq N_G(H)$.

By the definition of the normalizer, every element of $N_G(H)$ conjugates H onto itself. Therefore $H \trianglelefteq N_G(H)$.

Finally, suppose $L \leq G$ and $H \trianglelefteq L$. Then for every $l \in L$, we have $lHl^{-1} = H$, so $l \in N_G(H)$. Hence $L \subseteq N_G(H)$.

Thus $N_G(H)$ is the largest subgroup of G in which H is normal. ■

Corollary 2.5.16. *For a subgroup H of G,*

$$H \trianglelefteq G \quad \Longleftrightarrow \quad N_G(H) = G.$$

Proof. If $H \trianglelefteq G$, then $gHg^{-1} = H$ for all $g \in G$, so every element belongs to $N_G(H)$.

Conversely, if $N_G(H) = G$, then every $g \in G$ satisfies $gHg^{-1} = H$, proving that H is normal. ■

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Example 2.5.17. Let $G = S_3$ and $H = \{e, (12)\}$. Since H has order 2, it is normal in itself. However, it is not normal in S_3.

The only elements of S_3 that conjugate (12) to itself are e and (12). Therefore

$$N_{S_3}(H) = H.$$

Thus H is its own normalizer in S_3.

2.6 Permutation Groups

Permutation groups provide an important class of examples in group theory. They arise naturally in the study of symmetries, arrangements, and transformations of finite sets. More importantly, Cayley's theorem shows that every abstract group can be represented as a group of permutations.

Definition 2.6.1. Let X be a nonempty set. A *permutation* of X is a bijective mapping from X onto itself.

The set of all permutations of X, denoted by $\mathrm{Sym}(X)$, forms a group under composition of mappings. Indeed, the composition of two bijections is a bijection, composition is associative, the identity mapping is an identity element, and every bijection possesses an inverse.

When $X = \{1, 2, \ldots, n\}$, this group is called the *symmetric group of degree n*, denoted by S_n.

Since the images of $1, 2, \ldots, n$ must be distinct, there are $n!$ possible permutations. Consequently,

$$|S_n| = n!.$$

2.6.1 Representation of Permutations

A permutation $\sigma \in S_n$ may be represented by listing the images of the elements of $\{1, 2, \ldots, n\}$.

For example,

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix}$$

means that $\sigma(1) = 3$, $\sigma(2) = 1$, $\sigma(3) = 4$, and $\sigma(4) = 2$.

This representation is called *two-line notation*. Although convenient for describing an arbitrary permutation, it is often more useful to express permutations as products of cycles.

Definition 2.6.2. Let $a_1, a_2, \ldots, a_r$ be distinct elements of $\{1, 2, \ldots, n\}$. The permutation

$$(a_1\ a_2\ \cdots\ a_r)$$

that sends a_1 to a_2, a_2 to a_3, and so on, sends a_r to a_1, and fixes every other element is called a *cycle of length r*, or an *r-cycle*.

A cycle of length two is called a *transposition*. A cycle of length one fixes its only entry and may be omitted from the cycle notation.

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Example 2.6.3. The cycle $(1\ 3\ 5\ 2)$ sends

$$1 \mapsto 3, \quad 3 \mapsto 5, \quad 5 \mapsto 2, \quad 2 \mapsto 1,$$

and fixes every other element.

The cycles $(1\ 3\ 5\ 2)$, $(3\ 5\ 2\ 1)$, and $(5\ 2\ 1\ 3)$ represent the same permutation, since each describes the same cyclic movement of elements.

Throughout this chapter, products of permutations are evaluated from right to left. Thus, in the product $\sigma\tau$, the permutation τ is applied first, followed by σ.

Example 2.6.4. In S_3, let $\sigma = (12)$ and $\tau = (13)$. Applying τ first and then σ, we obtain

$$\sigma\tau = (132).$$

On the other hand,

$$\tau\sigma = (123).$$

Thus $\sigma\tau \neq \tau\sigma$, proving that S_3 is nonabelian.

2.6.2 Disjoint Cycles

Definition 2.6.5. Two cycles are said to be *disjoint* if they have no symbols in common.

For example, $(1\ 4\ 3)$ and $(2\ 5)$ are disjoint, whereas $(1\ 2\ 3)$ and $(3\ 4)$ are not.

A fundamental property of permutations is that every permutation of a finite set can be expressed as a product of disjoint cycles.

Theorem 2.6.6. *Every permutation of a finite set can be expressed as a product of pairwise disjoint cycles. This decomposition is unique up to the order of the cycles and cyclic rearrangement of the entries within each cycle, with cycles of length one omitted if desired.*

Proof. Let σ be a permutation of the finite set X.

Choose an element $a \in X$ and consider the sequence

$$a,\ \sigma(a),\ \sigma^2(a), \ldots.$$

Since X is finite, some two terms must be equal. Because σ is bijective, applying a suitable inverse power shows that there exists a positive integer r such that $\sigma^r(a) = a$.

Choose r to be the least such positive integer. Then

$$a, \sigma(a), \ldots, \sigma^{r-1}(a)$$

are distinct and form the cycle

$$(a\ \sigma(a)\ \cdots\ \sigma^{r-1}(a)).$$

If these elements exhaust X, the required decomposition is obtained. Otherwise, choose an element b not occurring in this cycle and repeat the construction.

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The new cycle is disjoint from the first. Indeed, if an iterate of b belonged to the first cycle, applying suitable powers of σ and σ^{-1} would imply that b itself belonged to the first cycle, a contradiction.

Continuing in this way, and using the finiteness of X, we eventually obtain a collection of pairwise disjoint cycles whose union contains every element of X. Their product agrees with σ on all elements of X.

For uniqueness, observe that the set

$$\{\sigma^k(a) : k \in \mathbb{Z}\}$$

is uniquely determined by σ and a. Thus the elements occurring in each cycle are uniquely determined. Their cyclic order is determined by the action of σ, up to the choice of the first entry.

Hence the decomposition is unique in the stated sense. ■

Example 2.6.7. Let

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 1 & 6 & 2 & 4 \end{pmatrix}.$$

Starting with 1, we have $1 \mapsto 3 \mapsto 1$, giving the cycle (13).

Next, $2 \mapsto 5 \mapsto 2$, giving (25). Finally, $4 \mapsto 6 \mapsto 4$, giving (46).

Therefore

$$\sigma = (13)(25)(46).$$

Since these cycles are disjoint transpositions, $\sigma^2 = e$, and hence $\operatorname{ord}(\sigma) = 2$.

Proposition 2.6.8. *Disjoint cycles commute.*

Proof. Let α and β be disjoint cycles, and let A and B denote the sets of symbols moved by them.

If $x \in A$, then $\beta(x) = x$ and $\alpha(x) \in A$. Since β fixes every element of A,

$$(\alpha\beta)(x) = \alpha(x) = (\beta\alpha)(x).$$

Similarly, if $x \in B$, then both products send x to $\beta(x)$.

If $x \notin A \cup B$, both cycles fix x. Thus $\alpha\beta$ and $\beta\alpha$ agree on every element, proving that $\alpha\beta = \beta\alpha$. ■

2.6.3 Order of a Permutation

The disjoint-cycle decomposition provides a convenient method for determining the order of a permutation.

Theorem 2.6.9. *Let $\sigma \in S_n$ be the product of pairwise disjoint cycles of lengths $r_1, r_2, \ldots, r_t$. Then*

$$\operatorname{ord}(\sigma) = \operatorname{lcm}(r_1, r_2, \ldots, r_t).$$

Proof. Write $\sigma = \sigma_1\sigma_2 \cdots \sigma_t$, where the σ_i are pairwise disjoint cycles of lengths r_i.

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Since disjoint cycles commute,

$$\sigma^m = \sigma_1^m \sigma_2^m \cdots \sigma_t^m.$$

Because the cycles act on disjoint sets, this product is the identity precisely when every σ_i^m is the identity.

An r_i-cycle has order r_i, so $\sigma_i^m = e$ if and only if $r_i \mid m$.

Therefore $\sigma^m = e$ precisely when m is divisible by every r_i. The least positive such integer is their least common multiple, proving the result. ■

Example 2.6.10. Find the order of

$$\tau = (1\,4\,3)(2\,6\,5\,7)$$

in S_7.

The two cycles are disjoint and have lengths 3 and 4. Hence

$$\operatorname{ord}(\tau) = \operatorname{lcm}(3,4) = 12.$$

Example 2.6.11. Consider $\sigma = (123)(45)(6789)$ in S_9. Its disjoint cycles have lengths 3, 2, and 4. Consequently,

$$\operatorname{ord}(\sigma) = \operatorname{lcm}(3,2,4) = 12.$$

2.6.4 Transpositions and Parity of Permutations

Every permutation can be expressed as a product of transpositions. This observation leads to the classification of permutations as even or odd.

Theorem 2.6.12. *Every permutation in S_n can be expressed as a product of transpositions.*

Proof. By the disjoint-cycle decomposition theorem, every permutation is a product of cycles. It is therefore sufficient to show that every cycle is a product of transpositions.

Let $\sigma = (a_1\ a_2\ \cdots\ a_r)$. Then

$$\sigma = (a_1\ a_r)(a_1\ a_{r-1}) \cdots (a_1\ a_2).$$

Indeed, applying the transpositions from right to left sends a_i to a_{i+1} for $1 \le i < r$, sends a_r to a_1, and fixes all other elements.

Thus every cycle, and hence every permutation, is a product of transpositions. ■

Example 2.6.13. The cycle $(1\,2\,3\,4)$ can be written as

$$(1\,2\,3\,4) = (14)(13)(12).$$

Similarly, $(1\,2\,3) = (13)(12)$.

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Definition 2.6.14. A permutation is called *even* if it can be expressed as a product of an even number of transpositions, and *odd* if it can be expressed as a product of an odd number of transpositions.

This definition requires justification: a permutation may admit many different expressions as products of transpositions. We must show that their parity is always the same.

Theorem 2.6.15. *If a permutation is expressed as a product of transpositions in two different ways, then the numbers of transpositions occurring in the two expressions have the same parity.*

Proof. For $\sigma \in S_n$, define its inversion number by

$$I(\sigma) = \big|\{(i,j) : 1 \le i < j \le n,\ \sigma(i) > \sigma(j)\}\big|.$$

We show that composing a permutation with a transposition changes the parity of its inversion number.

First consider a transposition that interchanges two adjacent positions in the one-line representation of σ. Only the relative order of the two interchanged entries changes; their combined contribution to inversions with any other entry remains unchanged. Thus the inversion number changes by exactly 1 or -1, and hence its parity is reversed.

Every transposition can be expressed as a product of an odd number of adjacent transpositions. Indeed, interchanging positions i and j, where $i < j$, can be accomplished using $2(j-i)-1$ adjacent transpositions.

Consequently, multiplication by any transposition reverses the parity of the inversion number.

The identity permutation has inversion number 0. Therefore, if σ is a product of r transpositions, then

$$I(\sigma) \equiv r \pmod 2.$$

Hence the parity of r depends only on σ and not on its representation as a product of transpositions. This proves the theorem. ■

Definition 2.6.16. The *sign* of a permutation σ is

$$\operatorname{sgn}(\sigma) = (-1)^{I(\sigma)}.$$

Thus $\operatorname{sgn}(\sigma) = 1$ for an even permutation and $\operatorname{sgn}(\sigma) = -1$ for an odd permutation.

Proposition 2.6.17. *For $\sigma, \tau \in S_n$,*

$$\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau).$$

Proof. Suppose σ and τ are products of r and s transpositions, respectively. Then $\sigma\tau$ is a product of $r+s$ transpositions.

By the parity theorem,

$$\operatorname{sgn}(\sigma\tau) = (-1)^{r+s} = (-1)^r(-1)^s = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau).$$

■

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Corollary 2.6.18. *An r-cycle has sign* $(-1)^{r-1}$.

Proof. Every r-cycle is a product of $r-1$ transpositions. The result follows from the definition of sign. ■

2.6.5 The Alternating Group

Definition 2.6.19. The set of all even permutations in S_n is called the *alternating group of degree n*, denoted by A_n.

Theorem 2.6.20. *For every* $n \geq 2$, A_n *is a normal subgroup of* S_n, *and*

$$|A_n| = \frac{n!}{2}.$$

Proof. The sign mapping

$$\text{sgn} : S_n \longrightarrow \{1, -1\}$$

is a group homomorphism. It is surjective for $n \geq 2$, since the identity has sign 1 and every transposition has sign -1.

Its kernel is precisely the set of even permutations:

$$\ker(\text{sgn}) = A_n.$$

Since the kernel of a homomorphism is a normal subgroup,

$$A_n \trianglelefteq S_n.$$

By the First Isomorphism Theorem,

$$S_n / A_n \cong \{1, -1\}.$$

Thus $[S_n : A_n] = 2$. Since $|S_n| = n!$, Lagrange's theorem gives $|A_n| = n!/2$. ■

Example 2.6.21. The alternating group A_3 consists of

$$A_3 = \{e, (123), (132)\}.$$

It is cyclic of order 3, generated by (123).

2.6.6 Cayley's Theorem

Cayley's theorem establishes a fundamental connection between abstract groups and permutation groups. It shows that every group can be represented faithfully as permutations of an appropriate set.

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Theorem 2.6.22 (Cayley's Theorem). *Every group G is isomorphic to a subgroup of the permutation group* $\mathrm{Sym}(G)$.
In particular, every finite group of order n is isomorphic to a subgroup of S_n.

Proof. For each $g \in G$, define a mapping $L_g : G \to G$ by

$$L_g(x) = gx.$$

This is called left translation by g.

Since

$$L_{g^{-1}}(L_g(x)) = g^{-1}(gx) = x,$$

and similarly $L_g(L_{g^{-1}}(x)) = x$, the mapping L_g is bijective, with inverse $L_{g^{-1}}$. Thus $L_g \in \mathrm{Sym}(G)$.

Now define $\Phi : G \to \mathrm{Sym}(G)$ by $\Phi(g) = L_g$.

For $g, h, x \in G$,

$$(L_g \circ L_h)(x) = g(hx) = (gh)x = L_{gh}(x).$$

Thus

$$\Phi(gh) = L_{gh} = L_g \circ L_h = \Phi(g)\Phi(h).$$

Hence Φ is a homomorphism.

To prove injectivity, suppose $\Phi(g) = \Phi(h)$. Then $L_g = L_h$. Evaluating both maps at the identity e gives

$$g = L_g(e) = L_h(e) = h.$$

Therefore Φ is injective.

Consequently, G is isomorphic to its image $\Phi(G)$, which is a subgroup of $\mathrm{Sym}(G)$.

If $|G| = n$, then $\mathrm{Sym}(G) \cong S_n$, and the final assertion follows. ■

Example 2.6.23. Let $G = \mathbb{Z}_3 = \{[0], [1], [2]\}$ under addition.

Left translation by $[1]$ sends

$$[0] \mapsto [1], \qquad [1] \mapsto [2], \qquad [2] \mapsto [0].$$

Labeling $[0], [1], [2]$ by $1, 2, 3$, respectively, this translation corresponds to the cycle (123).

Similarly, translation by $[2]$ corresponds to (132), while translation by $[0]$ is the identity.

Thus Cayley's theorem realizes $\mathbb{Z}_3$ as the subgroup

$$\{e, (123), (132)\} \leq S_3.$$

2.7 The Alternating Group A_4

The alternating group A_4 provides an important example in finite group theory. It consists of the even permutations of four symbols and has order

$$|A_4| = \frac{4!}{2} = 12.$$

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Its elements are the identity, eight 3-cycles, and three products of two disjoint transpositions. More explicitly,

$$A_4 = \{e, (123), (132), (124), (142), \\ (134), (143), (234), (243), \\ (12)(34), (13)(24), (14)(23)\}.$$

The group A_4 is particularly useful for showing that the converse of Lagrange's theorem does not hold in general.

Theorem 2.7.1. *The alternating group A_4 has no subgroup of order 6.*

Proof. Suppose, to the contrary, that H is a subgroup of A_4 with $|H| = 6$.

Since $|A_4| = 12$, Lagrange's theorem gives

$$[A_4 : H] = \frac{12}{6} = 2.$$

Every subgroup of index two is normal, so $H \trianglelefteq A_4$.

The quotient group A_4/H therefore exists and has order 2. Every element of a group of order 2 has square equal to the identity. Consequently, for every $g \in A_4$,

$$(gH)^2 = H,$$

which implies $g^2H = H$, and hence $g^2 \in H$.

Now consider the eight 3-cycles in A_4. The square of a 3-cycle is its inverse, which is again a 3-cycle. As the squaring operation permutes these eight 3-cycles, the condition $g^2 \in H$ forces all eight 3-cycles to belong to H.

But H has only six elements, giving a contradiction.

Therefore A_4 has no subgroup of order 6. ■

2.8 Dihedral Groups

Dihedral groups arise from the rigid symmetries of regular polygons. They provide important geometric examples of nonabelian groups and illustrate the interaction between rotations and reflections.

Definition 2.8.1. For $n \geq 3$, the *dihedral group* D_n is the group of all rigid symmetries of a regular n-gon under composition.

It consists of n rotations and n reflections, so

$$|D_n| = 2n.$$

Let r denote counterclockwise rotation through an angle $2\pi/n$, and let s denote reflection in a fixed symmetry axis.

The elements r and s satisfy

$$r^n = e, \qquad s^2 = e, \qquad srs = r^{-1}.$$

Equivalently, $sr = r^{-1}s$.

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Every symmetry can be written uniquely in one of the forms r^k or sr^k, where $0 \leq k < n$. Thus

$$D_n = \{e, r, \ldots, r^{n-1}, s, sr, \ldots, sr^{n-1}\}.$$

These relations are summarized by the presentation

$$D_n = \langle r, s \mid r^n = e,\ s^2 = e,\ srs = r^{-1} \rangle.$$

Remark 2.8.2. We use D_n for the dihedral group of order $2n$. Some textbooks use the notation D_{2n} for the same group.

2.8.1 Algebraic Properties of Dihedral Groups

The relation $sr = r^{-1}s$ is fundamental for simplifying products in a dihedral group.

Proposition 2.8.3. *In D_n, for all integers i, j,*

(i) $sr^i = r^{-i}s$;

(ii) $(sr^i)(sr^j) = r^{j-i}$;

(iii) $(sr^i)^2 = e$.

Proof. The defining relation $sr = r^{-1}s$ implies, by induction and the inverse laws, that $sr^i = r^{-i}s$ for every integer i.

Consequently,

$$(sr^i)(sr^j) = sr^i sr^j = r^{-i}ssr^j = r^{j-i}.$$

Taking $j = i$ gives

$$(sr^i)^2 = r^{i-i} = e.$$

Thus every reflection has order 2. ∎

Proposition 2.8.4. *For every $n \geq 3$, the dihedral group D_n is nonabelian.*

Proof. From the defining relations, $sr = r^{-1}s$. If D_n were abelian, we would also have $sr = rs$.

Therefore $r^{-1}s = rs$, and right cancellation would give $r^{-1} = r$, so $r^2 = e$.

But r has order $n \geq 3$, a contradiction. Thus D_n is nonabelian. ∎

2.8.2 The Dihedral Group D_3

The group D_3 is the symmetry group of an equilateral triangle. It has six elements: three rotations and three reflections.

Let r be rotation through 120° and s a reflection. Then

$$D_3 = \{e, r, r^2, s, sr, sr^2\},$$

with defining relations

$$r^3 = e, \qquad s^2 = e, \qquad srs = r^{-1}.$$

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Theorem 2.8.5. *The dihedral group D_3 is isomorphic to S_3.*

Proof. Label the vertices of an equilateral triangle by $1, 2, 3$.

Every symmetry permutes these vertices, giving a mapping

$$\varphi : D_3 \to S_3.$$

Since the action of a composition of symmetries on the vertices is the composition of their corresponding permutations, φ is a homomorphism.

If a symmetry fixes all three vertices, it must be the identity symmetry. Hence $\ker \varphi = \{e\}$, so φ is injective.

Since $|D_3| = 6 = |S_3|$, an injective mapping between these finite sets is also surjective.

Therefore φ is an isomorphism, and $D_3 \cong S_3$. ■

2.8.3 The Dihedral Group D_4

The group D_4 is the symmetry group of a square. It has eight elements:

$$D_4 = \{e, r, r^2, r^3, s, sr, sr^2, sr^3\},$$

where r denotes rotation through 90° and s is a reflection.

The defining relations are

$$r^4 = e, \qquad s^2 = e, \qquad srs = r^{-1}.$$

Example 2.8.6. Compute $(sr^2)(sr^3)$.

Using the relation $sr^i s = r^{-i}$, we obtain

$$(sr^2)(sr^3) = sr^2sr^3 = r^{-2}r^3 = r.$$

Thus the composition of these two reflections is rotation through 90°.

Example 2.8.7. Find the order of sr^3 in D_4.

Since sr^3 is a reflection, the preceding proposition gives

$$(sr^3)^2 = e.$$

As $sr^3 \neq e$, its order is 2.

2.8.4 The Rotation Subgroup

The rotations of a regular polygon form a cyclic subgroup of the corresponding dihedral group.

Theorem 2.8.8. *Let D_n be the dihedral group of order $2n$, with rotation generator r. Then*

$$\langle r \rangle \trianglelefteq D_n$$

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and

$$D_n/\langle r\rangle \cong \mathbb{Z}_2.$$

Proof. The subgroup

$$\langle r\rangle = \{e, r, \dots, r^{n-1}\}$$

has order n.

Since $|D_n| = 2n$, its index is

$$[D_n : \langle r\rangle] = \frac{2n}{n} = 2.$$

Every subgroup of index two is normal, so $\langle r\rangle \trianglelefteq D_n$.

The quotient group has exactly two elements,

$$D_n/\langle r\rangle = \{\langle r\rangle, s\langle r\rangle\}.$$

Every group of order two is cyclic and isomorphic to $\mathbb{Z}_2$. Hence

$$D_n/\langle r\rangle \cong \mathbb{Z}_2.$$

■

2.9 Exercises

The following exercises are arranged to progress from routine computations to proofs. Unless stated otherwise, all groups are written multiplicatively.

1. Show directly that every subgroup of $(\mathbb{Z}, +)$ is normal.

2. Determine whether $\{e, (123), (132)\}$ is normal in S_3.

3. Prove that the intersection of any family of normal subgroups is normal.

4. Give an example of a subgroup of S_3 that is not normal, and exhibit an explicit conjugate outside it.

5. List all cosets of $3\mathbb{Z}$ in $\mathbb{Z}$ and construct the addition table of $\mathbb{Z}/3\mathbb{Z}$.

6. Find the order of $(\mathbb{Z}/18\mathbb{Z})/\langle\overline{6}\rangle$.

7. Prove that a quotient group G/N is abelian if and only if $aba^{-1}b^{-1} \in N$ for every $a, b \in G$.

8. Determine the kernel and image of $f : \mathbb{Z} \to \mathbb{Z}$, $f(k) = 5k$.

9. Determine the kernel and image of $f : \mathbb{Z} \to \mathbb{Z}/15\mathbb{Z}$, $f(k) = \overline{6k}$.

10. Show that a homomorphism carries inverses to inverses and that its kernel is trivial exactly when it is injective.

11. Use the first isomorphism theorem to identify $\mathbb{Z}/8\mathbb{Z}$ as the image of a homomorphism from $\mathbb{Z}$.

12. Prove that $\mathrm{SL}_2(\mathbb{R})$ is normal in $\mathrm{GL}_2(\mathbb{R})$.

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13. Apply the third isomorphism theorem to $(\mathbb{Z}/24\mathbb{Z})/(6\mathbb{Z}/24\mathbb{Z})$.

14. Prove that HK is a subgroup whenever $H \trianglelefteq G$ and $K \leq G$.

15. Compute $C_{S_3}((123))$ and compare it with $Z(S_3)$.

16. Determine the normalizer of $\langle (12) \rangle$ in S_3.

17. Write $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 1 & 5 & 3 \end{pmatrix}$ in disjoint cycles and find its order.

18. Express (12345) as a product of transpositions and determine its parity.

19. Find the order and sign of $(135)(2467)$ as a permutation of seven symbols.

20. Prove that disjoint cycles commute, and give two non-disjoint cycles that do not commute.

21. Use Cayley's construction to embed $\mathbb{Z}/4\mathbb{Z}$ into S_4.

22. List the elements of A_4 by their cycle types and verify that it has order 12.

23. Explain why A_4 disproves the converse of Lagrange's theorem.

24. In D_4, simplify sr^3s, r^2sr, and $(sr)^2$.

25. Prove that the centre of D_3 is trivial and that r^2 belongs to the centre of D_4.

26. Show that D_n has exactly n rotations and n reflections.

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CHAPTER
3

RINGS, HOMOMORPHISMS AND IDEALS

In the preceding chapters, we studied groups, which are algebraic structures equipped with a single binary operation. We now turn to rings, in which two binary operations are connected by the distributive laws.

The ring of integers is the most familiar example. Other important examples include polynomial rings, matrix rings, and rings of integers modulo a positive integer. Although these structures differ in their elements and operations, they share many fundamental algebraic properties.

Ring theory investigates these common properties and the ways in which they differ from ordinary integer arithmetic. In particular, the existence of zero divisors, the possibility of multiplicative inverses, and the role of commutativity lead to several important classes of rings.

An especially significant concept in ring theory is that of an *ideal*. Ideals provide the appropriate framework for constructing quotient rings, just as normal subgroups permit the construction of quotient groups. They also play a central role in understanding ring homomorphisms and the structure of commutative rings.

3.1 Rings: Definitions and First Examples

Definition 3.1.1. A *ring* is a nonempty set R equipped with two binary operations, called *addition* and *multiplication*, satisfying the following conditions:

(R1) $(R, +)$ is an abelian group.

(R2) Multiplication is associative; that is,

$$(ab)c = a(bc)$$

for all $a, b, c \in R$.

(R3) Multiplication is distributive over addition on both sides; that is,

$$a(b + c) = ab + ac, \qquad (a + b)c = ac + bc$$

for all $a, b, c \in R$.

Thus, in a ring, addition is associative and commutative, there is an additive identity 0,

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and every element a has an additive inverse $-a$. Multiplication is associative, but it need not be commutative, and multiplicative inverses need not exist.

The additive identity and additive inverses are unique because $(R, +)$ is a group.

Definition 3.1.2. A ring R is called *commutative* if $ab = ba$ for all $a, b \in R$.

A ring R is said to have a *multiplicative identity*, or *unity*, if there exists an element $1_R \in R$ such that

$$1_R a = a 1_R = a$$

for every $a \in R$.

Unless otherwise stated, rings in the initial sections of this chapter are not assumed to be commutative or to possess a multiplicative identity. In the study of principal, prime, and maximal ideals, the relevant hypotheses will be stated explicitly.

It is important to distinguish the additive identity 0 from a multiplicative identity 1_R. Every ring has an additive identity, whereas not every ring has a multiplicative identity.

Example 3.1.3. (i) The sets $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are commutative rings with identity under ordinary addition and multiplication.

(ii) For $n \geq 2$, the set $\mathbb{Z}_n$ of residue classes modulo n is a commutative ring with identity $[1]$, where

$$[a] + [b] = [a + b], \qquad [a][b] = [ab].$$

Both operations are well-defined because congruence modulo n is compatible with addition and multiplication.

(iii) The set $M_n(\mathbb{R})$ of all $n \times n$ real matrices is a ring under matrix addition and multiplication. Its multiplicative identity is I_n. For $n \geq 2$, matrix multiplication is not commutative in general, so $M_n(\mathbb{R})$ is a noncommutative ring.

(iv) The set $2\mathbb{Z}$ of even integers forms a commutative ring under ordinary addition and multiplication. It has no multiplicative identity.

(v) The polynomial ring $\mathbb{Z}[x]$ is a commutative ring with identity. Similarly, $\mathbb{Z}[x, y]$ is the ring of polynomials in two commuting indeterminates with integer coefficients.

(vi) The singleton set $\{0\}$ forms a ring under $0 + 0 = 0$ and $0 \cdot 0 = 0$. It is called the *zero ring*.

Remark 3.1.4. If a ring has an identity and $1_R = 0$, then for every $a \in R$,

$$a = 1_R a = 0a = 0.$$

Consequently, $R = \{0\}$. Thus the condition $1_R \neq 0$ excludes the zero ring.

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3.1.1 Examples of Specially Defined Operations

A ring need not arise from the usual multiplication of numbers. However, its operations must satisfy all the ring axioms.

Example 3.1.5. Let $R = \mathbb{Z}$ with ordinary addition, and define multiplication by

$$a * b = 0$$

for all $a, b \in \mathbb{Z}$.

Since ordinary addition makes $\mathbb{Z}$ an abelian group, only the multiplicative properties need to be verified.

For $a, b, c \in \mathbb{Z}$,

$$(a * b) * c = 0 = a * (b * c),$$

so multiplication is associative.

Also,

$$a * (b + c) = 0 = a * b + a * c,$$

and similarly

$$(a + b) * c = 0 = a * c + b * c.$$

Thus both distributive laws hold.

Therefore $\mathbb{Z}$, with ordinary addition and zero multiplication, is a ring. It has no multiplicative identity.

Example 3.1.6. Let $R = \mathbb{Z}$ with ordinary addition, and define $a * b = a + b$.

Although this operation is associative and commutative, it does not satisfy distributivity. Indeed,

$$a * (b + c) = a + b + c,$$

whereas

$$a * b + a * c = (a + b) + (a + c) = 2a + b + c.$$

These expressions are not equal for arbitrary integers a, b, c.

Hence the given operations do not make $\mathbb{Z}$ into a ring.

3.1.2 Elementary Properties of Rings

The ring axioms imply several identities familiar from elementary arithmetic. These hold in every ring, including noncommutative rings and rings without unity.

Theorem 3.1.7. *Let R be a ring. For all $a, b, c \in R$, the following identities hold:*

(i) $a0 = 0a = 0$;

(ii) $a(-b) = (-a)b = -(ab)$;

(iii) $(-a)(-b) = ab$;

(iv) $a(b - c) = ab - ac$;

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(v) $(a-b)c = ac - bc$.

If R has a multiplicative identity, then $(-1_R)a = a(-1_R) = -a$ for every $a \in R$.

Proof. For (i), distributivity gives

$$a0 = a(0+0) = a0 + a0.$$

Adding the additive inverse of $a0$ to both sides yields $a0 = 0$. Similarly,

$$0a = (0+0)a = 0a + 0a$$

implies $0a = 0$.

For (ii), observe that

$$ab + a(-b) = a(b + (-b)) = a0 = 0.$$

Thus $a(-b)$ is the additive inverse of ab, giving $a(-b) = -(ab)$.

Similarly,

$$ab + (-a)b = (a + (-a))b = 0b = 0,$$

so $(-a)b = -(ab)$.

Applying (ii) twice gives

$$(-a)(-b) = -\big(a(-b)\big) = -\big(-(ab)\big) = ab,$$

proving (iii).

For (iv), distributivity and (ii) yield

$$a(b-c) = ab + a(-c) = ab - ac.$$

Similarly,

$$(a-b)c = ac + (-b)c = ac - bc,$$

proving (v).

Finally, suppose R has identity 1_R. Then

$$a + (-1_R)a = (1_R + (-1_R))a = 0a = 0.$$

Hence $(-1_R)a = -a$. The identity $a(-1_R) = -a$ follows analogously. ■

Proposition 3.1.8. *If a ring has a multiplicative identity, then it is unique.*

Proof. Suppose u and v are multiplicative identities of R. Since u is an identity, $uv = v$. Since v is also an identity, $uv = u$. Therefore $u = v$. ■

Definition 3.1.9. Let R be a ring. An element $a \in R$ is called *idempotent* if $a^2 = a$.

An element $a \in R$ is called *nilpotent* if $a^n = 0$ for some positive integer n.

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Example 3.1.10. In $\mathbb{Z}_6$, the elements $[3]$ and $[4]$ are nontrivial idempotents because

$$[3]^2 = [3], \qquad [4]^2 = [4].$$

In $\mathbb{Z}_8$, the element $[2]$ is nilpotent, since

$$[2]^3 = [0],$$

although $[2] \neq [0]$.

Proposition 3.1.11. *Every nonzero nilpotent element of a ring is a zero divisor.*

Proof. Let $a \neq 0$ be nilpotent, and let $n \geq 2$ be the least positive integer such that $a^n = 0$. Then $a^{n-1} \neq 0$, but

$$a \cdot a^{n-1} = a^n = 0.$$

Thus a is a zero divisor. ■

3.2 Zero Divisors, Integral Domains and Fields

The existence of multiplicative inverses and zero divisors distinguishes several important classes of rings. In ordinary integer arithmetic, the product of two nonzero integers is nonzero, and cancellation by a nonzero factor is valid. These properties need not hold in an arbitrary ring.

We begin by examining zero divisors and units.

3.2.1 Zero Divisors and Units

Definition 3.2.1. Let R be a ring. A nonzero element $a \in R$ is called a *left zero divisor* if there exists a nonzero element $b \in R$ such that $ab = 0$.

A nonzero element $a \in R$ is called a *right zero divisor* if there exists a nonzero element $b \in R$ such that $ba = 0$.

An element that is either a left or a right zero divisor is called a *zero divisor*.

In a commutative ring, the distinction between left and right zero divisors disappears.

Example 3.2.2. In the ring $\mathbb{Z}_6$, we have

$$[2][3] = [6] = [0].$$

Since $[2]$ and $[3]$ are nonzero, both are zero divisors.

Thus, unlike in $\mathbb{Z}$, the product of two nonzero elements of $\mathbb{Z}_6$ can be zero.

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Example 3.2.3. Consider the matrices

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Both are nonzero, but

$$AB = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Therefore A and B are zero divisors in $M_2(\mathbb{R})$.

Definition 3.2.4. Let R be a ring with identity $1_R \neq 0$. An element $u \in R$ is called a *unit* if there exists $v \in R$ such that

$$uv = vu = 1_R.$$

The element v is called the *multiplicative inverse* of u, denoted by u^{-1}.

The collection of all units of R is denoted by $U(R)$ or $R^\times$.

Proposition 3.2.5. *The set $U(R)$ of units of a ring with identity forms a group under multiplication.*

Proof. Let $u, v \in U(R)$. Since u and v possess multiplicative inverses,

$$(uv)(v^{-1}u^{-1}) = u(vv^{-1})u^{-1} = 1_R,$$

and similarly

$$(v^{-1}u^{-1})(uv) = 1_R.$$

Thus uv is a unit, with inverse $v^{-1}u^{-1}$.

Multiplication is associative, 1_R is the identity, and the inverse of every unit is again a unit. Therefore $U(R)$ is a group. ■

Example 3.2.6. The only units of $\mathbb{Z}$ are 1 and -1, whereas every nonzero element of $\mathbb{Q}$ is a unit.

In $\mathbb{Z}_{10}$, the element $[3]$ is a unit because

$$[3][7] = [21] = [1].$$

Thus $[3]^{-1} = [7]$. By contrast, $[2]$ is a zero divisor because $[2][5] = [0]$.

Proposition 3.2.7. *In a nonzero ring with identity, no unit is a left or right zero divisor.*

Proof. Let u be a unit with inverse u^{-1}. If $ub = 0$, then multiplying on the left by u^{-1} gives

$$b = u^{-1}(ub) = u^{-1}0 = 0.$$

Thus u is not a left zero divisor.

Similarly, if $bu = 0$, multiplication on the right by u^{-1} gives $b = 0$. Hence u is not a right zero divisor. ■

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3.2.2 Integral Domains

Definition 3.2.8. An *integral domain* is a commutative ring R with identity $1_R \neq 0$ that has no zero divisors.

Equivalently, a commutative ring with identity $1_R \neq 0$ is an integral domain if

$$ab = 0 \quad \Longrightarrow \quad a = 0 \text{ or } b = 0$$

for all $a, b \in R$.

Example 3.2.9. The rings $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are integral domains.
The polynomial ring $\mathbb{Z}[x]$ is also an integral domain, whereas $\mathbb{Z}_6$ is not, because $[2][3] = [0]$ with both factors nonzero.

An important consequence of the absence of zero divisors is the cancellation law.

Theorem 3.2.10. *Let R be an integral domain. If $a, b, c \in R$, $a \neq 0$, and $ab = ac$, then $b = c$.*

Proof. Since $ab = ac$, subtracting ac from both sides gives

$$ab - ac = 0.$$

By distributivity, $a(b - c) = 0$.
As R is an integral domain and $a \neq 0$, we must have $b - c = 0$. Therefore $b = c$. ■

Theorem 3.2.11. *Let R be a nonzero commutative ring with identity. Then R is an integral domain if and only if multiplication by every nonzero element satisfies the cancellation law.*

Proof. If R is an integral domain, cancellation follows from the preceding theorem.
Conversely, suppose cancellation holds whenever the common factor is nonzero. Let $ab = 0$ with $a \neq 0$.
Since $a0 = 0$, we have $ab = a0$. Cancellation by a gives $b = 0$. Thus R has no zero divisors and is an integral domain. ■

Theorem 3.2.12. *If R is an integral domain, then the polynomial ring $R[x]$ is also an integral domain.*

Proof. The ring $R[x]$ is commutative and has identity because R has these properties.
Let $f(x), g(x) \in R[x]$ be nonzero polynomials of degrees m and n, with leading coefficients a_m and b_n, respectively.
The coefficient of x^{m+n} in $f(x)g(x)$ is $a_m b_n$. Since R has no zero divisors and both leading coefficients are nonzero,

$$a_m b_n \neq 0.$$

Therefore $f(x)g(x) \neq 0$.
Hence $R[x]$ has no zero divisors and is an integral domain. ■

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3.2.3 Division Rings and Fields

Definition 3.2.13. A *division ring*, also called a *skew field*, is a nonzero ring with identity in which every nonzero element has a multiplicative inverse.

Multiplication in a division ring need not be commutative.

Definition 3.2.14. A *field* is a commutative division ring. Thus a field is a nonzero commutative ring with identity in which every nonzero element is a unit.

The relationship among these structures may be summarized as follows: every field is a commutative division ring and hence an integral domain, but an integral domain need not be a field.

Example 3.2.15. The rings $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are fields. The ring $\mathbb{Z}$ is an integral domain but not a field because 2 has no multiplicative inverse in $\mathbb{Z}$.

The real quaternion algebra $\mathbb{H}$ is a noncommutative division ring. Thus a division ring need not be a field.

Theorem 3.2.16. *Every division ring has no zero divisors.*

Proof. Let R be a division ring and suppose $ab = 0$ for some $a, b \in R$.

If $a = 0$, there is nothing to prove. Suppose $a \neq 0$. Since a is invertible, multiplying $ab = 0$ on the left by a^{-1} gives

$$b = a^{-1}(ab) = a^{-1}0 = 0.$$

Therefore $ab = 0$ implies $a = 0$ or $b = 0$, so R has no zero divisors. ■

Corollary 3.2.17. *Every field is an integral domain.*

Proof. A field is a commutative division ring with identity $1 \neq 0$. By the preceding theorem, it has no zero divisors. Hence it is an integral domain. ■

The converse is false in general, as the ring $\mathbb{Z}$ demonstrates. However, finiteness makes a significant difference.

Theorem 3.2.18. *Every finite integral domain is a field.*

Proof. Let R be a finite integral domain, and let $a \in R$ be nonzero.

Define $f : R \to R$ by $f(r) = ar$.

Suppose $f(r) = f(s)$. Then $ar = as$. Since $a \neq 0$ and R is an integral domain, the cancellation law gives $r = s$. Thus f is injective.

Since R is finite, every injective mapping from R to itself is surjective. Hence f is surjective.

In particular, there exists $b \in R$ such that

$$f(b) = ab = 1_R.$$

Since multiplication is commutative, $ba = 1_R$ as well. Therefore a is invertible.

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As a was an arbitrary nonzero element of R, every nonzero element is a unit. Hence R is a field. ■

3.2.4 Fields of Integers Modulo n

We now investigate when the ring $\mathbb{Z}_n$ is an integral domain or a field.

Theorem 3.2.19. *Let $n \geq 2$. The following statements are equivalent:*

(i) n is a prime number;

(ii) $\mathbb{Z}_n$ is an integral domain;

(iii) $\mathbb{Z}_n$ is a field.

Proof. Suppose first that $n = p$ is prime. Let $[a], [b] \in \mathbb{Z}_p$ satisfy $[a][b] = [0]$. Then $p \mid ab$.

By Euclid's lemma, $p \mid a$ or $p \mid b$. Hence $[a] = [0]$ or $[b] = [0]$, showing that $\mathbb{Z}_p$ has no zero divisors. Thus $\mathbb{Z}_p$ is an integral domain.

Since $\mathbb{Z}_p$ is finite, the preceding theorem implies that it is a field.

Every field is an integral domain, so (iii) implies (ii).

Finally, suppose $\mathbb{Z}_n$ is an integral domain. If n were composite, we could write $n = rs$ with $1 < r, s < n$. Then $[r]$ and $[s]$ would be nonzero elements of $\mathbb{Z}_n$ satisfying

$$[r][s] = [rs] = [n] = [0].$$

This contradicts the absence of zero divisors. Therefore n must be prime.

Hence all three statements are equivalent. ■

Example 3.2.20. The rings $\mathbb{Z}_2$, $\mathbb{Z}_3$, $\mathbb{Z}_5$, and $\mathbb{Z}_7$ are fields because $2, 3, 5$, and 7 are prime numbers.

In contrast, $\mathbb{Z}_4$, $\mathbb{Z}_6$, $\mathbb{Z}_8$, and $\mathbb{Z}_9$ are not fields because their moduli are composite.

3.2.5 Units and Zero Divisors in $\mathbb{Z}_n$

The units of $\mathbb{Z}_n$ admit a simple characterization using the greatest common divisor.

Theorem 3.2.21. *Let $n \geq 2$. A residue class $[a] \in \mathbb{Z}_n$ is a unit if and only if*

$$\gcd(a, n) = 1.$$

Proof. Suppose $[a]$ is a unit. Then there exists $[b] \in \mathbb{Z}_n$ such that $[a][b] = [1]$. Hence $ab \equiv 1 \pmod{n}$.

Therefore $ab - 1 = kn$ for some integer k, giving

$$ab - kn = 1.$$

Every common divisor of a and n must divide the left side, and hence must divide 1. Thus $\gcd(a, n) = 1$.

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Conversely, suppose $\gcd(a, n) = 1$. By Bézout's identity, there exist integers u, v such that

$$au + nv = 1.$$

Reducing modulo n gives $[a][u] = [1]$. Thus $[u]$ is a multiplicative inverse of $[a]$, proving that $[a]$ is a unit. ■

Theorem 3.2.22. *Let $n \geq 2$. A nonzero element $[a] \in \mathbb{Z}_n$ is a zero divisor if and only if*

$$\gcd(a, n) > 1.$$

Proof. Suppose $[a] \neq [0]$ and put $d = \gcd(a, n) > 1$.

Write $a = da_1$ and $n = dn_1$. Since $d > 1$, we have $1 \leq n_1 < n$, so $[n_1] \neq [0]$ in $\mathbb{Z}_n$.

Moreover,

$$[a][n_1] = [an_1] = [da_1n_1] = [a_1n] = [0].$$

Thus $[a]$ is a zero divisor.

Conversely, suppose $[a]$ is a zero divisor. Then there exists $[b] \neq [0]$ such that $[a][b] = [0]$.

If $\gcd(a, n) = 1$, the preceding theorem would imply that $[a]$ is a unit. Multiplying $[a][b] = [0]$ by its inverse would then give $[b] = [0]$, a contradiction.

Therefore $\gcd(a, n) > 1$. ■

Corollary 3.2.23. *Every nonzero element of $\mathbb{Z}_n$ is either a unit or a zero divisor, but not both.*

Proof. For any nonzero $[a] \in \mathbb{Z}_n$, either $\gcd(a, n) = 1$ or $\gcd(a, n) > 1$. The preceding two theorems show that $[a]$ is respectively a unit or a zero divisor.

The two possibilities are mutually exclusive. ■

Example 3.2.24. We determine all units and nonzero zero divisors of $\mathbb{Z}_{12}$.

By the unit criterion, $[a]$ is invertible precisely when $\gcd(a, 12) = 1$. The corresponding integers are $1, 5, 7, 11$. Hence

$$U(\mathbb{Z}_{12}) = \{[1], [5], [7], [11]\}.$$

Indeed, the multiplicative inverses are

$$[1]^{-1} = [1], \quad [5]^{-1} = [5], \quad [7]^{-1} = [7], \quad [11]^{-1} = [11].$$

The remaining nonzero residue classes are

$$[2], [3], [4], [6], [8], [9], [10].$$

Each has greatest common divisor with 12 greater than 1, so these are precisely the nonzero zero divisors of $\mathbb{Z}_{12}$.

For example,

$$[2][6] = [0], \qquad [3][4] = [0], \qquad [8][3] = [0].$$

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Corollary 3.2.25. *For every $n \geq 2$, the group of units $U(\mathbb{Z}_n)$ has order $\varphi(n)$, where φ denotes Euler's totient function.*

Proof. By the unit criterion, $U(\mathbb{Z}_n)$ consists precisely of the residue classes $[a]$ for which $\gcd(a, n) = 1$.

The number of such residue classes is $\varphi(n)$. Hence $|U(\mathbb{Z}_n)| = \varphi(n)$. ■

3.3 Subrings and Ideals

A subset of a ring may itself form a ring under the same addition and multiplication. Such a subset is called a *subring*. However, the construction of quotient rings requires a stronger concept, namely that of an *ideal*.

Ideals are distinguished from arbitrary subrings by an additional property: multiplying an element of an ideal by any element of the ambient ring produces another element of the ideal. This property, called *absorption*, is fundamental in the theory of quotient rings and ring homomorphisms.

3.3.1 Subrings

Definition 3.3.1. Let R be a ring. A nonempty subset S of R is called a *subring* of R if S is itself a ring under the operations of addition and multiplication inherited from R.

A subring necessarily contains the additive identity 0 of R, but it need not contain the multiplicative identity 1_R, even when R possesses one.

Since associativity and distributivity are inherited from the ambient ring, it is unnecessary to verify these properties again. The following result gives a convenient criterion for identifying subrings.

Theorem 3.3.2. *A nonempty subset S of a ring R is a subring of R if and only if*

$$a - b \in S, \qquad ab \in S$$

for all $a, b \in S$.

Proof. Suppose S is a subring of R. Then $(S, +)$ is an additive subgroup of $(R, +)$, so $a - b \in S$ whenever $a, b \in S$. Also, S is closed under multiplication, giving $ab \in S$.

Conversely, suppose S is nonempty and satisfies the stated conditions.

Since $a - b \in S$ for all $a, b \in S$, the subgroup test shows that $(S, +)$ is a subgroup of $(R, +)$. In particular, $0 \in S$, every element of S has its additive inverse in S, and S is closed under addition.

By assumption, S is also closed under multiplication. Associativity of multiplication and both distributive laws hold in S because they hold in R.

Therefore S is a ring under the inherited operations, and hence a subring of R. ■

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Example 3.3.3. (i) The integers $\mathbb{Z}$ form a subring of $\mathbb{Q}$, and $\mathbb{Q}$ is a subring of $\mathbb{R}$.

(ii) The even integers $2\mathbb{Z}$ form a subring of $\mathbb{Z}$. Indeed, the difference and product of any two even integers are even.

(iii) The Gaussian integers

$$\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$$

form a subring of $\mathbb{C}$. For $a + bi, c + di \in \mathbb{Z}[i]$, their difference is

$$(a - c) + (b - d)i \in \mathbb{Z}[i],$$

and their product is

$$(ac - bd) + (ad + bc)i \in \mathbb{Z}[i].$$

(iv) The set of diagonal matrices

$$D_n(\mathbb{R}) = \{\mathrm{diag}(a_1, \ldots, a_n) : a_i \in \mathbb{R}\}$$

is a subring of $M_n(\mathbb{R})$.

(v) For every ring R, the subsets $\{0\}$ and R are subrings of R.

Remark 3.3.4. Some authors require a subring of a ring with identity to contain the same multiplicative identity. Under that convention, $2\mathbb{Z}$ is not a unital subring of $\mathbb{Z}$.

Throughout this chapter, we use the more general definition, under which a subring need not contain 1_R.

3.3.2 Ideals

A subring is closed under multiplication of its own elements. An ideal satisfies the stronger requirement that it absorbs multiplication by arbitrary elements of the ambient ring.

Definition 3.3.5. Let R be a ring. A nonempty subset I of R is called a *two-sided ideal*, or simply an *ideal*, if

(i) $a - b \in I$ for all $a, b \in I$;

(ii) $ra \in I$ and $ar \in I$ for every $a \in I$ and $r \in R$.

The first condition says that $(I, +)$ is a subgroup of $(R, +)$. The second condition is called the *absorption property*.

If R is commutative, the conditions $ra \in I$ and $ar \in I$ are identical.

In a noncommutative ring, it is also possible to consider left ideals and right ideals, where only one of the absorption conditions is imposed. Unless stated otherwise, the word *ideal* will mean a two-sided ideal.

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Example 3.3.6. For any integer n, the subset

$$n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$$

is an ideal of $\mathbb{Z}$.

Indeed, if $na, nb \in n\mathbb{Z}$, then

$$na - nb = n(a - b) \in n\mathbb{Z}.$$

Furthermore, if $r \in \mathbb{Z}$, then

$$r(na) = n(ra) \in n\mathbb{Z}.$$

Thus $n\mathbb{Z}$ satisfies the ideal conditions.

Example 3.3.7. Let $R = \mathbb{R}[x]$ and consider

$$I = x\mathbb{R}[x] = \{xf(x) : f(x) \in \mathbb{R}[x]\}.$$

The set I consists precisely of the polynomials whose constant term is zero.

If $xf(x), xg(x) \in I$, then

$$xf(x) - xg(x) = x(f(x) - g(x)) \in I.$$

Also, for $h(x) \in R$,

$$h(x) \cdot xf(x) = x(h(x)f(x)) \in I.$$

Therefore I is an ideal of $\mathbb{R}[x]$.

Example 3.3.8. The ring $\mathbb{Z}$ is a subring of $\mathbb{Q}$, but it is not an ideal of $\mathbb{Q}$.

Indeed, $1 \in \mathbb{Z}$ and $\frac{1}{2} \in \mathbb{Q}$, yet

$$\frac{1}{2} \cdot 1 = \frac{1}{2} \notin \mathbb{Z}.$$

Thus $\mathbb{Z}$ fails the absorption property.

Proposition 3.3.9. *Every ideal of a ring is a subring, but the converse need not hold.*

Proof. Let I be an ideal of R. By definition, I is nonempty and closed under subtraction.

If $a, b \in I$, then $a \in I$ and $b \in R$, so the absorption property gives $ab \in I$. Hence I is also closed under multiplication.

By the subring test, I is a subring of R.

The converse fails, as shown by the subring $\mathbb{Z}$ of $\mathbb{Q}$, which is not an ideal. ■

3.3.3 Elementary Properties of Ideals

Every ring has at least two ideals: the zero ideal and the entire ring.

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Definition 3.3.10. The ideals $\{0\}$ and R are called the *trivial ideals* of R. An ideal I is called a *proper ideal* if $I \neq R$.

In a ring with identity, proper ideals are characterized by the absence of the identity element.

Theorem 3.3.11. *Let R be a ring with identity 1_R, and let I be an ideal of R. Then*

$$I = R \iff 1_R \in I.$$

Consequently, if I contains a unit of R, then $I = R$.

Proof. If $I = R$, then clearly $1_R \in I$.

Conversely, suppose $1_R \in I$. For any $r \in R$, the absorption property gives

$$r = r1_R \in I.$$

Hence every element of R belongs to I, and therefore $I = R$.

Now suppose I contains a unit u. Since $u^{-1} \in R$ and $u \in I$, the absorption property gives

$$1_R = u^{-1}u \in I.$$

By the first part, $I = R$. ■

Corollary 3.3.12. *In a field F, the only ideals are $\{0\}$ and F.*

Proof. Let I be a nonzero ideal of F. Choose $0 \neq a \in I$.

Since F is a field, a is a unit. By the unit criterion, $I = F$.

Thus the only ideals of F are $\{0\}$ and F. ■

3.3.4 Intersections, Sums and Products of Ideals

Ideals can be combined to construct new ideals. The most important operations are intersection, sum, and product.

Theorem 3.3.13. *Let $\{I_\lambda : \lambda \in \Lambda\}$ be a nonempty family of ideals of a ring R. Then*

$$\bigcap_{\lambda \in \Lambda} I_\lambda$$

is an ideal of R.

Proof. Every ideal contains 0, so the intersection is nonempty.

Let a, b belong to the intersection. Then $a, b \in I_\lambda$ for every $\lambda \in \Lambda$. Since each I_λ is an ideal, $a - b \in I_\lambda$ for every λ. Hence $a - b$ belongs to the intersection.

Similarly, for any $r \in R$, both ra and ar belong to every I_λ, and therefore to their intersection.

Thus the intersection is an ideal. ■

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Definition 3.3.14. Let I and J be ideals of R. Their *sum* is

$$I + J = \{i + j : i \in I,\ j \in J\}.$$

Theorem 3.3.15. *If I and J are ideals of R, then $I + J$ is an ideal of R. Moreover, it is the smallest ideal containing both I and J.*

Proof. Since $0 = 0 + 0 \in I + J$, the sum is nonempty.

Let $a = i_1 + j_1$ and $b = i_2 + j_2$ belong to $I + J$. Then

$$a - b = (i_1 - i_2) + (j_1 - j_2) \in I + J,$$

because $i_1 - i_2 \in I$ and $j_1 - j_2 \in J$.

For $r \in R$,

$$ra = ri_1 + rj_1 \in I + J,$$

and similarly

$$ar = i_1 r + j_1 r \in I + J.$$

Therefore $I + J$ is an ideal.

Clearly $I \subseteq I + J$ and $J \subseteq I + J$, because both ideals contain 0.

If K is any ideal containing both I and J, then $i + j \in K$ for every $i \in I$ and $j \in J$. Hence $I + J \subseteq K$.

Thus $I + J$ is the smallest ideal containing both I and J. ■

Definition 3.3.16. Let I and J be ideals of R. Their *product* is

$$IJ = \left\{ \sum_{k=1}^{m} i_k j_k : m \geq 1,\ i_k \in I,\ j_k \in J \right\}.$$

Thus IJ consists of all finite sums of products of elements of I and J.

Remark 3.3.17. The set $\{ij : i \in I,\ j \in J\}$ need not be closed under addition. For this reason, the definition of the product ideal requires finite sums of such products.

Theorem 3.3.18. *If I and J are two-sided ideals of R, then IJ is a two-sided ideal. Moreover,*

$$IJ \subseteq I \cap J.$$

Proof. The set IJ is nonempty because $0 = 0 \cdot 0 \in IJ$.

Let

$$a = \sum_{k=1}^{m} i_k j_k, \qquad b = \sum_{\ell=1}^{t} i'_\ell j'_\ell$$

be elements of IJ.

Then

$$a - b = \sum_{k=1}^{m} i_k j_k + \sum_{\ell=1}^{t} (-i'_\ell) j'_\ell.$$

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Since $-i'_\ell \in I$, this expression belongs to IJ.

For $r \in R$,

$$ra = \sum_{k=1}^{m} (ri_k)j_k \in IJ,$$

because $ri_k \in I$. Similarly,

$$ar = \sum_{k=1}^{m} i_k(j_k r) \in IJ,$$

because $j_k r \in J$.

Thus IJ is an ideal.

Finally, every product ij, with $i \in I$ and $j \in J$, belongs to I because I absorbs multiplication on the right, and belongs to J because J absorbs multiplication on the left.

Hence $ij \in I \cap J$. Since $I \cap J$ is closed under addition, every finite sum of such products also belongs to $I \cap J$. Therefore

$$IJ \subseteq I \cap J.$$

■

Example 3.3.19. In $\mathbb{Z}$, let $I = 4\mathbb{Z}$ and $J = 6\mathbb{Z}$. Then

$$I + J = 2\mathbb{Z}, \qquad I \cap J = 12\mathbb{Z}, \qquad IJ = 24\mathbb{Z}.$$

The sum consists of multiples of $\gcd(4,6) = 2$, while the intersection consists of multiples of $\operatorname{lcm}(4,6) = 12$.

The product is $24\mathbb{Z}$, since every finite sum of products of multiples of 4 and 6 is a multiple of 24, and 24 itself is such a product.

3.4 Principal Ideals and Principal Ideal Domains

We now turn to ideals generated by individual elements. These are called *principal ideals*. They are especially important because divisibility of elements can be expressed through containment of principal ideals.

Throughout this section, R denotes a commutative ring with identity $1_R \neq 0$, unless otherwise stated.

3.4.1 Ideals Generated by Elements

Definition 3.4.1. Let $a_1, \ldots, a_t \in R$. The ideal generated by $a_1, \ldots, a_t$ is

$$(a_1, \ldots, a_t) = \{r_1 a_1 + \cdots + r_t a_t : r_1, \ldots, r_t \in R\}.$$

The elements $a_1, \ldots, a_t$ are called *generators* of the ideal.

Definition 3.4.2. An ideal I of R is called *principal* if there exists $a \in R$ such that

$$I = (a) = \{ra : r \in R\}.$$

The element a is called a generator of the principal ideal (a).

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The zero ideal is principal, since $(0) = \{0\}$, and the whole ring is principal, since $(1_R) = R$.

Theorem 3.4.3. *Let $a_1, \ldots, a_t \in R$. Then $(a_1, \ldots, a_t)$ is the smallest ideal of R containing all the elements $a_1, \ldots, a_t$.*

In particular, (a) is the smallest ideal containing a.

Proof. Let

$$I = (a_1, \ldots, a_t) = \left\{ \sum_{i=1}^{t} r_i a_i : r_i \in R \right\}.$$

Clearly $0 \in I$. If $x = \sum r_i a_i$ and $y = \sum s_i a_i$ belong to I, then

$$x - y = \sum_{i=1}^{t} (r_i - s_i) a_i \in I.$$

If $r \in R$, then

$$rx = \sum_{i=1}^{t} (r r_i) a_i \in I.$$

Since R is commutative, the absorption property holds on both sides. Hence I is an ideal.

For each j, taking $r_j = 1_R$ and all other coefficients equal to 0 gives $a_j \in I$.

Now let J be any ideal containing all $a_1, \ldots, a_t$. Since J absorbs multiplication by arbitrary elements of R, it contains every $r_i a_i$, and therefore every finite sum $\sum r_i a_i$.

Thus $I \subseteq J$, proving that I is the smallest ideal containing the specified elements. ■

Proposition 3.4.4. *Let $a, b \in R$. Then*

$$(a) \subseteq (b) \quad \Longleftrightarrow \quad b \mid a.$$

Proof. Suppose $(a) \subseteq (b)$. Since $a \in (a)$, we have $a \in (b)$. Therefore $a = br$ for some $r \in R$, showing that $b \mid a$.

Conversely, suppose $b \mid a$. Then $a = br$ for some $r \in R$. Every element of (a) has the form $sa = sbr$, which belongs to (b). Thus $(a) \subseteq (b)$. ■

Remark 3.4.5. Containment of principal ideals reverses the direction of divisibility. If b divides a, then every multiple of a is a multiple of b, so $(a) \subseteq (b)$.

For example,

$$(12) \subseteq (6) \subseteq (3)$$

in $\mathbb{Z}$, because $3 \mid 6 \mid 12$.

Proposition 3.4.6. *If R is an integral domain and $a, b \in R$ are nonzero, then*

$$(a) = (b) \quad \Longleftrightarrow \quad a = ub$$

for some unit $u \in R$.

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Proof. If $a = ub$ with u a unit, then $(a) \subseteq (b)$. Also $b = u^{-1}a$, so $(b) \subseteq (a)$. Hence $(a) = (b)$.

Conversely, suppose $(a) = (b)$. Then $a = rb$ and $b = sa$ for some $r, s \in R$.

Substituting gives $a = rsa$, so

$$(1_R - rs)a = 0.$$

Since R is an integral domain and $a \neq 0$, we obtain $rs = 1_R$.

Thus r is a unit, and $a = rb$ gives the required relation. ■

3.4.2 Principal Ideal Domains

Definition 3.4.7. An integral domain R is called a *principal ideal domain*, abbreviated *PID*, if every ideal of R is principal.

Thus, in a PID, every ideal can be generated by a single element. The rings $\mathbb{Z}$ and $F[x]$, where F is a field, are the most important elementary examples.

Theorem 3.4.8. *Every ideal I of $\mathbb{Z}$ has the form*

$$I = n\mathbb{Z} = (n)$$

for a unique integer $n \geq 0$. Consequently, $\mathbb{Z}$ is a PID.

Proof. If $I = \{0\}$, then $I = 0\mathbb{Z}$, and the conclusion holds with $n = 0$.

Suppose $I \neq \{0\}$. Since I contains a nonzero integer and is closed under additive inverses, it contains a positive integer.

By the well-ordering principle, there exists a least positive integer $n \in I$.

Since I is an ideal and $n \in I$, every multiple of n belongs to I. Therefore $n\mathbb{Z} \subseteq I$.

Conversely, let $a \in I$. By the division algorithm, there exist integers q, r such that

$$a = qn + r, \qquad 0 \leq r < n.$$

Since $a, n \in I$ and I is an ideal, we have

$$r = a - qn \in I.$$

By the minimality of n, the only possibility is $r = 0$. Hence $a = qn \in n\mathbb{Z}$.

Thus $I \subseteq n\mathbb{Z}$, and therefore $I = n\mathbb{Z}$.

The integer n is unique because, for a nonzero ideal, it is the least positive element of I. ■

Example 3.4.9. The ideals of $\mathbb{Z}$ include

$$(0) = \{0\}, \quad (1) = \mathbb{Z}, \quad (2) = 2\mathbb{Z}, \quad (3) = 3\mathbb{Z},$$

and so on.

Every ideal of $\mathbb{Z}$ is obtained in this manner. In particular, there are no nonprincipal ideals in $\mathbb{Z}$.

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3.4.3 Ideals in Polynomial Rings

The polynomial ring over a field is another important example of a principal ideal domain.

Theorem 3.4.10. *If F is a field, then $F[x]$ is a principal ideal domain.*

Proof. Let I be an ideal of $F[x]$.

If $I = \{0\}$, then $I = (0)$ is principal.

Suppose $I \neq \{0\}$. Choose a nonzero polynomial $f(x) \in I$ of least degree. Such a polynomial exists because the degrees of nonzero polynomials in I form a nonempty set of nonnegative integers.

Since $f(x) \in I$, every multiple of $f(x)$ belongs to I, so $(f) \subseteq I$.

Conversely, let $g(x) \in I$. By the polynomial division algorithm over the field F, there exist $q(x), r(x) \in F[x]$ such that

$$g(x) = q(x)f(x) + r(x),$$

where either $r(x) = 0$ or $\deg r < \deg f$.

Since $g(x) \in I$ and $q(x)f(x) \in I$, their difference $r(x)$ belongs to I.

If $r(x) \neq 0$, its degree would be smaller than that of $f(x)$, contradicting the choice of $f(x)$. Therefore $r(x) = 0$.

Thus $g(x) = q(x)f(x) \in (f)$, proving $I \subseteq (f)$.

Consequently $I = (f)$, so every ideal of $F[x]$ is principal. ■

Example 3.4.11. In $\mathbb{R}[x]$, the set

$$I = \{f(x) \in \mathbb{R}[x] : f(2) = 0\}$$

is the principal ideal $(x - 2)$.

Indeed, the factor theorem states that $f(2) = 0$ if and only if $x - 2$ divides $f(x)$.

3.4.4 Sums and Intersections of Principal Ideals

In a principal ideal domain, sums and intersections of principal ideals have a direct interpretation in terms of greatest common divisors and least common multiples.

Theorem 3.4.12. *Let R be a PID, and let $a, b \in R$ be nonzero. Then*

$$(a) + (b) = (d),$$

where d is a greatest common divisor of a and b.

Proof. Since R is a PID, the ideal $(a) + (b)$ is principal. Write

$$(a) + (b) = (d).$$

Because $a, b \in (d)$, the element d divides both a and b.

Now let c be any common divisor of a and b. Then c divides every linear combination $ra + sb$, where $r, s \in R$.

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Since $d \in (a) + (b)$, there exist $r, s \in R$ such that

$$d = ra + sb.$$

Hence $c \mid d$.

Thus d is a greatest common divisor of a and b. ■

Corollary 3.4.13. *If R is a PID and d is a greatest common divisor of $a, b \in R$, then there exist $r, s \in R$ such that*

$$d = ra + sb.$$

Proof. By the preceding theorem, $(a) + (b) = (d)$. Since $d \in (a) + (b)$, it has the required representation. ■

Theorem 3.4.14. *Let R be a PID, and let $a, b \in R$ be nonzero. Then*

$$(a) \cap (b) = (m),$$

where m is a least common multiple of a and b.

Proof. Since R is a PID, the ideal $(a) \cap (b)$ is principal. Write

$$(a) \cap (b) = (m).$$

Because $m \in (a)$ and $m \in (b)$, both a and b divide m. Thus m is a common multiple.

Now let c be any common multiple of a and b. Then $c \in (a)$ and $c \in (b)$, so $c \in (a) \cap (b) = (m)$.

Therefore $m \mid c$. Hence m is a least common multiple of a and b. ■

Example 3.4.15. Find the sum, intersection, and product of the ideals (18) and (30) in $\mathbb{Z}$.

Since

$$\gcd(18, 30) = 6, \qquad \operatorname{lcm}(18, 30) = 90,$$

we have

$$(18) + (30) = (6)$$

and

$$(18) \cap (30) = (90).$$

The product ideal is

$$(18)(30) = (540).$$

Indeed, every product of a multiple of 18 and a multiple of 30 is divisible by 540, and $540 = 18 \cdot 30$ belongs to the product ideal.

Thus

$$(18) + (30) = (6), \quad (18) \cap (30) = (90), \quad (18)(30) = (540).$$

Proposition 3.4.16. *If R is a commutative ring with identity and $a, b \in R$, then*

$$(a)(b) = (ab).$$

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Proof. Every element of $(a)(b)$ is a finite sum of terms $(ra)(sb)$, where $r, s \in R$. By commutativity,

$$(ra)(sb) = (rs)(ab) \in (ab).$$

Hence $(a)(b) \subseteq (ab)$.

Conversely, ab is the product of $a \in (a)$ and $b \in (b)$, so $ab \in (a)(b)$. Since $(a)(b)$ is an ideal, it contains the smallest ideal generated by ab, namely (ab).

Thus $(ab) \subseteq (a)(b)$, proving equality. ■

3.4.5 Nonprincipal Ideals

Not every integral domain is a principal ideal domain. The following examples illustrate why certain ideals cannot be generated by a single element.

Theorem 3.4.17. *The ideal $(2, x)$ of $\mathbb{Z}[x]$ is not principal. Consequently, $\mathbb{Z}[x]$ is not a PID.*

Proof. Consider

$$I = (2, x) = \{2f(x) + xg(x) : f(x), g(x) \in \mathbb{Z}[x]\}.$$

Suppose, for a contradiction, that $I = (h(x))$ for some $h(x) \in \mathbb{Z}[x]$.

Since $2 \in I$, the polynomial $h(x)$ divides the constant polynomial 2. Since $\mathbb{Z}[x]$ is an integral domain, degree considerations show that $h(x)$ must be a nonzero constant polynomial.

Moreover, the constant integer $h(x)$ must divide 2 in $\mathbb{Z}$. Hence

$$h(x) \in \{1, -1, 2, -2\}.$$

If $h(x) = \pm 1$, then $I = \mathbb{Z}[x]$. This is impossible because every polynomial in I has an even constant term, so $1 \notin I$.

If $h(x) = \pm 2$, then $I = (2)$. This is also impossible because $x \in I$ but $x \notin (2)$.

Both possibilities lead to contradictions. Therefore I is not principal.

Hence $\mathbb{Z}[x]$ is not a principal ideal domain. ■

Example 3.4.18.] Let F be a field and consider the ideal

$$I = (x, y)$$

of $F[x, y]$.

Suppose $I = (f)$ for some polynomial $f \in F[x, y]$. Since both x and y belong to I, the polynomial f must divide both x and y.

But x and y have no common nonunit polynomial factor. Consequently, f must be a unit, which would imply $I = F[x, y]$.

This is impossible because every polynomial in (x, y) has zero constant term, whereas the constant polynomial 1 does not.

Therefore (x, y) is not principal, and hence $F[x, y]$ is not a PID.

3.5 Quotient Rings and Ring Homomorphisms

In group theory, normal subgroups allow us to construct quotient groups. A similar construction exists in ring theory, where ideals play the role of normal subgroups.

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Every ideal of a ring is an additive subgroup, so its additive cosets partition the ring. The additional absorption property ensures that these cosets can be multiplied consistently. The resulting algebraic structure is called a quotient ring.

Ring homomorphisms provide another fundamental tool. They describe mappings that preserve both addition and multiplication and establish important connections between ideals and quotient rings.

3.5.1 Construction of Quotient Rings

Let R be a ring and let I be a two-sided ideal of R. Since $(I, +)$ is a subgroup of the abelian group $(R, +)$, we may consider the additive cosets

$$a + I = \{a + i : i \in I\}, \qquad a \in R.$$

Two elements $a, b \in R$ determine the same coset precisely when $a - b \in I$. Thus

$$a + I = b + I \quad \Longleftrightarrow \quad a - b \in I.$$

The collection of all these cosets is denoted by

$$R/I = \{a + I : a \in R\}.$$

We define addition and multiplication of cosets by

$$(a + I) + (b + I) = (a + b) + I$$

and

$$(a + I)(b + I) = ab + I.$$

Before treating these operations as ring operations, it is necessary to establish that their values do not depend on the representatives chosen for the cosets.

Theorem 3.5.1. *Let I be a two-sided ideal of a ring R. Then R/I is a ring under the operations*

$$(a + I) + (b + I) = (a + b) + I$$

and

$$(a + I)(b + I) = ab + I.$$

Its additive identity is $I = 0 + I$, and the additive inverse of $a + I$ is $-a + I$.

If R is commutative, then R/I is commutative. If R has a multiplicative identity 1_R, then $1_R + I$ is a multiplicative identity of R/I.

Proof. We first verify that the operations are well-defined.

Suppose

$$a + I = a' + I, \qquad b + I = b' + I.$$

Then $a' - a \in I$ and $b' - b \in I$. Write $a' = a + i$ and $b' = b + j$, where $i, j \in I$.

For addition,

$$(a' + b') - (a + b) = i + j \in I.$$

Hence $(a' + b') + I = (a + b) + I$, proving that addition is independent of representatives.

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For multiplication, we have

$$\begin{aligned} a'b' - ab &= (a+i)(b+j) - ab \\ &= aj + ib + ij. \end{aligned}$$

Since I is a two-sided ideal, all three terms aj, ib, and ij belong to I. Thus $a'b' - ab \in I$, so

$$a'b' + I = ab + I.$$

Therefore multiplication is also well-defined.

The additive group axioms follow from those of $(R, +)$. In particular, addition is associative and commutative, I is the additive identity, and

$$(a + I) + (-a + I) = I.$$

For multiplication, associativity follows from

$$\begin{aligned} ((a+I)(b+I))(c+I) &= ((ab)c) + I \\ &= (a(bc)) + I \\ &= (a+I)((b+I)(c+I)). \end{aligned}$$

The distributive laws follow from the corresponding laws in R. For example,

$$\begin{aligned} (a+I)((b+I)+(c+I)) &= a(b+c) + I \\ &= (ab+ac) + I \\ &= (ab+I) + (ac+I). \end{aligned}$$

The right distributive law is verified similarly.

Therefore R/I is a ring.

If R is commutative, then

$$(a+I)(b+I) = ab + I = ba + I = (b+I)(a+I).$$

Finally, if R has identity 1_R, then

$$(1_R + I)(a+I) = a + I = (a+I)(1_R + I),$$

so $1_R + I$ is the multiplicative identity of R/I. ■

Remark 3.5.2. The requirement that I be a two-sided ideal is essential for multiplication of cosets to be well-defined. An arbitrary additive subgroup need not satisfy the absorption property.

Notice also that if R has identity and $I = R$, then R/I is the zero ring, whose multiplicative identity coincides with its additive identity.

If I is proper, then $1_R + I \neq I$, so the quotient has distinct additive and multiplicative identities.

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Example 3.5.3. Let $R = \mathbb{Z}$ and $I = n\mathbb{Z}$, where $n \geq 2$. Then

$$\mathbb{Z}/n\mathbb{Z} = \{0 + n\mathbb{Z}, 1 + n\mathbb{Z}, \ldots, (n-1) + n\mathbb{Z}\}.$$

The quotient operations are

$$(a + n\mathbb{Z}) + (b + n\mathbb{Z}) = (a + b) + n\mathbb{Z}$$

and

$$(a + n\mathbb{Z})(b + n\mathbb{Z}) = ab + n\mathbb{Z}.$$

Thus $\mathbb{Z}/n\mathbb{Z}$ is precisely the familiar ring $\mathbb{Z}_n$ of integers modulo n.
For example, in $\mathbb{Z}/7\mathbb{Z}$,

$$(5 + 7\mathbb{Z}) + (6 + 7\mathbb{Z}) = 4 + 7\mathbb{Z}$$

and

$$(5 + 7\mathbb{Z})(6 + 7\mathbb{Z}) = 2 + 7\mathbb{Z}.$$

Example 3.5.4. Consider the ideal (x) of $\mathbb{Z}[x]$.
Two polynomials belong to the same coset modulo (x) if and only if they have the same constant term. Indeed, $f(x) - g(x) \in (x)$ precisely when $f(0) = g(0)$.
Thus every coset contains a unique constant polynomial, and the quotient $\mathbb{Z}[x]/(x)$ has the same ring structure as $\mathbb{Z}$.
We shall establish this formally using a homomorphism below.

3.5.2 Ring Homomorphisms

A ring homomorphism is a mapping that preserves both of the operations defining a ring.

Definition 3.5.5. Let R and S be rings. A mapping $\varphi : R \to S$ is called a *ring homomorphism* if, for all $a, b \in R$,

$$\varphi(a + b) = \varphi(a) + \varphi(b)$$

and

$$\varphi(ab) = \varphi(a)\varphi(b).$$

If R and S have multiplicative identities, the homomorphism is called *unital* if

$$\varphi(1_R) = 1_S.$$

A ring homomorphism need not be injective or surjective. A bijective ring homomorphism is called a *ring isomorphism*. If an isomorphism exists, we write $R \cong S$.

Throughout this section, homomorphisms between rings with identity are not automatically assumed to preserve the identity; this additional condition will be stated whenever needed.

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Example 3.5.6. Define $\varphi : \mathbb{Z} \to \mathbb{Z}_n$ by $\varphi(a) = [a]$.

For $a, b \in \mathbb{Z}$,

$$\varphi(a+b) = [a+b] = [a] + [b]$$

and

$$\varphi(ab) = [ab] = [a][b].$$

Hence φ is a ring homomorphism. It is unital and surjective.

Example 3.5.7. Let F be a field and let $a \in F$. Define $\mathrm{ev}_a : F[x] \to F$ by

$$\mathrm{ev}_a(f(x)) = f(a).$$

For any $f, g \in F[x]$,

$$(f+g)(a) = f(a) + g(a)$$

and

$$(fg)(a) = f(a)g(a).$$

Therefore ev_a is a ring homomorphism.

It is surjective because every element of F is the image of the corresponding constant polynomial. Moreover, $\mathrm{ev}_a(1) = 1$, so it is unital.

Example 3.5.8. Define $f : \mathbb{Z} \to \mathbb{Z}$ by $f(a) = a + 1$.

Then

$$f(a+b) = a + b + 1,$$

whereas

$$f(a) + f(b) = a + b + 2.$$

Thus f does not preserve addition and is not a ring homomorphism.

3.5.3 Elementary Properties of Ring Homomorphisms

The defining conditions of a ring homomorphism imply several useful identities.

Proposition 3.5.9. *Let $\varphi : R \to S$ be a ring homomorphism. Then, for every $a, b \in R$ and every positive integer n,*

(i) $\varphi(0_R) = 0_S$;

(ii) $\varphi(-a) = -\varphi(a)$;

(iii) $\varphi(a-b) = \varphi(a) - \varphi(b)$;

(iv) $\varphi(a^n) = \varphi(a)^n$.

Proof. Since $0_R + 0_R = 0_R$,

$$\varphi(0_R) = \varphi(0_R) + \varphi(0_R).$$

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Additive cancellation in S gives $\varphi(0_R) = 0_S$.

Next,

$$\varphi(a) + \varphi(-a) = \varphi(a - a) = \varphi(0_R) = 0_S.$$

Therefore $\varphi(-a) = -\varphi(a)$, proving (ii).

Using (ii),

$$\varphi(a - b) = \varphi(a + (-b)) = \varphi(a) - \varphi(b),$$

which proves (iii).

Finally, repeated application of multiplicativity gives

$$\varphi(a^n) = \underbrace{\varphi(a) \cdots \varphi(a)}_{n \text{ factors}} = \varphi(a)^n.$$

■

Remark 3.5.10. If φ is unital, the power formula also holds for $n = 0$. Without the unital condition, one cannot generally assert that $\varphi(1_R) = 1_S$.

3.5.4 Kernel and Image

As in group theory, the kernel and image of a homomorphism provide fundamental information about the mapping.

Definition 3.5.11. Let $\varphi : R \to S$ be a ring homomorphism. Its *kernel* and *image* are defined by

$$\ker \varphi = \{r \in R : \varphi(r) = 0_S\}$$

and

$$\operatorname{Im} \varphi = \{\varphi(r) : r \in R\},$$

respectively.

The kernel consists of the elements mapped to zero, whereas the image consists of all elements actually attained by the homomorphism.

Theorem 3.5.12. *Let $\varphi : R \to S$ be a ring homomorphism. Then*

(i) $\ker \varphi$ *is a two-sided ideal of* R*;*

(ii) $\operatorname{Im} \varphi$ *is a subring of* S*.*

Proof. Put $K = \ker \varphi$.

Since $\varphi(0_R) = 0_S$, we have $0_R \in K$.

If $a, b \in K$, then

$$\varphi(a - b) = \varphi(a) - \varphi(b) = 0_S,$$

so $a - b \in K$.

Let $r \in R$ and $a \in K$. Then

$$\varphi(ra) = \varphi(r)\varphi(a) = 0_S$$

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and

$$\varphi(ar) = \varphi(a)\varphi(r) = 0_S.$$

Hence $ra, ar \in K$, proving that K is a two-sided ideal of R.

Now let $T = \operatorname{Im}\varphi$. Since $0_S = \varphi(0_R)$, the set T is nonempty.

If $x, y \in T$, write $x = \varphi(a)$ and $y = \varphi(b)$. Then

$$x - y = \varphi(a - b) \in T$$

and

$$xy = \varphi(ab) \in T.$$

By the subring test, T is a subring of S. ■

Theorem 3.5.13. *A ring homomorphism $\varphi : R \to S$ is injective if and only if*

$$\ker\varphi = \{0_R\}.$$

Proof. Suppose φ is injective. If $a \in \ker\varphi$, then

$$\varphi(a) = 0_S = \varphi(0_R).$$

Injectivity gives $a = 0_R$. Thus $\ker\varphi = \{0_R\}$.

Conversely, suppose $\ker\varphi = \{0_R\}$. If $\varphi(a) = \varphi(b)$, then

$$\varphi(a - b) = 0_S.$$

Hence $a - b \in \ker\varphi$, so $a - b = 0_R$ and therefore $a = b$.

Thus φ is injective. ■

Example 3.5.14. Consider $\varphi : \mathbb{Z} \to \mathbb{Z}_n$, defined by $\varphi(a) = [a]$.

An integer a belongs to the kernel precisely when $[a] = [0]$, that is, when $n \mid a$. Therefore

$$\ker\varphi = n\mathbb{Z}.$$

Every residue class has an integer representative, so $\operatorname{Im}\varphi = \mathbb{Z}_n$. Thus the mapping is surjective but not injective.

Example 3.5.15. Let $\mathrm{ev}_a : F[x] \to F$ be evaluation at $a \in F$.

Then

$$\ker(\mathrm{ev}_a) = \{f(x) \in F[x] : f(a) = 0\}.$$

By the factor theorem, $f(a) = 0$ if and only if $(x - a) \mid f(x)$. Consequently,

$$\ker(\mathrm{ev}_a) = (x - a).$$

3.5.5 The Natural Projection

Every quotient ring comes equipped with a particularly important homomorphism.

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Theorem 3.5.16. *Let I be a two-sided ideal of a ring R. The mapping $\pi : R \to R/I$, defined by*

$$\pi(r) = r + I,$$

is a surjective ring homomorphism with

$$\ker \pi = I.$$

If R has identity, then π is unital.

Proof. For $a, b \in R$,

$$\pi(a+b) = (a+b) + I = (a+I) + (b+I)$$

and

$$\pi(ab) = ab + I = (a+I)(b+I).$$

Thus π is a ring homomorphism.

It is surjective because every element of R/I has the form $r + I = \pi(r)$ for some $r \in R$.

Furthermore,

$$\ker \pi = \{r \in R : r + I = I\} = I.$$

If R has identity, then $\pi(1_R) = 1_R + I$, the identity of R/I. Hence π is unital. ■

Corollary 3.5.17. *A subset I of a ring R is a two-sided ideal if and only if it is the kernel of some ring homomorphism defined on R.*

Proof. Every kernel is a two-sided ideal by the kernel and image theorem.

Conversely, if I is a two-sided ideal, the natural projection $\pi : R \to R/I$ is a ring homomorphism with kernel I. ■

3.5.6 The First Isomorphism Theorem

The First Isomorphism Theorem explains the precise relationship between the kernel and image of a ring homomorphism.

Theorem 3.5.18 (First Isomorphism Theorem for Rings)**.** *Let $\varphi : R \to S$ be a ring homomorphism. Then*

$$R/\ker \varphi \cong \operatorname{Im} \varphi.$$

Proof. Put $K = \ker \varphi$. Since K is a two-sided ideal of R, the quotient ring R/K exists.

Define

$$\overline{\varphi} : R/K \longrightarrow \operatorname{Im} \varphi$$

by

$$\overline{\varphi}(r + K) = \varphi(r).$$

We first show that $\overline{\varphi}$ is well-defined. Suppose $r + K = s + K$. Then $r - s \in K$, and hence

$$\varphi(r) - \varphi(s) = \varphi(r - s) = 0_S.$$

Therefore $\varphi(r) = \varphi(s)$.

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For $r, s \in R$,

$$\begin{aligned}\overline{\varphi}((r+K)+(s+K)) &= \overline{\varphi}((r+s)+K)\\ &= \varphi(r+s)\\ &= \varphi(r)+\varphi(s),\end{aligned}$$

and

$$\begin{aligned}\overline{\varphi}((r+K)(s+K)) &= \overline{\varphi}(rs+K)\\ &= \varphi(rs)\\ &= \varphi(r)\varphi(s).\end{aligned}$$

Thus $\overline{\varphi}$ preserves addition and multiplication.

It is surjective onto $\operatorname{Im}\varphi$ by definition.

Finally, suppose $\overline{\varphi}(r+K) = 0_S$. Then $\varphi(r) = 0_S$, so $r \in K$. Consequently $r + K = K$, the zero element of R/K.

Thus the kernel of $\overline{\varphi}$ is the zero ideal, so $\overline{\varphi}$ is injective.

Therefore $\overline{\varphi}$ is a ring isomorphism, and

$$R/\ker\varphi \cong \operatorname{Im}\varphi.$$

■

Example 3.5.19.] Let F be a field and let $a \in F$. The evaluation homomorphism $\mathrm{ev}_a : F[x] \to F$ is surjective, and

$$\ker(\mathrm{ev}_a) = (x-a).$$

By the First Isomorphism Theorem,

$$F[x]/(x-a) \cong F.$$

In particular, taking $a = 0$ gives

$$F[x]/(x) \cong F.$$

Example 3.5.20.] Define $\varphi : \mathbb{Z}[x] \to \mathbb{Z}$ by $\varphi(f(x)) = f(0)$.

This is a surjective ring homomorphism. Its kernel consists precisely of polynomials with zero constant term, so

$$\ker\varphi = (x).$$

Hence

$$\mathbb{Z}[x]/(x) \cong \mathbb{Z}.$$

Example 3.5.21. Define $\varphi : \mathbb{Z}[x] \to \mathbb{Z}_2$ by

$$\varphi(f(x)) = [f(0)].$$

The mapping evaluates at 0 and reduces the result modulo 2, so it is a surjective unital ring homomorphism.

Its kernel consists of polynomials whose constant terms are even. Every such polynomial can be written as

$$f(x) = 2m + xg(x)$$

for some $m \in \mathbb{Z}$ and $g(x) \in \mathbb{Z}[x]$. Therefore

$$\ker \varphi = (2, x).$$

By the First Isomorphism Theorem,

$$\mathbb{Z}[x]/(2, x) \cong \mathbb{Z}_2.$$

3.5.7 The Second Isomorphism Theorem

The Second Isomorphism Theorem relates a subring, an ideal, and their sum.

Theorem 3.5.22 (Second Isomorphism Theorem for Rings). *Let S be a subring of a ring R, and let I be a two-sided ideal of R. Then $S + I$ is a subring of R, $S \cap I$ is a two-sided ideal of S, and I is a two-sided ideal of $S + I$.*

Moreover,

$$S/(S \cap I) \cong (S + I)/I.$$

Proof. We first show that $S + I$ is a subring of R.

Let $s_1 + i_1, s_2 + i_2 \in S + I$. Their difference is

$$(s_1 + i_1) - (s_2 + i_2) = (s_1 - s_2) + (i_1 - i_2) \in S + I.$$

Also,

$$(s_1 + i_1)(s_2 + i_2) = s_1 s_2 + s_1 i_2 + i_1 s_2 + i_1 i_2.$$

Since $s_1 s_2 \in S$ and the remaining terms belong to I, the product lies in $S + I$.

Thus $S + I$ is a subring.

The intersection $S \cap I$ is an additive subgroup of S. If $s \in S$ and $a \in S \cap I$, then sa, as belong to both S and I. Hence $S \cap I$ is an ideal of S.

Since $I \subseteq S + I$ and I is an ideal of R, it is also an ideal of $S + I$.

Now define

$$\varphi : S \longrightarrow (S + I)/I$$

by $\varphi(s) = s + I$.

This is a ring homomorphism. Every coset of I in $S + I$ has the form

$$(s + i) + I = s + I$$

for some $s \in S$ and $i \in I$. Hence φ is surjective.

Moreover,

$$\begin{aligned} \ker \varphi &= \{s \in S : s + I = I\} \\ &= \{s \in S : s \in I\} \\ &= S \cap I. \end{aligned}$$

By the First Isomorphism Theorem,

$$S/(S \cap I) \cong (S + I)/I.$$

■

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Example 3.5.23. Let $R = \mathbb{Z}$, $S = 4\mathbb{Z}$, and $I = 6\mathbb{Z}$.

Then

$$S + I = 4\mathbb{Z} + 6\mathbb{Z} = 2\mathbb{Z}$$

and

$$S \cap I = 4\mathbb{Z} \cap 6\mathbb{Z} = 12\mathbb{Z}.$$

The Second Isomorphism Theorem gives

$$4\mathbb{Z}/12\mathbb{Z} \cong 2\mathbb{Z}/6\mathbb{Z}.$$

Both quotient rings have three elements. The isomorphism sends

$$4k + 12\mathbb{Z} \longmapsto 4k + 6\mathbb{Z}.$$

3.5.8 The Third Isomorphism Theorem

Successive quotient constructions can often be combined into a single quotient.

Theorem 3.5.24 (Third Isomorphism Theorem for Rings)**.** *Let I and J be two-sided ideals of a ring R such that $I \subseteq J$. Then J/I is a two-sided ideal of R/I, and*

$$(R/I)/(J/I) \cong R/J.$$

Proof. Define

$$\psi : R/I \longrightarrow R/J$$

by

$$\psi(r + I) = r + J.$$

To show that ψ is well-defined, suppose $r + I = s + I$. Then $r - s \in I \subseteq J$, so $r + J = s + J$.

For $r, s \in R$,

$$\psi((r + s) + I) = (r + s) + J = (r + J) + (s + J)$$

and

$$\psi(rs + I) = rs + J = (r + J)(s + J).$$

Thus ψ is a ring homomorphism.

It is surjective because every coset $r + J$ is the image of $r + I$.

Its kernel is

$$\begin{aligned} \ker \psi &= \{r + I : r + J = J\} \\ &= \{r + I : r \in J\} \\ &= J/I. \end{aligned}$$

Therefore J/I is an ideal of R/I. By the First Isomorphism Theorem,

$$(R/I)/(J/I) \cong R/J.$$

■

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Example 3.5.25. Consider the ideals

$$12\mathbb{Z} \subseteq 4\mathbb{Z} \subseteq \mathbb{Z}.$$

By the Third Isomorphism Theorem,

$$(\mathbb{Z}/12\mathbb{Z})/(4\mathbb{Z}/12\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}.$$

Thus quotienting first by $12\mathbb{Z}$ and then by $4\mathbb{Z}/12\mathbb{Z}$ has the same effect, up to isomorphism, as quotienting directly by $4\mathbb{Z}$.

3.5.9 Correspondence of Ideals

The following theorem describes all ideals of a quotient ring in terms of ideals of the original ring.

Theorem 3.5.26 (Correspondence Theorem for Ideals)**.** *Let I be a two-sided ideal of a ring R. There is a one-to-one correspondence between the two-sided ideals of R/I and the two-sided ideals J of R satisfying $I \subseteq J$.*

The correspondence is given by

$$J \longmapsto J/I,$$

with inverse

$$L \longmapsto \pi^{-1}(L),$$

where $\pi : R \to R/I$ is the natural projection.

Proof. Let J be an ideal of R containing I. Then J/I is an ideal of R/I by the Third Isomorphism Theorem.

Conversely, let L be an ideal of R/I. The inverse image

$$\pi^{-1}(L) = \{r \in R : r + I \in L\}$$

is an ideal of R. Moreover, since $\pi(i) = I$ for every $i \in I$ and every ideal contains zero, we have $I \subseteq \pi^{-1}(L)$.

These constructions are inverse to one another.

Indeed, if J contains I, then

$$\pi^{-1}(J/I) = J.$$

Also, since π is surjective, for every ideal L of R/I,

$$\pi(\pi^{-1}(L)) = L.$$

Hence the stated correspondence is bijective. ∎

3.6 Prime Ideals

We now study a distinguished class of ideals in commutative ring theory. Prime ideals extend the fundamental divisibility property of prime numbers and are closely connected with integral domains.

Throughout this section, R denotes a commutative ring with multiplicative identity $1_R \neq 0$.

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3.6.1 Definition and First Examples

Recall that a prime integer p has the property that if p divides a product ab, then p divides at least one of the factors.

Prime ideals generalize this property by replacing divisibility by membership in an ideal.

Definition 3.6.1. A proper ideal P of R is called a *prime ideal* if, for all $a, b \in R$,

$$ab \in P \quad \Longrightarrow \quad a \in P \text{ or } b \in P.$$

The requirement that P be proper is essential. The whole ring cannot be a prime ideal.

Example 3.6.2. In $\mathbb{Z}$, the ideal (5) is prime. Indeed, if $ab \in (5)$, then $5 \mid ab$. Since 5 is a prime integer, Euclid's lemma gives $5 \mid a$ or $5 \mid b$. Hence $a \in (5)$ or $b \in (5)$.

In contrast, (6) is not prime because

$$2 \cdot 3 = 6 \in (6),$$

although neither 2 nor 3 belongs to (6).

Example 3.6.3. The zero ideal (0) of $\mathbb{Z}$ is prime. Indeed, if $ab \in (0)$, then $ab = 0$, which implies $a = 0$ or $b = 0$.

However, the zero ideal of $\mathbb{Z}_6$ is not prime, since $[2][3] = [0]$ while both factors are nonzero.

3.6.2 Prime Ideals and Integral Domains

The most important characterization of prime ideals is expressed in terms of quotient rings.

Theorem 3.6.4. *Let P be an ideal of a commutative ring R with identity $1_R \neq 0$. Then*

$$P \text{ is prime} \quad \Longleftrightarrow \quad R/P \text{ is an integral domain.}$$

Proof. Suppose first that P is a prime ideal. Since P is proper, $1_R \notin P$, and hence

$$1_R + P \neq P.$$

Therefore R/P is a nonzero commutative ring with identity.

Suppose

$$(a + P)(b + P) = P.$$

By the definition of quotient multiplication, this means $ab + P = P$, or equivalently, $ab \in P$.

Since P is prime, $a \in P$ or $b \in P$. Hence

$$a + P = P \quad \text{or} \quad b + P = P.$$

Thus R/P has no zero divisors and is an integral domain.

Conversely, suppose R/P is an integral domain. Since an integral domain is nonzero, $P \neq R$, so P is proper.

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Let $a, b \in R$ satisfy $ab \in P$. Then

$$(a + P)(b + P) = ab + P = P.$$

Since R/P is an integral domain, at least one of the factors must be zero. Hence

$$a + P = P \quad \text{or} \quad b + P = P.$$

Equivalently, $a \in P$ or $b \in P$.

Therefore P is a prime ideal. ■

Corollary 3.6.5. *The zero ideal (0) of R is prime if and only if R is an integral domain.*

Proof. Since $R/(0) \cong R$, the result follows immediately from the quotient characterization. ■

Example 3.6.6.] Consider the ideal (x) in $\mathbb{Z}[x]$.

By evaluation at zero,

$$\mathbb{Z}[x]/(x) \cong \mathbb{Z}.$$

Since $\mathbb{Z}$ is an integral domain, (x) is a prime ideal of $\mathbb{Z}[x]$.

Example 3.6.7.] Reducing the coefficients of a polynomial modulo 2 gives a surjective ring homomorphism

$$\varphi : \mathbb{Z}[x] \longrightarrow \mathbb{F}_2[x].$$

Its kernel is (2), the set of polynomials with even integer coefficients.

By the First Isomorphism Theorem,

$$\mathbb{Z}[x]/(2) \cong \mathbb{F}_2[x].$$

Since $\mathbb{F}_2[x]$ is an integral domain, the ideal (2) is prime in $\mathbb{Z}[x]$.

Example 3.6.8. Consider (x^2) in $F[x]$, where F is a field.

We have

$$x \cdot x = x^2 \in (x^2),$$

but $x \notin (x^2)$.

Therefore (x^2) is not prime. Equivalently, the nonzero coset $x + (x^2)$ has square zero in $F[x]/(x^2)$, so the quotient is not an integral domain.

3.6.3 The Ideal-Product Characterization

Prime ideals can also be characterized using products of ideals instead of products of individual elements.

Theorem 3.6.9. *Let P be a proper ideal of a commutative ring R with identity. Then P is prime if*

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and only if, for every pair of ideals I, J of R,

$$IJ \subseteq P \quad \Longrightarrow \quad I \subseteq P \text{ or } J \subseteq P.$$

Proof. Suppose P is prime. Let I, J be ideals of R such that $IJ \subseteq P$.
Assume that neither I nor J is contained in P. Then there exist

$$a \in I \setminus P, \qquad b \in J \setminus P.$$

Since $a \in I$ and $b \in J$, their product belongs to IJ, so

$$ab \in IJ \subseteq P.$$

Primality of P implies $a \in P$ or $b \in P$, contradicting the choice of a and b.
Hence $I \subseteq P$ or $J \subseteq P$.
Conversely, suppose the ideal-product condition holds.
Let $a, b \in R$ satisfy $ab \in P$. Consider the principal ideals (a) and (b).
Since R is commutative,

$$(a)(b) = (ab) \subseteq P.$$

By assumption,

$$(a) \subseteq P \quad \text{or} \quad (b) \subseteq P.$$

Since $a \in (a)$ and $b \in (b)$, we obtain $a \in P$ or $b \in P$.
Thus P is prime. ■

Corollary 3.6.10. *Let P be a prime ideal of R. If $I_1, \ldots, I_n$ are ideals satisfying*

$$I_1 I_2 \cdots I_n \subseteq P,$$

then $I_j \subseteq P$ for at least one $j \in \{1, \ldots, n\}$.

Proof. The result follows by induction on n. For $n = 2$, it is the preceding theorem.
For $n > 2$, apply the ideal-product criterion to $(I_1 \cdots I_{n-1})I_n \subseteq P$. Either $I_n \subseteq P$ or $I_1 \cdots I_{n-1} \subseteq P$. In the latter case, apply the induction hypothesis. ■

3.6.4 Prime Ideals of the Integers

The prime ideals of $\mathbb{Z}$ can be classified completely.

Theorem 3.6.11. *The prime ideals of $\mathbb{Z}$ are precisely*

$$(0) \quad \text{and} \quad (p),$$

where p is a positive prime integer.

Proof. Since $\mathbb{Z}$ is a principal ideal domain, every ideal has the form (n) for a unique integer $n \geq 0$.
The zero ideal (0) is prime because $\mathbb{Z}$ is an integral domain.
The ideal $(1) = \mathbb{Z}$ is not prime because it is not proper.

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Now suppose $n \geq 2$. By the quotient characterization, (n) is prime if and only if $\mathbb{Z}/(n)$ is an integral domain.

But

$$\mathbb{Z}/(n) \cong \mathbb{Z}_n,$$

and $\mathbb{Z}_n$ is an integral domain precisely when n is prime.

Therefore the nonzero prime ideals of $\mathbb{Z}$ are exactly the ideals (p), where p is prime. ■

Example 3.6.12. The ideals (2), (3), (5), and (7) are prime ideals of $\mathbb{Z}$.

The ideals (4), (6), and (9) are not prime.

For example, $2 \cdot 2 = 4 \in (4)$, while $2 \notin (4)$.

3.6.5 Prime Ideals in Polynomial Rings

Polynomial rings provide further important examples of prime ideals.

Theorem 3.6.13. *Let F be a field and let $f(x) \in F[x]$ be a nonzero, nonunit polynomial. Then $(f(x))$ is a prime ideal of $F[x]$ if and only if $f(x)$ is irreducible over F.*

Proof. Suppose $f(x)$ is irreducible. Since $F[x]$ is a principal ideal domain, irreducible elements are prime elements.

Hence, whenever $f(x)$ divides $g(x)h(x)$, it divides $g(x)$ or $h(x)$.

Now suppose

$$g(x)h(x) \in (f(x)).$$

Then $f(x)$ divides $g(x)h(x)$, so $f(x)$ divides $g(x)$ or $h(x)$. Equivalently,

$$g(x) \in (f(x)) \quad \text{or} \quad h(x) \in (f(x)).$$

Since $f(x)$ is a nonunit, $(f(x))$ is proper, and therefore prime.

Conversely, suppose $(f(x))$ is prime. If $f(x)$ were reducible, we could write

$$f(x) = g(x)h(x),$$

where both $g(x)$ and $h(x)$ have positive degree strictly less than $\deg f$.

Then $g(x)h(x) \in (f(x))$, so primality gives $g(x) \in (f(x))$ or $h(x) \in (f(x))$.

But either inclusion would imply that $f(x)$ divides a polynomial of smaller positive degree, which is impossible.

Therefore $f(x)$ is irreducible. ■

Example 3.6.14. In $\mathbb{R}[x]$, the polynomial $x^2 + 1$ is irreducible because it has no real root and has degree two.

Therefore the ideal $(x^2 + 1)$ is prime in $\mathbb{R}[x]$.

In contrast, $x^2 - 1$ is reducible because

$$x^2 - 1 = (x - 1)(x + 1).$$

Hence $(x^2 - 1)$ is not a prime ideal.

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Remark 3.6.15. The assumption that f be a nonzero nonunit is necessary. For example, the zero ideal of $F[x]$ is prime because $F[x]$ is an integral domain, even though the zero polynomial is not irreducible.

3.6.6 Preimages of Prime Ideals

Prime ideals behave particularly well under inverse images of ring homomorphisms.

Theorem 3.6.16. *Let $\varphi : R \to S$ be a unital ring homomorphism between commutative rings with identities $1_R \neq 0$ and $1_S \neq 0$.*
If Q is a prime ideal of S, then

$$\varphi^{-1}(Q) = \{r \in R : \varphi(r) \in Q\}$$

is a prime ideal of R.

Proof. Put $P = \varphi^{-1}(Q)$.
Since $\varphi(0_R) = 0_S \in Q$, we have $0_R \in P$.
If $a, b \in P$, then $\varphi(a), \varphi(b) \in Q$. Since Q is an ideal,

$$\varphi(a - b) = \varphi(a) - \varphi(b) \in Q.$$

Thus $a - b \in P$.
If $r \in R$ and $a \in P$, then

$$\varphi(ra) = \varphi(r)\varphi(a) \in Q,$$

so $ra \in P$. Therefore P is an ideal.
Since φ is unital,

$$\varphi(1_R) = 1_S.$$

Because Q is proper, $1_S \notin Q$. Hence $1_R \notin P$, proving that P is proper.
Finally, suppose $ab \in P$. Then

$$\varphi(a)\varphi(b) = \varphi(ab) \in Q.$$

Since Q is prime, $\varphi(a) \in Q$ or $\varphi(b) \in Q$. Therefore $a \in P$ or $b \in P$.
Thus P is a prime ideal of R. ■

Example 3.6.17. Let $\varphi : \mathbb{Z}[x] \to \mathbb{Z}$ be evaluation at zero, defined by $\varphi(f(x)) = f(0)$.
The ideal (2) is prime in $\mathbb{Z}$. Its inverse image is

$$\varphi^{-1}((2)) = (2, x),$$

because a polynomial has even constant term precisely when it belongs to $(2, x)$.
Therefore $(2, x)$ is a prime ideal of $\mathbb{Z}[x]$.
In fact,

$$\mathbb{Z}[x]/(2, x) \cong \mathbb{Z}_2,$$

so the quotient characterization also establishes its primality.

Remark 3.6.18. The inverse image of a prime ideal under a unital homomorphism is always prime. By contrast, the image of a prime ideal under an arbitrary ring homomorphism need not be a prime ideal of the codomain.

For example, the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ maps the prime ideal (2) of $\mathbb{Z}$ to $2\mathbb{Z}$, which is not even an ideal of $\mathbb{Q}$.

3.6.7 Prime Ideals and Maximal Ideals

The theory of prime ideals is closely connected with that of maximal ideals, which will be studied in the next section.

The essential distinction is reflected in their quotient rings:

$$P \text{ prime} \iff R/P \text{ is an integral domain},$$
$$M \text{ maximal} \iff R/M \text{ is a field}.$$

Since every field is an integral domain, every maximal ideal is prime. The converse, however, does not hold in general.

For instance, (x) is prime in $\mathbb{Z}[x]$ because $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, but it is not maximal because $\mathbb{Z}$ is not a field.

These observations will be established formally in the next section.

3.7 Maximal Ideals

Prime ideals are characterized by the property that their quotient rings are integral domains. We now study a stronger class of ideals, called *maximal ideals*, whose quotient rings are fields. These ideals occupy an important position in commutative algebra because they describe the largest proper ideals of a ring and provide a systematic way of constructing fields from more general rings.

Throughout this section, R denotes a commutative ring with identity $1_R \neq 0$.

Definition 3.7.1. A proper ideal M of R is called a *maximal ideal* if there is no ideal J of R satisfying $M \subsetneq J \subsetneq R$.

Thus, a maximal ideal is a proper ideal that cannot be enlarged without obtaining the entire ring. Equivalently, M is maximal if whenever J is an ideal satisfying $M \subseteq J \subseteq R$, then $J = M$ or $J = R$.

Example 3.7.2. The ideal (5) is maximal in $\mathbb{Z}$. Indeed, if $(5) \subseteq (n) \subseteq \mathbb{Z}$, then $n \mid 5$. Since 5 is prime, $n = \pm 1$ or $n = \pm 5$. Hence $(n) = \mathbb{Z}$ or $(n) = (5)$.

On the other hand, (6) is not maximal because $(6) \subsetneq (2) \subsetneq \mathbb{Z}$.

3.7.1 Characterization of Maximal Ideals

The following theorem provides the fundamental characterization of maximal ideals and is one of the most useful results in elementary commutative algebra.

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Theorem 3.7.3. *Let M be an ideal of R. Then M is maximal if and only if R/M is a field.*

Proof. Suppose first that M is maximal. Since M is proper, $1_R \notin M$, and hence R/M is a nonzero commutative ring with identity $1_R + M$.

Let $a + M$ be a nonzero element of R/M. Then $a \notin M$. Consider the ideal $M + (a)$. Since $a \in M + (a)$ but $a \notin M$, we have $M \subsetneq M + (a)$. By maximality, $M + (a) = R$.

In particular, $1_R \in M + (a)$, so there exist $m \in M$ and $r \in R$ such that

$$1_R = m + ra.$$

Passing to the quotient gives

$$1_R + M = (m + M) + (ra + M) = (r + M)(a + M).$$

Thus $r + M$ is a multiplicative inverse of $a + M$. Since every nonzero element of R/M is invertible, R/M is a field.

Conversely, suppose R/M is a field. Since a field is nonzero, M is a proper ideal of R.

Let J be an ideal satisfying $M \subseteq J \subseteq R$. By the ideal correspondence theorem, J/M is an ideal of R/M. Since the only ideals of a field are the zero ideal and the whole field, either $J/M = \{M\}$ or $J/M = R/M$.

In the first case, $J = M$, while in the second case, $J = R$. Therefore there is no ideal properly between M and R, proving that M is maximal. ■

Corollary 3.7.4. *Every maximal ideal of a commutative ring with identity is prime.*

Proof. Let M be a maximal ideal of R. By the preceding theorem, R/M is a field. Every field is an integral domain, so R/M is an integral domain. By the quotient characterization of prime ideals, M is prime. ■

Proposition 3.7.5. *A proper ideal M of R is maximal if and only if, for every $a \in R \setminus M$, there exist $m \in M$ and $r \in R$ such that $1_R = m + ra$.*

Proof. Suppose M is maximal and let $a \notin M$. Then $M + (a)$ is an ideal properly containing M, so $M + (a) = R$. Consequently, $1_R = m + ra$ for some $m \in M$ and $r \in R$.

Conversely, suppose the stated condition holds. Let J be an ideal satisfying $M \subsetneq J \subseteq R$. Choose $a \in J \setminus M$. By assumption, $1_R = m + ra$ for some $m \in M$ and $r \in R$. Since $m, a \in J$ and J is an ideal, $1_R \in J$. Hence $J = R$, proving that M is maximal. ■

3.7.2 Maximal Ideals of the Integers

The maximal ideals of $\mathbb{Z}$ admit a complete classification.

Theorem 3.7.6. *The maximal ideals of $\mathbb{Z}$ are precisely the ideals (p), where p is a positive prime integer.*

Proof. Every ideal of $\mathbb{Z}$ has the form (n) for a unique integer $n \geq 0$.

The ideal $(1) = \mathbb{Z}$ is not proper and hence is not maximal. The zero ideal (0) is not maximal because $\mathbb{Z}/(0) \cong \mathbb{Z}$, which is not a field.

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For $n \geq 2$, the quotient ring $\mathbb{Z}/(n)$ is isomorphic to $\mathbb{Z}_n$. By the quotient characterization, (n) is maximal if and only if $\mathbb{Z}_n$ is a field.

We have already established that $\mathbb{Z}_n$ is a field precisely when n is prime. Therefore the maximal ideals of $\mathbb{Z}$ are exactly the ideals (p), where p is a positive prime integer. ■

Example 3.7.7. The ideals (2), (3), (5) and (7) are maximal in $\mathbb{Z}$, whereas (4), (6) and (9) are not.

For example, (4) is properly contained in (2), and (2) is properly contained in $\mathbb{Z}$. Therefore (4) cannot be maximal.

3.7.3 Maximal Ideals in Polynomial Rings

The relationship between maximal ideals and irreducible polynomials is especially important in polynomial rings over fields.

Theorem 3.7.8. *] Let F be a field and let $f(x) \in F[x]$ be a nonzero nonunit polynomial. Then the following conditions are equivalent:*

(i) $f(x)$ is irreducible over F;

(ii) $(f(x))$ is a maximal ideal of $F[x]$;

(iii) $(f(x))$ is a prime ideal of $F[x]$.

Proof. Suppose first that $f(x)$ is irreducible. Let J be an ideal of $F[x]$ satisfying $(f) \subseteq J \subseteq F[x]$.

Since $F[x]$ is a principal ideal domain, there exists $g(x) \in F[x]$ such that $J = (g)$. The inclusion $(f) \subseteq (g)$ implies that g divides f.

Because f is irreducible, either g is a unit or g is an associate of f. In the first case, $J = F[x]$, while in the second case, $J = (f)$. Therefore (f) is maximal.

Every maximal ideal is prime, so (ii) implies (iii).

Finally, suppose (f) is prime and write $f(x) = g(x)h(x)$ for some $g, h \in F[x]$. Since $gh = f \in (f)$, primality gives $g \in (f)$ or $h \in (f)$.

If $g \in (f)$, then $g = fu$ for some $u \in F[x]$. Substituting into $f = gh$ gives $f = fuh$. Since $F[x]$ is an integral domain and $f \neq 0$, cancellation yields $uh = 1$. Hence h is a unit.

Similarly, if $h \in (f)$, then g is a unit. Thus every factorization of f has a unit factor, proving that f is irreducible. ■

Corollary 3.7.9. *Every nonzero prime ideal of $F[x]$ is maximal.*

Proof. Let P be a nonzero prime ideal of $F[x]$. Since $F[x]$ is a PID, $P = (f)$ for some nonzero polynomial f. Since P is proper, f is not a unit. By the preceding theorem, (f) is maximal. ■

Remark 3.7.10. The zero ideal of $F[x]$ is prime because $F[x]$ is an integral domain, but it is not maximal because $F[x]$ is not a field. Thus the nonzero hypothesis in the preceding corollary cannot be omitted.

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Example 3.7.11.] Consider the ideal $I = (x^2 + 1)$ of $\mathbb{R}[x]$. Since $x^2 + 1$ has no real root and has degree two, it is irreducible over $\mathbb{R}$. Therefore I is maximal and hence prime.

In fact, the quotient is isomorphic to the complex field:

$$\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}.$$

To see this, define $\varphi : \mathbb{R}[x] \to \mathbb{C}$ by $\varphi(f(x)) = f(i)$. Every complex number $a + bi$ is the image of $a + bx$, so φ is surjective. By polynomial division, $\ker \varphi = (x^2 + 1)$. The First Isomorphism Theorem gives the result.

Example 3.7.12. Consider $(x^2 - 1)$ in $\mathbb{R}[x]$. Since $x^2 - 1 = (x - 1)(x + 1)$, the polynomial is reducible. Therefore $(x^2 - 1)$ is neither prime nor maximal.

Indeed, neither $x - 1$ nor $x + 1$ belongs to $(x^2 - 1)$, although their product does.

3.7.4 Existence of Maximal Ideals

The preceding results identify maximal ideals in familiar rings. We now state a general existence theorem.

Theorem 3.7.13 (Existence of Maximal Ideals)**.** *Let R be a nonzero commutative ring with identity. Then every proper ideal of R is contained in a maximal ideal. In particular, every such ring possesses at least one maximal ideal.*

Proof. Let I be a proper ideal of R, and consider the collection

$$\mathcal{S} = \{J : J \text{ is an ideal of } R,\ I \subseteq J \subsetneq R\},$$

ordered by inclusion.

The set $\mathcal{S}$ is nonempty because $I \in \mathcal{S}$.

Let $\mathcal{C}$ be a nonempty chain in $\mathcal{S}$ and put $K = \bigcup_{J \in \mathcal{C}} J$. Since the ideals in $\mathcal{C}$ are linearly ordered by inclusion, any two elements of K belong to a common ideal in the chain. Consequently, K is closed under subtraction and absorbs multiplication by elements of R, so K is an ideal.

Moreover, $I \subseteq K$. Since every $J \in \mathcal{C}$ is proper, no such J contains 1_R. Hence $1_R \notin K$, and therefore K is proper. Thus $K \in \mathcal{S}$ and is an upper bound for $\mathcal{C}$.

The empty chain also has an upper bound, namely I. Therefore every chain in $\mathcal{S}$ has an upper bound in $\mathcal{S}$. By Zorn's lemma, $\mathcal{S}$ contains a maximal element M.

If $M \subsetneq J \subsetneq R$ for some ideal J, then $J \in \mathcal{S}$, contradicting the maximality of M in $\mathcal{S}$. Hence M is a maximal ideal of R containing I.

Taking $I = (0)$ proves that every nonzero commutative ring with identity has a maximal ideal. ∎

Remark 3.7.14. The preceding theorem uses Zorn's lemma, which is equivalent to the axiom of choice. It is more advanced than the elementary results of this chapter but is included to emphasize that maximal ideals exist in complete generality for nonzero commutative rings with identity.

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3.7.5 Maximal Ideals and Quotient Rings

Recall that if I is an ideal of R, the ideal correspondence theorem gives a bijection between ideals of R/I and ideals of R containing I.

This correspondence also preserves primality and maximality.

Proposition 3.7.15. *Let $I \subseteq J$ be ideals of R. Then:*

(i) J is prime in R if and only if J/I is prime in R/I;

(ii) J is maximal in R if and only if J/I is maximal in R/I.

Proof. By the Third Isomorphism Theorem,

$$(R/I)/(J/I) \cong R/J.$$

Consequently, $(R/I)/(J/I)$ is an integral domain if and only if R/J is an integral domain. By the quotient characterization of prime ideals, this is equivalent to J/I being prime in R/I if and only if J is prime in R.

Similarly, $(R/I)/(J/I)$ is a field if and only if R/J is a field. By the quotient characterization of maximal ideals, this is equivalent to J/I being maximal in R/I if and only if J is maximal in R. ■

Example 3.7.16. We determine all prime and maximal ideals of $\mathbb{Z}_{12}$.

Since $\mathbb{Z}_{12} \cong \mathbb{Z}/12\mathbb{Z}$, the ideal correspondence theorem identifies its prime ideals with prime ideals of $\mathbb{Z}$ containing (12).

The prime ideals of $\mathbb{Z}$ are (0) and (p), where p is prime. The containment $(12) \subseteq (p)$ holds precisely when $p \mid 12$. Hence the only possibilities are $p = 2$ and $p = 3$.

Therefore the prime ideals of $\mathbb{Z}_{12}$ are

$$([2]) = \{[0], [2], [4], [6], [8], [10]\}$$

and

$$([3]) = \{[0], [3], [6], [9]\}.$$

Their quotient rings are

$$\mathbb{Z}_{12}/([2]) \cong \mathbb{Z}_2, \qquad \mathbb{Z}_{12}/([3]) \cong \mathbb{Z}_3.$$

Since both quotient rings are fields, these ideals are maximal as well. Thus $\mathbb{Z}_{12}$ has exactly two prime ideals, and both are maximal.

Example 3.7.17. Let F be a field and let $a, b \in F$. We show that the ideal $I = (x - a, y - b)$ is maximal in $F[x, y]$.

Define $\varphi : F[x, y] \to F$ by $\varphi(f(x, y)) = f(a, b)$. This is a surjective ring homomorphism because every element of F is the image of a constant polynomial.

Clearly $x - a$ and $y - b$ belong to $\ker \varphi$, so $I \subseteq \ker \varphi$.

Conversely, suppose $f(a, b) = 0$. Then

$$f(x, y) = \big(f(x, y) - f(a, y)\big) + \big(f(a, y) - f(a, b)\big).$$

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Regarding $f(x,y)$ as a polynomial in x with coefficients in $F[y]$, the first difference is divisible by $x-a$. Similarly, the second difference is divisible by $y-b$. Hence $f(x,y) \in (x-a, y-b)$.

Thus $\ker \varphi = I$, and the First Isomorphism Theorem gives

$$F[x,y]/(x-a, y-b) \cong F.$$

Since the quotient is a field, $(x-a, y-b)$ is maximal.

3.8 Boolean Rings

We conclude our discussion of rings with a class of rings in which every element is idempotent. These rings arise naturally in set theory and mathematical logic.

Definition 3.8.1. A ring R is called a *Boolean ring* if

$$a^2 = a$$

for every $a \in R$.

The definition does not require the ring to be commutative or to possess a multiplicative identity. Nevertheless, the Boolean property forces multiplication to be commutative and every element to be its own additive inverse.

Theorem 3.8.2. *Let R be a Boolean ring. Then:*

(i) $a + a = 0$ *for every* $a \in R$;

(ii) $ab = ba$ *for all* $a, b \in R$.

Consequently, every Boolean ring is commutative, and every nonzero Boolean ring has characteristic 2.

Proof. Let $a \in R$. Since R is Boolean, $a^2 = a$ and $(-a)^2 = -a$. On the other hand, the ring identities give $(-a)^2 = a^2 = a$. Therefore $-a = a$, and hence

$$a + a = 0.$$

This proves (i).

Now let $a, b \in R$. By the Boolean property,

$$(a+b)^2 = a + b.$$

Expanding the left side gives

$$a^2 + ab + ba + b^2 = a + b.$$

Since $a^2 = a$ and $b^2 = b$, additive cancellation yields

$$ab + ba = 0.$$

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By (i), every element is its own additive inverse, so $ba = -ba$. Hence $ab = ba$, proving (ii).

Thus R is commutative. If R is nonzero, the equality $a + a = 0$ for every $a \in R$ shows that its additive group has exponent 2 and its characteristic is 2. ■

Example 3.8.3. The ring $\mathbb{Z}_2 = \{[0], [1]\}$ is Boolean, because

$$[0]^2 = [0], \qquad [1]^2 = [1].$$

Example 3.8.4. Let X be a set and let $\mathcal{P}(X)$ denote its power set. Define addition by symmetric difference and multiplication by intersection:

$$A + B = A \triangle B, \qquad AB = A \cap B.$$

Then $\mathcal{P}(X)$ is a commutative ring. Its additive identity is the empty set $\varnothing$, and every element is its own additive inverse because $A \triangle A = \varnothing$. Multiplication is associative and distributive over symmetric difference.

Moreover, $A^2 = A \cap A = A$ for every $A \subseteq X$. Hence $\mathcal{P}(X)$ is a Boolean ring. If X is nonempty, its multiplicative identity is X.

Proposition 3.8.5. *A Boolean ring has no nonzero nilpotent elements.*

Proof. Let R be a Boolean ring and suppose $a^n = 0$ for some positive integer n. Since $a^2 = a$, induction gives $a^n = a$ for every positive integer n. Therefore $a = 0$. ■

Proposition 3.8.6. *Every Boolean integral domain is isomorphic to $\mathbb{Z}_2$.*

Proof. Let R be a Boolean integral domain. For every $a \in R$, the Boolean property gives $a^2 = a$, so

$$a(a - 1_R) = 0.$$

Since R is an integral domain, either $a = 0$ or $a = 1_R$.

Thus R consists of exactly two elements, 0 and 1_R. The map $\mathbb{Z}_2 \to R$ defined by $[0] \mapsto 0$ and $[1] \mapsto 1_R$ is therefore a ring isomorphism. ■

Corollary 3.8.7. *Every field that is a Boolean ring is isomorphic to $\mathbb{Z}_2$.*

Proof. Every field is an integral domain, so the result follows immediately from the preceding proposition. ■

3.9 Summary and Guided Review

The principal concepts developed in this chapter are rings, integral domains, fields, subrings, ideals, quotient rings, ring homomorphisms, and principal, prime, and maximal ideals. The following table summarizes the most important characterizations for commutative rings with identity $1_R \neq 0$.

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Concept	Characterization
Integral domain	A nonzero commutative ring with identity and no nonzero zero divisors.
Field	A nonzero commutative ring with identity in which every nonzero element is invertible.
Principal ideal (a)	$(a) = \{ra : r \in R\}$, the smallest ideal containing a.
Principal ideal domain	An integral domain in which every ideal is principal.
Prime ideal P	$P \neq R$ and $ab \in P$ implies $a \in P$ or $b \in P$; equivalently, R/P is an integral domain.
Maximal ideal M	$M \neq R$ and there is no ideal strictly between M and R; equivalently, R/M is a field.
Boolean ring	A ring satisfying $a^2 = a$ for every element a.

The relationship between prime and maximal ideals is particularly important. Every maximal ideal is prime, but a prime ideal need not be maximal. In $\mathbb{Z}$ and $F[x]$, every nonzero prime ideal is maximal, whereas in $\mathbb{Z}[x]$ this need not be the case.

When examining a given ideal, it is usually advantageous to identify the corresponding quotient ring. For example, the primality of (x) in $\mathbb{Z}[x]$ follows from $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, while the maximality of (x^2+1) in $\mathbb{R}[x]$ follows from $\mathbb{R}[x]/(x^2+1) \cong \mathbb{C}$. These examples illustrate how quotient rings transform questions about ideals into questions about integral domains and fields.

3.10 Exercises

Basic concepts and computations.

1. Verify directly that $3\mathbb{Z}$ is a subring and an ideal of $\mathbb{Z}$.
2. Find all units and all nonzero zero divisors in $\mathbb{Z}_8$ and $\mathbb{Z}_{15}$.
3. Prove that a unit cannot be a zero divisor in a nonzero commutative ring.
4. Show that an integral domain has no nonzero nilpotent elements.
5. Determine $(24) + (36)$, $(24) \cap (36)$ and $(24)(36)$ in $\mathbb{Z}$.
6. Prove that $(a) = (b)$ in an integral domain if and only if $a = ub$ for some unit u (allow $a = b = 0$).
7. Show that $(x^2 - 4, x - 2) = (x - 2)$ in $\mathbb{Q}[x]$.
8. Use the division algorithm to prove $(x^2 + 1, x + 1) = (1)$ in $\mathbb{Q}[x]$.
9. Describe the elements of $\mathbb{Z}[x]/(2, x)$ and identify this quotient ring.
10. Decide whether (4) is a prime ideal of $\mathbb{Z}$; justify with a product outside the ideal.
11. Decide whether (0) is prime and/or maximal in $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{Z}_6$.
12. List all ideals of $\mathbb{Z}_{18}$. Which are prime? Which are maximal?

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Proof-based exercises.

13. Prove that the intersection of an arbitrary family of ideals is an ideal.
14. Prove that the sum of two ideals is the smallest ideal containing both.
15. Prove that $IJ \subseteq I \cap J$, and give an example where the inclusion is strict.
16. Let P be prime. Prove that if $a_1 \cdots a_n \in P$, then some $a_i \in P$.
17. Prove that every prime ideal of a finite commutative ring with identity is maximal.
18. Show that the union of two ideals is an ideal if and only if one is contained in the other.
19. Show that (x) is prime but not maximal in $\mathbb{Z}[x]$.
20. Prove that $(x^2 + 2)$ is maximal in $\mathbb{Q}[x]$.
21. Find the kernel and image of $\mathbb{Z}[x] \to \mathbb{Z}_5$, $f \mapsto [f(2)]_5$.
22. Show that $\mathbb{Z}[x]/(x^2 + 1)$ is not a field, although $\mathbb{R}[x]/(x^2 + 1)$ is.
23. Prove that if $I + J = R$, then $I \cap J = IJ$ for commutative R.
24. For $R = \mathbb{Z} \times \mathbb{Z}$, prove that $\mathbb{Z} \times \{0\}$ is prime but not maximal.
25. Explain why the statement “the preimage of every maximal ideal is maximal” requires extra hypotheses, and find a counterexample using an inclusion of rings.

Chapter 4

Tutorial: Structures, Examples and Applications

The purpose of this chapter is to consolidate the theory of groups and rings through carefully chosen computations, counterexamples, and proofs. A good example in abstract algebra does more than illustrate a definition: it shows which hypotheses matter, suggests a theorem, and explains why apparently similar structures behave differently. The first four sections correspond to the prescribed tutorial component; the final section develops a few natural extensions for interested students and instructors. Unless stated otherwise, rings are associative and need not be commutative; when the existence of a multiplicative identity is needed, it is explicitly assumed. We write $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ and $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$.

4.1 Semigroups, Monoids and Groups

Recognizing algebraic structures

A *binary operation* on a nonempty set S is a mapping $S \times S \to S$. Closure is therefore built into the meaning of a binary operation. The operation is *associative* if $(ab)c = a(bc)$ for all $a, b, c \in S$ and *commutative* if $ab = ba$ for all $a, b \in S$. These conditions are logically independent.

Definition 4.1.1. A *semigroup* is a nonempty set with an associative binary operation. A *monoid* is a semigroup with an identity element e satisfying $ea = ae = a$ for every a. A *group* is a monoid in which each element a has a two-sided inverse a^{-1}.

The identity, when it exists, is unique: if e and f are identities, then $e = ef = f$. Likewise inverses in a monoid are unique whenever they exist. Indeed, if $ba = ab = e$ and $ca = ac = e$, then $b = b(ac) = (ba)c = c$. A monoid may have some invertible elements without being a group.

Example 4.1.2 (A progression of familiar structures). (i) $(\mathbb{Z}_{>0}, +)$ is a commutative semigroup but not a monoid: the additive identity 0 is absent.

(ii) $(\mathbb{N}_0, +)$ is a commutative monoid but not a group: the element 1 has no additive inverse in $\mathbb{N}_0$.

(iii) $(\mathbb{Z}, \cdot)$ is a commutative monoid but not a group: its only invertible elements are 1 and -1.

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(iv) $(\mathbb{Z}, +)$ is an abelian group, while $(\mathbb{Q}^\times, \cdot)$ is an abelian group, where $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$.

(v) The real matrices $M_n(\mathbb{R})$ form a monoid under multiplication, with identity I_n. The invertible matrices form the group $GL_n(\mathbb{R})$.

(vi) If Σ is an alphabet, the nonempty words Σ^+ form a semigroup under concatenation. Including the empty word ε gives the monoid Σ^*. For an alphabet with at least two letters, concatenation is noncommutative.

Example 4.1.3 (A semigroup of transformations)**.** Let $X = \{1, 2, 3\}$ and let $T(X)$ be the set of all functions $X \to X$, with composition as multiplication. Because function composition is associative and the identity map belongs to $T(X)$, it is a monoid. Its elements need not be invertible: the constant function $c(x) = 1$ has no inverse. Its units are precisely the bijections, forming the symmetric group S_3. Thus the familiar group of permutations occurs naturally inside a much larger monoid.

Example 4.1.4 (An unusual operation)**.** On $\mathbb{Z}$, define $a \star b = a + b + 2$. Then $(a \star b) \star c = a + b + c + 4 = a \star (b \star c)$, and $a \star (-2) = (-2) \star a = a$. The inverse of a relative to $\star$ is $-a - 4$. Thus $(\mathbb{Z}, \star)$ is an abelian group whose identity is -2, not 0. The map $f(a) = a + 2$ is an isomorphism from $(\mathbb{Z}, \star)$ to $(\mathbb{Z}, +)$, since $f(a \star b) = f(a) + f(b)$. This example shows that the identity depends on the operation rather than on the underlying set.

Example 4.1.5 (Associativity cannot be assumed)**.** Subtraction is closed on $\mathbb{Z}$, but $(5 - 3) - 1 = 1$, whereas $5 - (3 - 1) = 3$. Hence $(\mathbb{Z}, -)$ is not a semigroup. Division also fails to define a binary operation on $\mathbb{Z}$ because quotients of integers need not be integers.

Proposition 4.1.6. *Let M be a monoid. The set $U(M)$ of all invertible elements of M is a group under the operation inherited from M.*

Proof. The identity belongs to $U(M)$. If $a, b \in U(M)$, then ab is invertible, with $(ab)^{-1} = b^{-1}a^{-1}$. Associativity is inherited from M, and the inverse of an invertible element remains invertible. Hence $U(M)$ is a group. ∎

Example 4.1.7 (Units modulo an integer)**.** An element $[a] \in \mathbb{Z}_n$ is invertible under multiplication precisely when $\gcd(a, n) = 1$. For example, $U(10) = \{[1], [3], [7], [9]\}$, and $[3]^{-1} = [7]$ because $3 \cdot 7 \equiv 1 \pmod{10}$. In contrast, $[2]$ is not a unit in $\mathbb{Z}_{10}$: an even number cannot be congruent to 1 modulo 10.

Proposition 4.1.8. *If G is a group, then $ab = ac$ implies $b = c$, and $ba = ca$ implies $b = c$.*

Proof. Multiply $ab = ac$ on the left by a^{-1}, and use associativity. The second assertion follows by multiplying on the right by a^{-1}. ∎

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Cancellation may fail in monoids: in $(\mathbb{Z}, \cdot)$ one has $0 \cdot 2 = 0 \cdot 3$ although $2 \neq 3$. Conversely, a finite cancellative monoid is a group.

Theorem 4.1.9. *A finite monoid in which both left and right cancellation hold is a group. In fact, left cancellation alone suffices for a finite monoid.*

Proof. Fix $a \in M$. Left cancellation says that the map $L_a(x) = ax$ is injective. Because M is finite, it is surjective. Thus there is $b \in M$ such that $ab = e$. Now $a(ba) = (ab)a = a = ae$, and left cancellation yields $ba = e$. Thus every a is invertible. ■

Commutativity and the role of counterexamples

A group is *abelian* if $ab = ba$ for all a, b. The groups $(\mathbb{Z}, +)$, $(\mathbb{Z}_n, +)$, $(\mathbb{R}^\times, \cdot)$, and $\mathbb{Z}_2 \times \mathbb{Z}_2$ are abelian. A cyclic group must be abelian, but the converse fails, as the Klein four-group demonstrates.

Example 4.1.10 (Two nonabelian groups)**.** In S_3, with permutations composed from right to left, $(12)(13) = (132)$ while $(13)(12) = (123)$. Thus S_3 is nonabelian. Similarly, take

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Both lie in $GL_2(\mathbb{R})$, but $AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ and $BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$. Therefore $GL_2(\mathbb{R})$ is nonabelian.

Example 4.1.11 (The quaternion group)**.** The group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ has the rules $i^2 = j^2 = k^2 = -1$, $ij = k$, and $ji = -k$. Since $ij \neq ji$, it is nonabelian. Nevertheless, all six elements of order 4 have the same square, namely -1. The orders of individual elements alone do not generally determine the isomorphism type of a group.

Proposition 4.1.12. *A group G is abelian if and only if $(ab)^{-1} = a^{-1}b^{-1}$ for all $a, b \in G$.*

Proof. In every group $(ab)^{-1} = b^{-1}a^{-1}$. Thus the stated identity is equivalent to $a^{-1}b^{-1} = b^{-1}a^{-1}$ for all a, b. Since inverses range over all elements of G, this is equivalent to commutativity. ■

Proposition 4.1.13. *If $(ab)^2 = a^2b^2$ for all a, b in a group G, then G is abelian.*

Proof. The assumption gives $abab = aabb$. Left cancellation by a yields $bab = abb$, and right cancellation by b gives $ba = ab$. ■

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Practice and discussion

1. Classify $(\mathbb{N}_0, \max)$ and $(\mathbb{N}_0, \min)$ as semigroups or monoids. Determine their identities when they exist.

2. On $\mathbb{R}$, consider $a \star b = a + b + ab$. Prove associativity, determine the identity, and find exactly which elements are invertible.

3. Construct a semigroup that is not commutative and has no identity.

4. Determine the units of $(\mathbb{Z}_{12}, \cdot)$ and their multiplicative orders.

5. Prove that a group cannot have two distinct identity elements or two distinct inverses of the same element.

6. Show that $U(M_n(\mathbb{R})) = GL_n(\mathbb{R})$.

7. Explain why left cancellation in an infinite monoid does not necessarily imply that every element is invertible; use $(\mathbb{N}_0, +)$.

8. Verify directly that the product of two invertible elements of any monoid is invertible, and give its inverse.

4.2 Cyclic Groups, Generators and Permutations

Cyclic and noncyclic groups

Definition 4.2.1. A group G is *cyclic* if $G = \langle a \rangle = \{a^k : k \in \mathbb{Z}\}$ for some $a \in G$. Such an element is a *generator*. The *order* $\operatorname{ord}(a)$ is the least positive m satisfying $a^m = e$, if such an integer exists, and is infinite otherwise.

The group $(\mathbb{Z}, +)$ is generated by 1 and by -1; $(\mathbb{Z}_n, +)$ is generated by $[1]$; and the group of complex nth roots of unity is generated by $e^{2\pi i/n}$. Noncyclic groups include $\mathbb{Z}_2 \times \mathbb{Z}_2$, S_3, and $(\mathbb{Q}, +)$. In the Klein group, every nonidentity element has order 2, so no element can generate the entire group of order 4.

Theorem 4.2.2. *If a has finite order n, then for every integer k,*

$$\operatorname{ord}(a^k) = \frac{n}{\gcd(n, k)}.$$

In particular, a^k generates $\langle a \rangle$ if and only if $\gcd(n, k) = 1$.

Proof. Put $d = \gcd(n, k)$ and write $n = dn_0$, $k = dk_0$, where $\gcd(n_0, k_0) = 1$. Now $(a^k)^m = e$ exactly when $n \mid km$, equivalently $n_0 \mid k_0 m$, equivalently $n_0 \mid m$. Its least positive such exponent is n_0. ∎

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Example 4.2.3 (Finding generators and subgroup orders)**.** In $(\mathbb{Z}_{12}, +)$ the generators are $[1], [5], [7], [11]$, since these are exactly the residues coprime to 12. The order of $[8]$ is $12/\gcd(12, 8) = 3$, and $\langle [8] \rangle = \{[0], [8], [4]\}$. In $(\mathbb{Z}_{18}, +)$, the element $[12]$ has order 3 and generates $\{[0], [12], [6]\}$.

Theorem 4.2.4. *Every subgroup of a cyclic group is cyclic. If G has order n, then for each positive divisor d of n there is exactly one subgroup of order d, namely $\langle a^{n/d} \rangle$ when $G = \langle a \rangle$.*

Proof. Let $H \leq \langle a \rangle$. If $H = \{e\}$ it is cyclic. Otherwise, choose the least positive m such that $a^m \in H$. For $a^t \in H$, divide $t = qm + r$ with $0 \leq r < m$. Since $a^r = a^t (a^m)^{-q} \in H$, minimality forces $r = 0$. Hence $H = \langle a^m \rangle$. In the finite case the unique subgroup of order d consists of the powers of $a^{n/d}$, by the order-of-a-power formula. ■

Example 4.2.5 (Complete subgroup description)**.** The subgroups of $(\mathbb{Z}_{12}, +)$ correspond to the divisors $1, 2, 3, 4, 6, 12$. They are $\langle [0] \rangle$, $\langle [6] \rangle$, $\langle [4] \rangle$, $\langle [3] \rangle$, $\langle [2] \rangle$, and $\langle [1] \rangle$, with orders $1, 2, 3, 4, 6, 12$, respectively. This also explains why a cyclic group has precisely as many subgroups as its order has positive divisors.

Theorem 4.2.6. *Every infinite cyclic group is isomorphic to $(\mathbb{Z}, +)$, and every cyclic group of order n is isomorphic to $(\mathbb{Z}_n, +)$.*

Proof. If $G = \langle a \rangle$, the map $k \mapsto a^k$ from $(\mathbb{Z}, +)$ to G is a surjective homomorphism. Its kernel is $\{0\}$ when a has infinite order and $n\mathbb{Z}$ when a has order n. The first is an isomorphism; the second induces $[k] \mapsto a^k$ from $\mathbb{Z}_n$ to G. ■

Example 4.2.7 (Computing generators efficiently)**.** The group $\mathbb{Z}_{30}$ has $\varphi(30) = 30(1 - 1/2)(1 - 1/3)(1 - 1/5) = 8$ generators: $[1], [7], [11], [13], [17], [19], [23], [29]$. The totient formula is efficient when n has several prime factors; manually listing all powers of every candidate is unnecessary.

Proposition 4.2.8. *For positive integers m, n, the group $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic if and only if $\gcd(m, n) = 1$.*

Proof. The order of (x, y) equals $\operatorname{lcm}(\operatorname{ord}(x), \operatorname{ord}(y))$, which divides $\operatorname{lcm}(m, n)$. If $\gcd(m, n) > 1$, then $\operatorname{lcm}(m, n) < mn$, so there is no element of order mn. If $\gcd(m, n) = 1$, the element $([1], [1])$ has order $\operatorname{lcm}(m, n) = mn$ and generates the group. ■

Example 4.2.9. The group $\mathbb{Z}_3 \times \mathbb{Z}_4$ is cyclic of order 12, whereas $\mathbb{Z}_4 \times \mathbb{Z}_6$ is not cyclic. The latter has order 24, but every element has order at most $\operatorname{lcm}(4, 6) = 12$. Notice that equal group orders do not imply isomorphism: $\mathbb{Z}_4$ and $\mathbb{Z}_2 \times \mathbb{Z}_2$ are different groups of order 4.

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Permutation calculations and parity

A permutation is a bijection of a finite set onto itself. We use right-to-left composition, so $\sigma\tau$ means apply τ first and then σ. Disjoint cycles commute, and an r-cycle has order r.

Example 4.2.10 (Converting to disjoint cycles)**.** Consider the permutation of $\{1,2,3,4,5,6\}$ given by

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 1 & 6 & 2 & 4 \end{pmatrix}.$$

Tracing the images gives $1 \mapsto 3 \mapsto 1$, $2 \mapsto 5 \mapsto 2$, and $4 \mapsto 6 \mapsto 4$. Hence $\sigma = (13)(25)(46)$, and $\operatorname{ord}(\sigma) = 2$. It is odd, since its disjoint-cycle decomposition contains three transpositions.

Example 4.2.11 (Multiplying permutations)**.** Let $\sigma = (123)$ and $\tau = (12)$ in S_3. Then $\sigma\tau = (13)$ while $\tau\sigma = (23)$. For instance, under $\sigma\tau$ we have $1 \mapsto 2 \mapsto 3$, $3 \mapsto 3 \mapsto 1$, and $2 \mapsto 1 \mapsto 2$. Working symbol by symbol removes ambiguities about the convention for composition.

Theorem 4.2.12. *If σ is the product of disjoint cycles of lengths $r_1, \dots, r_t$, then* $\operatorname{ord}(\sigma) = \operatorname{lcm}(r_1, \dots, r_t)$.

Proof. Disjoint cycles commute. The power σ^m is the identity exactly when each cycle raised to m is the identity, which happens exactly when every r_i divides m. The least positive such integer is their least common multiple. ∎

Example 4.2.13. For $\sigma = (1234)(567)(89)$, the order is $\operatorname{lcm}(4,3,2) = 12$. Since a cycle of length r is a product of $r-1$ transpositions, its parity is $(-1)^{r-1}$. Thus the sign of σ is $(-1)^{3+2+1} = 1$, so σ is even.

Definition 4.2.14. A *transposition* interchanges two symbols. A permutation is *even* if it is a product of an even number of transpositions, and *odd* otherwise. The sign function is $\operatorname{sgn} : S_n \to \{1, -1\}$, with value 1 on even permutations and -1 on odd permutations.

Theorem 4.2.15. *The parity of the number of transpositions in a permutation is independent of the chosen decomposition. Moreover,*

$$\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau).$$

Proof. For $\sigma \in S_n$, let $\operatorname{inv}(\sigma)$ be the number of pairs $i < j$ with $\sigma(i) > \sigma(j)$. Interchanging two adjacent entries changes this number by exactly one. Every permutation can be sorted to the identity by adjacent interchanges, and an arbitrary transposition can be written as an odd number of adjacent interchanges. Therefore every transposition reverses $(-1)^{\operatorname{inv}(\sigma)}$. Since the identity has sign 1, any expression of σ as a product of t transpositions satisfies

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$(-1)^t = (-1)^{\mathrm{inv}(\sigma)}$, proving well-defined parity. Concatenating expressions for σ and τ gives the multiplication rule. ■

Example 4.2.16 (Two ways to compute sign)**.** For $\sigma = (123)(45)(6789)$, the disjoint cycles have lengths $3, 2, 4$ and use $2 + 1 + 3 = 6$ transpositions; hence σ is even. In two-line notation, one may instead count inversions. For example, $(3, 1, 4, 2)$ has three inversions, namely $(3, 1), (3, 2), (4, 2)$, and therefore represents an odd permutation of four symbols.

Proposition 4.2.17. *For $n \geq 2$, exactly half the permutations of S_n are even. Thus the alternating group A_n has order $n!/2$.*

Proof. Fix the transposition $t = (12)$. The map $\sigma \mapsto t\sigma$ pairs every even permutation with a unique odd permutation and vice versa, because multiplication by t reverses parity. Since S_n has $n!$ elements, each class has $n!/2$ elements. ■

Example 4.2.18 (A compound generator problem)**.** Find all generators of the subgroup $H = \langle [6] \rangle$ of $(\mathbb{Z}_{30}, +)$. Since $\mathrm{ord}([6]) = 30/\gcd(30, 6) = 5$, the generators are $[6]^k$ in additive notation, that is, $[6k]$ with $\gcd(k, 5) = 1$. They are $[6], [12], [18], [24]$. The point is to compute the order of the subgroup first; there is no need to test all thirty residue classes.

1. Determine the generators of $\mathbb{Z}_{16}$ and $\mathbb{Z}_{20}$, and verify the totals using Euler's totient function.

2. Find all elements of order 6 in $\mathbb{Z}_{18}$.

3. List every subgroup of $\mathbb{Z}_{24}$ and give a generator for each.

4. Show that every quotient of a cyclic group is cyclic.

5. Give two nonisomorphic groups of order 6 and explain how commutativity distinguishes them.

6. Find the order and parity of $(12345)(678)(9\,10)$.

7. Express $(1\,4\,3\,2)$ as a product of transpositions and determine its sign.

8. Compute $(123)(24)$ and $(24)(123)$ using right-to-left composition.

9. Prove that a cycle of even length is an odd permutation.

10. Is $\mathbb{Z}_8 \times \mathbb{Z}_{15}$ cyclic? Is $\mathbb{Z}_6 \times \mathbb{Z}_{10}$ cyclic? Justify your answers.

11. Prove that if a group has precisely one element of order 2, then that element lies in its centre.

12. Explain why $(\mathbb{Q}, +)$ is not cyclic.

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4.3 Rings, Subrings and One-Sided Ideals

Commutative and noncommutative rings

A ring R has an abelian group structure under addition and an associative multiplication satisfying both distributive laws. In a *commutative ring*, $ab = ba$ for all elements. The familiar rings $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}_n$, and polynomial rings in commuting indeterminates are commutative. Matrix rings $M_n(F)$ for $n \geq 2$ are generally noncommutative.

Example 4.3.1 (Noncommutativity in a matrix ring)**.** Let E_{ij} be the matrix with entry 1 in position (i, j) and zeros elsewhere. In $M_2(\mathbb{R})$, one has $E_{12}E_{21} = E_{11}$ while $E_{21}E_{12} = E_{22}$. Hence $M_2(\mathbb{R})$ is noncommutative. More generally, $E_{ij}E_{kl} = \delta_{jk}E_{il}$, a simple identity that streamlines many matrix calculations.

Example 4.3.2 (Upper triangular matrices)**.** The set $T_2(\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$ is a ring under matrix operations. It is closed under addition, negatives, and multiplication, but multiplication is not commutative. Unlike $M_2(\mathbb{R})$, it has a proper nonzero two-sided ideal, namely the strictly upper triangular matrices $N = \left\{ \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} \right\}$. Indeed, $N^2 = \{0\}$.

Example 4.3.3 (Zero divisors do not mean noncommutativity)**.** In the commutative ring $\mathbb{Z}_6$, $[2] \neq [0]$ and $[3] \neq [0]$, but $[2][3] = [0]$. Thus $\mathbb{Z}_6$ has nonzero zero divisors. By contrast, $\mathbb{Z}_5$ is a field. In $M_2(\mathbb{R})$, the nonzero matrix E_{12} satisfies $E_{12}^2 = 0$, and hence is a zero divisor (using the two-sided convention). The existence of zero divisors and noncommutativity are separate phenomena.

Proposition 4.3.4. *A nonzero residue $[a] \in \mathbb{Z}_n$ is a zero divisor if and only if* $\gcd(a, n) > 1$.

Proof. If $d = \gcd(a, n) > 1$, take $b = n/d$. Then $[b] \neq [0]$ and $n \mid ab$, so $[a][b] = [0]$. Conversely, suppose $[a][b] = [0]$ for some $[b] \neq [0]$. If $\gcd(a, n) = 1$, Bezout's identity gives integers u, v with $ua + vn = 1$, so multiplying by b shows $n \mid b$, a contradiction. ∎

Corollary 4.3.5. *For $n \geq 2$, the ring $\mathbb{Z}_n$ is a field if and only if n is prime.*

Proof. If n is prime, every nonzero residue is coprime to n and hence invertible. If n is composite, write $n = ab$ with $1 < a, b < n$; then $[a][b] = [0]$ with both factors nonzero, so $\mathbb{Z}_n$ is not a field. ∎

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Subrings and the importance of the ambient ring

Definition 4.3.6. A nonempty subset S of a ring R is a *subring* if it is itself a ring under the inherited addition and multiplication. The standard subring test asks whether $a - b \in S$ and $ab \in S$ whenever $a, b \in S$.

Here subrings are not required to share the identity of R. If a course adopts the convention that subrings must contain 1_R, then this extra condition must be checked and some examples below should be called nonunital subrings.

Example 4.3.7. The set $2\mathbb{Z}$ is a subring of $\mathbb{Z}$ but does not contain 1. The set $\mathbb{Z}$ is a subring of $\mathbb{Q}$ containing the same identity. The set of diagonal matrices in $M_2(\mathbb{R})$ is a commutative subring even though the ambient matrix ring is noncommutative.

Example 4.3.8 (A subset closed under products but not differences)**.** The subset $\mathbb{N}_0$ of $\mathbb{Z}$ is closed under multiplication and addition, but $1 - 2 = -1 \notin \mathbb{N}_0$. Consequently it is not a subring. Similarly, the strictly positive real numbers form a multiplicative group, but not a ring under the usual operations.

Definition 4.3.9. Let I be an additive subgroup of a ring R. It is a *left ideal* if $ra \in I$ for every $r \in R$ and $a \in I$; a *right ideal* if $ar \in I$ for all such r, a; and a *two-sided ideal* if it is both a left and a right ideal.

If R is commutative, every one-sided ideal is two-sided. Every ideal is closed under the multiplication of its own elements and hence is a subring under our convention, but a subring need not be an ideal. An ideal is required to absorb multiplication by elements of the entire ambient ring, not merely by elements of the subset.

Example 4.3.10 (A subring that is not an ideal)**.** The inclusion $\mathbb{Z} \subseteq \mathbb{Q}$ gives a subring, but not an ideal: $1 \in \mathbb{Z}$ and $1/2 \in \mathbb{Q}$, yet $(1/2) \cdot 1 = 1/2 \notin \mathbb{Z}$. Even inside $\mathbb{Z}[x]$, the subset $\mathbb{Z}$ is a subring but not an ideal, since $x \cdot 1 = x \notin \mathbb{Z}$.

Example 4.3.11 (A left ideal that is not a right ideal)**.** Set $R = M_2(\mathbb{R})$ and $I = \left\{ \begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix} : a, b \in \mathbb{R} \right\}$. Sums and additive inverses remain in I. Also,

$$\begin{pmatrix} r & s \\ t & u \end{pmatrix} \begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix} = \begin{pmatrix} ra + sb & 0 \\ ta + ub & 0 \end{pmatrix} \in I,$$

so I is a left ideal. But $E_{11} \in I$ and $E_{11}E_{12} = E_{12} \notin I$. Hence I is not a right ideal. Note that choosing matrices in the reverse order would test the wrong absorption condition.

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Example 4.3.12 (A right ideal that is not a left ideal)**.** Again let $R = M_2(\mathbb{R})$, but use $J = \left\{ \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} : a, b \in \mathbb{R} \right\}$. Right multiplication preserves the zero second row, so J is a right ideal. Yet $E_{11} \in J$ and $E_{21}E_{11} = E_{21} \notin J$. Therefore J is not a left ideal.

Example 4.3.13 (A proper two-sided ideal)**.** In the upper triangular ring $T_2(\mathbb{R})$, let N consist of the matrices bE_{12}. If $A = \begin{pmatrix} a & c \\ 0 & d \end{pmatrix}$, then $A(bE_{12}) = abE_{12}$ and $(bE_{12})A = dbE_{12}$. Hence N is a two-sided ideal of $T_2(\mathbb{R})$. The same set is not a two-sided ideal of $M_2(\mathbb{R})$. The ambient ring matters.

Proposition 4.3.14. *If R has an identity 1_R and an ideal I contains 1_R, then $I = R$. More generally, if a two-sided ideal of a unital ring contains any unit, it equals the whole ring.*

Proof. If $1_R \in I$, then $r = r1_R \in I$ for every $r \in R$. If $u \in I$ is invertible, then $1_R = u^{-1}u \in I$, and the first assertion applies. ■

Theorem 4.3.15. *Every ideal of $\mathbb{Z}$ is of the form $n\mathbb{Z}$ for a unique integer $n \geq 0$.*

Proof. The zero ideal is $0\mathbb{Z}$. If $I \neq \{0\}$, let n be its least positive element. Division gives $a = qn + r$, $0 \leq r < n$, for any $a \in I$. Because $r = a - qn \in I$, minimality forces $r = 0$. Hence $a \in n\mathbb{Z}$ and $I = n\mathbb{Z}$. Uniqueness follows from the least positive element. ■

Example 4.3.16 (Ideals in $\mathbb{Z}_{12}$)**.** Every ideal of $\mathbb{Z}_{12}$ is generated by one element $[d]$ with $d \mid 12$. Explicitly, the ideals are $([0])$, $([6])$, $([4])$, $([3])$, $([2])$, and $([1])$. Their cardinalities are $1, 2, 3, 4, 6, 12$, respectively. The ideal generated by $[8]$ equals the ideal generated by $[4]$, because both consist of $\{[0], [4], [8]\}$.

Practice and discussion

1. Determine all zero divisors and units in $\mathbb{Z}_{15}$.
2. Prove the subring test using the additive subgroup test.
3. Decide whether $\{a + b\sqrt{2} : a, b \in \mathbb{Z}\}$ is a subring of $\mathbb{R}$.
4. Determine whether the set of symmetric real 2×2 matrices is a subring of $M_2(\mathbb{R})$; if not, exhibit two symmetric matrices whose product is not symmetric.
5. Show that the trace-zero matrices in $M_2(\mathbb{R})$ are not generally a subring.
6. Explain carefully why $2\mathbb{Z}$ is a subring and an ideal of $\mathbb{Z}$, but not an ideal of $\mathbb{Q}$.
7. Show that the diagonal matrices form a subring but not a two-sided ideal of $M_2(\mathbb{R})$.
8. Prove that the intersection of two left ideals is a left ideal.

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9. In $T_2(\mathbb{R})$, verify directly that $N = \mathbb{R}E_{12}$ is a two-sided ideal and that $N^2 = 0$.

10. Classify all ideals of $\mathbb{Z}_{18}$ and determine their sizes.

4.4 Sums, Products and Intersections of Ideals

Definitions and fundamental inclusions

In this section I, J, K denote two-sided ideals of a ring R. Their *sum* and *intersection* are

$$I + J = \{i + j : i \in I,\ j \in J\}, \qquad I \cap J = \{x : x \in I \text{ and } x \in J\}.$$

The product ideal is not usually just the set of pairwise products $\{ij : i \in I, j \in J\}$; it consists of finite sums of these products:

$$IJ = \left\{ \sum_{k=1}^{m} i_k j_k : m \geq 1,\ i_k \in I,\ j_k \in J \right\}.$$

Since zero can be used as a summand, this definition includes the zero element.

Theorem 4.4.1. *For two-sided ideals I, J, K of R the following hold:*

(i) $I + J$, $I \cap J$, *and* IJ *are two-sided ideals;*

(ii) $I + J$ *is the least ideal containing* I *and* J, *while* $I \cap J$ *is the greatest ideal contained in both;*

(iii) $IJ \subseteq I \cap J$;

(iv) $I(J + K) = IJ + IK$ *and* $(I + J)K = IK + JK$;

(v) $(IJ)K = I(JK)$, *but* IJ *and* JI *need not be equal in a noncommutative ring.*

Proof. The sum and intersection satisfy the additive subgroup and absorption tests. For the product, closure under addition is built into the definition; additive inverses are obtained by negating one factor in each summand. If $r \in R$, then $r\sum i_k j_k = \sum (r i_k) j_k \in IJ$ and $(\sum i_k j_k) r = \sum i_k (j_k r) \in IJ$. Thus the product is an ideal. If $i \in I$ and $j \in J$, then $ij \in I$ by right absorption and $ij \in J$ by left absorption, proving $IJ \subseteq I \cap J$. The minimality and maximality statements are immediate from the definitions. Distributivity follows by expanding finite sums; associativity of products follows by reassociating each triple product. The final warning is illustrated below. ∎

Example 4.4.2 (Products can depend on order)**.** In the ring $T_2(\mathbb{R})$, let

$$I = \left\{ \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} : a, b \in \mathbb{R} \right\}, \qquad J = \left\{ \begin{pmatrix} 0 & b \\ 0 & c \end{pmatrix} : b, c \in \mathbb{R} \right\}.$$

Both are two-sided ideals. Their product is $IJ = \mathbb{R}E_{12}$, because $E_{11}E_{12} = E_{12}$ and every product lies in $\mathbb{R}E_{12}$. On the other hand, $JI = \{0\}$. Indeed, a matrix with zero first column multiplied by one with zero second row gives zero. Hence generally $IJ \neq JI$.

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Computations in $\mathbb{Z}$ and polynomial rings

Theorem 4.4.3. *For positive integers a, b,*

$$a\mathbb{Z} + b\mathbb{Z} = \gcd(a,b)\mathbb{Z}, \qquad a\mathbb{Z} \cap b\mathbb{Z} = \operatorname{lcm}(a,b)\mathbb{Z},$$

and $(a\mathbb{Z})(b\mathbb{Z}) = ab\mathbb{Z}$.

Proof. By Bezout's identity, $\gcd(a,b)$ is an integer linear combination of a, b, and it divides every such combination, proving the first formula. The common multiples of a, b are precisely the multiples of their least common multiple, proving the second. Every finite sum of terms $(ax)(by)$ is divisible by ab, while ab itself is such a product, proving the third. ■

Example 4.4.4 (Three different ideals)**.** For $I = 4\mathbb{Z}$ and $J = 6\mathbb{Z}$, one finds $I + J = 2\mathbb{Z}$, $I \cap J = 12\mathbb{Z}$, and $IJ = 24\mathbb{Z}$. The strict inclusion $IJ \subsetneq I \cap J$ is typical when the generators have a common factor.

Example 4.4.5 (A more substantial computation)**.** Let $I = 18\mathbb{Z}$ and $J = 30\mathbb{Z}$. Then $I + J = 6\mathbb{Z}$, $I \cap J = 90\mathbb{Z}$, and $IJ = 540\mathbb{Z}$. This demonstrates that the sum may be larger than either ideal, whereas the product may be much smaller than the intersection. Recall that $a\mathbb{Z} \subseteq b\mathbb{Z}$ exactly when $b \mid a$; ordinary numerical comparisons of a and b do not give the correct inclusion criterion.

Example 4.4.6 (Polynomial ideals)**.** In $\mathbb{Q}[x]$, let $I = (x^2)$ and $J = (x)$. Since $(x^2) \subseteq (x)$, one has $I + J = (x)$ and $I \cap J = (x^2)$, while $IJ = (x^3)$. On the other hand, if $I = (x)$ and $J = (x-1)$, then $1 = x - (x-1) \in I + J$, so $I + J = \mathbb{Q}[x]$ and $I \cap J = IJ = (x(x-1))$.

Proposition 4.4.7. *If R is commutative, $I + J = R$, and I, J are ideals, then $I \cap J = IJ$.*

Proof. We always have $IJ \subseteq I \cap J$. Conversely, write $1 = i + j$ with $i \in I$, $j \in J$. For $x \in I \cap J$, we have $x = xi + xj$. Because $x \in J$ and $i \in I$, $xi \in IJ$; similarly $x \in I$ and $j \in J$ give $xj \in IJ$. Hence $x \in IJ$. ■

Example 4.4.8 (Why the hypothesis matters)**.** For $I = 2\mathbb{Z}$ and $J = 3\mathbb{Z}$, Bezout's identity gives $I + J = \mathbb{Z}$ and therefore $I \cap J = IJ = 6\mathbb{Z}$. For $I = 4\mathbb{Z}$ and $J = 6\mathbb{Z}$, the sum is only $2\mathbb{Z}$, and the intersection $12\mathbb{Z}$ is strictly larger than the product $24\mathbb{Z}$. The equality is not automatic without comaximality.

Connections with quotient rings and homomorphisms

For a two-sided ideal I of a ring R, the quotient R/I has operations $(r + I) + (s + I) = (r + s) + I$ and $(r + I)(s + I) = rs + I$. Absorption by I is precisely what makes multiplication of cosets well-defined. If I were merely a left ideal, this multiplication would generally not be well-defined.

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Proposition 4.4.9. *Let $\varphi : R \to S$ be a ring homomorphism. Then* $\ker \varphi = \{r \in R : \varphi(r) = 0\}$ *is a two-sided ideal of R.*

Proof. The kernel is an additive subgroup. If $x \in \ker \varphi$ and $r \in R$, then $\varphi(rx) = \varphi(r)\varphi(x) = 0$ and $\varphi(xr) = \varphi(x)\varphi(r) = 0$. Thus it absorbs multiplication from both sides. ■

Example 4.4.10 (Reduction modulo n)**.** The mapping $\pi : \mathbb{Z} \to \mathbb{Z}_n$ given by $\pi(a) = [a]$ is a surjective ring homomorphism. Its kernel is $n\mathbb{Z}$. The equality $\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}_n$ illustrates how the ideal of multiples of n controls the ring of residue classes.

Example 4.4.11 (Evaluation of polynomials)**.** For a fixed $a \in \mathbb{R}$, the evaluation map $\mathrm{ev}_a : \mathbb{R}[x] \to \mathbb{R}$, defined by $f(x) \mapsto f(a)$, is a surjective ring homomorphism. By the factor theorem, its kernel is $(x - a)$. This is a concrete reason why the ideal $(x - a)$ is important: quotienting by it amounts to replacing a polynomial by its value at a.

Tutorial problems: from routine to conceptual

1. Compute the sum, intersection, and product of $8\mathbb{Z}$ and $12\mathbb{Z}$.

2. If $I = 15\mathbb{Z}$ and $J = 28\mathbb{Z}$, decide whether $I \cap J = IJ$.

3. Prove that for ideals I, J, K, $I \cap (J + K)$ contains $(I \cap J) + (I \cap K)$. Find an example showing equality can fail.

4. Prove that $I(J + K) = IJ + IK$ for two-sided ideals of any ring.

5. Give an example of distinct ideals I, J such that $IJ = \{0\}$.

6. In $\mathbb{Q}[x]$, compute $(x^2) + (x^3)$, $(x^2) \cap (x^3)$, and $(x^2)(x^3)$.

7. In $\mathbb{Q}[x]$, compute $(x^2 - 1) + (x - 1)$ and $(x^2 - 1) \cap (x - 1)$.

8. Show that (x, y) is an ideal of $\mathbb{R}[x, y]$, and explain why it contains all polynomials with zero constant term.

9. Let $\varphi : \mathbb{Z}[x] \to \mathbb{Z}$ be evaluation at $x = 2$. Find $\ker \varphi$.

10. Prove directly that a quotient ring R/I has a well-defined multiplication exactly when I is a two-sided ideal (with the usual additive cosets).

11. If $I \subseteq J$, prove $I + J = J$ and $I \cap J = I$. Determine whether $IJ = I$ must hold.

12. Suppose R is commutative with identity and $I + J = R$. Prove $I^m + J^n = R$ for all positive integers m, n.

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4.5 Beyond the Syllabus: Structural Connections

The results in this final section are enrichment material rather than prescribed tutorial requirements. They explain why the calculations developed above have lasting significance in algebra. The emphasis remains on proofs and examples that can be understood with the concepts already studied, rather than on introducing large amounts of new terminology.

The Chinese remainder theorem: arithmetic as a product of rings

Theorem 4.5.1 (Chinese remainder theorem for two integers)**.** *If m, n are relatively prime positive integers, then*

$$\Phi : \mathbb{Z}_{mn} \longrightarrow \mathbb{Z}_m \times \mathbb{Z}_n, \qquad [a]_{mn} \longmapsto ([a]_m, [a]_n)$$

is a ring isomorphism.

Proof. The map respects addition and multiplication. If $\Phi([a]) = (0,0)$, then both m and n divide a. Since they are coprime, $mn \mid a$, so $[a]_{mn} = 0$; hence the map is injective. Both rings have mn elements, so an injective map between them is also surjective. It is therefore an isomorphism. ■

Example 4.5.2 (Solving two congruences)**.** Solve $x \equiv 2 \pmod 5$ and $x \equiv 3 \pmod 7$. Write $x = 2 + 5t$. Modulo 7, this gives $5t \equiv 1 \pmod 7$. Since $5^{-1} \equiv 3 \pmod 7$, we get $t \equiv 3 \pmod 7$, hence $x \equiv 17 \pmod{35}$. The Chinese remainder theorem guarantees a unique solution modulo 35, not merely a lucky numerical answer.

Example 4.5.3 (An unexpected group consequence)**.** Since $\mathbb{Z}_{15} \cong \mathbb{Z}_3 \times \mathbb{Z}_5$ as rings, their groups of units satisfy $U(15) \cong U(3) \times U(5)$. Thus $|U(15)| = 2 \cdot 4 = 8$, recovering $\varphi(15) = 8$ from the factorization into coprime moduli. More generally, $\varphi(mn) = \varphi(m)\varphi(n)$ whenever $\gcd(m,n) = 1$.

Proposition 4.5.4. *Let R be a commutative ring with identity and I, J ideals satisfying $I + J = R$. Then $I \cap J = IJ$ and*

$$R/(I \cap J) \;\cong\; R/I \times R/J.$$

Proof. We proved $I \cap J = IJ$ in the preceding section. Define $f : R \to R/I \times R/J$ by $f(r) = (r + I, r + J)$. Its kernel is $I \cap J$. To show surjectivity, write $1 = i + j$ with $i \in I$ and $j \in J$. Given $a, b \in R$, set $r = aj + bi$. Then $r \equiv a \pmod I$ and $r \equiv b \pmod J$. The first isomorphism theorem gives the result. ■

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The centre of a group and the obstruction to commutativity

Definition 4.5.5. The *centre* of a group G is $Z(G) = \{z \in G : zg = gz \text{ for every } g \in G\}$. The group G is abelian exactly when $Z(G) = G$.

Proposition 4.5.6. *The centre $Z(G)$ is a normal subgroup of G.*

Proof. The identity commutes with every element, and if x, y commute with all elements then so do xy and x^{-1}; hence the centre is a subgroup. For every $g \in G$ and $z \in Z(G)$, we have $gzg^{-1} = z$, so the centre is normal. ■

Example 4.5.7. For S_3, the centre is $\{e\}$: a nonidentity transposition fails to commute with a suitable transposition, and a 3-cycle fails to commute with a transposition. For the quaternion group Q_8, the centre is $\{1, -1\}$. The difference between these centres gives information not visible merely from the group orders.

Theorem 4.5.8. *If $G/Z(G)$ is cyclic, then G is abelian.*

Proof. Suppose $G/Z(G)$ is generated by $gZ(G)$. Every $x, y \in G$ can be written as $x = g^a z_1$ and $y = g^b z_2$, with $z_1, z_2 \in Z(G)$. Central elements commute with everything, and powers of g commute with one another. Consequently $xy = g^{a+b} z_1 z_2 = g^{b+a} z_2 z_1 = yx$. ■

This result explains why a nonabelian group cannot have a cyclic quotient by its centre. It is especially useful when studying groups of small order.

Commutators: measuring the failure of elements to commute

For $a, b \in G$, define the *commutator* $[a, b] = aba^{-1}b^{-1}$. It equals the identity if and only if a and b commute. The subgroup generated by all commutators is the *commutator subgroup* G', also written $[G, G]$.

Proposition 4.5.9. *The commutator subgroup G' is normal in G, and the quotient G/G' is abelian. Moreover, if $N \trianglelefteq G$ and G/N is abelian, then $G' \subseteq N$.*

Proof. Conjugating a commutator by g gives $g[a, b]g^{-1} = [gag^{-1}, gbg^{-1}]$, so the subgroup generated by commutators is normal. In G/G', one has $(aG')(bG') = (bG')(aG')$, because $aba^{-1}b^{-1} \in G'$. Conversely, if G/N is abelian, all commutators map to the identity in G/N and thus belong to N. Therefore $G' \subseteq N$. ■

Example 4.5.10. The commutator subgroup of S_3 is A_3. Indeed, the sign homomorphism has abelian image $\{\pm 1\}$, hence $S_3' \subseteq A_3$. But S_3 is nonabelian, so $S_3' \neq \{e\}$; the only nontrivial subgroup of $A_3 \cong \mathbb{Z}_3$ is A_3 itself. Thus $S_3/S_3' \cong \mathbb{Z}_2$.

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The ideal lattice and a subtle failure of distributivity

Ideals are naturally ordered by inclusion. Their intersection is the largest ideal below both, while their sum is the smallest ideal above both. These operations resemble logical "and" and "or", but naive distributive identities do not always hold.

Example 4.5.11 (Strict containment)**.** Take $I = 2\mathbb{Z}$, $J = 3\mathbb{Z}$, and $K = 4\mathbb{Z}$. Then $J + K = \mathbb{Z}$, so $I \cap (J + K) = 2\mathbb{Z}$. But

$$(I \cap J) + (I \cap K) = 6\mathbb{Z} + 4\mathbb{Z} = 2\mathbb{Z}.$$

Here equality happens. To see failure, work in $R = \mathbb{R}[x, y]$ and choose $I = (x)$, $J = (y)$, $K = (x - y)$. Since $y + (x - y) = x$, one has $x \in I \cap (J + K)$. However $I \cap J = (xy)$ and $I \cap K = (x(x - y))$, so

$$(I \cap J) + (I \cap K) = \big(xy,\, x(x - y)\big) = x(x, y).$$

The element x is not in $x(x, y)$: if it were, cancellation in the integral domain $\mathbb{R}[x, y]$ would give $1 \in (x, y)$, which is impossible by evaluation at $(0, 0)$. Thus the inclusion is strict.

Proposition 4.5.12. *For ideals I, J, K with $I \subseteq K$, one has*

$$(I + J) \cap K = I + (J \cap K).$$

Proof. The right-hand side is contained in $(I + J) \cap K$. Conversely, let $x \in (I + J) \cap K$ and write $x = i + j$, where $i \in I$ and $j \in J$. Since $I \subseteq K$ and $x \in K$, we have $j = x - i \in K$. Thus $j \in J \cap K$ and $x \in I + (J \cap K)$. ∎

The modular law is an elegant and useful replacement for unrestricted distributivity. Its hypothesis $I \subseteq K$ is essential to the short proof.

Nilpotent elements and reduced rings

Definition 4.5.13. An element a of a ring is *nilpotent* if $a^n = 0$ for some positive integer n. A commutative ring is called *reduced* if 0 is its only nilpotent element.

Example 4.5.14. In $\mathbb{Z}_{12}$, $[6]^2 = [0]$ although $[6] \neq [0]$. Thus $\mathbb{Z}_{12}$ is not reduced. In $\mathbb{Z}_6$, no nonzero element is nilpotent, even though $[2][3] = [0]$; hence a reduced ring may still have zero divisors. In $T_2(\mathbb{R})$, the matrix E_{12} is nilpotent, and the ideal $\mathbb{R}E_{12}$ has square zero.

Proposition 4.5.15. *Every nonzero nilpotent element of a commutative ring is a zero divisor. Every integral domain is reduced.*

Proof. If $a^n = 0$ with $a \neq 0$, choose the smallest such $n \geq 2$. Then $a^{n-1} \neq 0$ but $a\, a^{n-1} = 0$, so a is a zero divisor. An integral domain has no nonzero zero divisors, so it has no nonzero

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nilpotents. ■

Example 4.5.16 (Polynomial quotients)**.** The quotient $\mathbb{R}[x]/(x^2)$ contains the nonzero class $\bar{x} = x + (x^2)$, yet $\bar{x}^2 = 0$. In contrast, $\mathbb{R}[x]/(x-2) \cong \mathbb{R}$ is a field and therefore reduced. An ideal therefore controls not just which elements become zero, but also whether a quotient introduces nilpotent behaviour.

Why these connections matter

The Chinese remainder theorem connects congruences with products of rings; the centre and commutator subgroup quantify different aspects of noncommutativity; the modular law clarifies the structure of ideals; and nilpotents reveal information that is invisible in ordinary additive calculations. Each construction begins with a familiar definition, then asks a more structural question. This is a productive approach to higher algebra: calculate in explicit examples, identify an invariant, and prove a statement explaining the calculation.

Enrichment problems

1. Solve simultaneously $x \equiv 3 \pmod 8$ and $x \equiv 4 \pmod 5$, and explain why the solution is unique modulo 40.
2. Determine whether the system $x \equiv 1 \pmod 6$, $x \equiv 2 \pmod 9$ has a solution. What changes when the moduli are not coprime?
3. Prove that $\mathbb{Z}_{35} \cong \mathbb{Z}_5 \times \mathbb{Z}_7$ as rings.
4. Prove that the centre of an abelian group is the whole group.
5. Find the centre of the dihedral group of symmetries of a square.
6. Determine the commutator subgroup of an abelian group.
7. Show that any homomorphism from a group G to an abelian group sends G' to the identity.
8. Verify the modular law for $I = 12\mathbb{Z}$, $J = 18\mathbb{Z}$, and $K = 4\mathbb{Z}$.
9. Find all nilpotent elements of $\mathbb{Z}_8$ and of $\mathbb{Z}_{10}$.
10. Prove that $\mathbb{Z}_n$ is reduced if and only if n is squarefree.
11. Let R be a commutative ring and I, J comaximal ideals. Prove $I^r + J^s = R$ for all positive integers r, s.
12. Investigate whether the set of nilpotent elements of a noncommutative ring must form an ideal; test your conjecture in $M_2(\mathbb{R})$.

Additional worked examples

The following examples illustrate connections among the algebraic structures developed in this chapter.

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Example 4.5.17 (A group operation hidden inside arithmetic)**.** Fix a nonzero real number c and let $G = \mathbb{R} \setminus \{-c\}$. Define $a \star b = a + b + ab/c$. Prove that $(G, \star)$ is an abelian group. Observe first that

$$1 + \frac{a \star b}{c} = \left(1 + \frac{a}{c}\right)\left(1 + \frac{b}{c}\right).$$

The factors on the right are nonzero for $a, b \in G$, so $a \star b \in G$. The map $f : G \to \mathbb{R}^\times$ given by $f(a) = 1 + a/c$ is bijective and satisfies $f(a \star b) = f(a)f(b)$. Associativity and commutativity of $\star$ follow from those of real multiplication. The identity is 0, since $a \star 0 = a$, and solving $a \star b = 0$ gives $b = -ac/(a + c)$, which belongs to G. Thus the apparently complicated law describes exactly the multiplicative group of nonzero real numbers in different coordinates. This example also illustrates a general method: transfer a familiar operation across a bijection.

Example 4.5.18 (A structure table and what it proves)**.** Let $G = \{e, a, b, c\}$, with multiplication given by

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

The table displays closure, the identity e, and inverses (each nonidentity element is its own inverse). Associativity cannot be inferred merely from the visual symmetry of a table. Here it follows by identifying $e = (0,0)$, $a = (1,0)$, $b = (0,1)$, and $c = (1,1)$ in $\mathbb{Z}_2 \times \mathbb{Z}_2$, with componentwise addition. This identification proves associativity and commutativity simultaneously. The group is not cyclic because it has order four but no element of order four. A useful teaching principle is that a well-chosen isomorphism can replace dozens of direct checks.

Example 4.5.19 (Powers in a finite group)**.** Suppose $a \in G$ has order 18. Determine the orders of a^6, a^8, a^{12}, and a^{25}. The power formula gives

$$\operatorname{ord}(a^6) = 3, \quad \operatorname{ord}(a^8) = 9, \quad \operatorname{ord}(a^{12}) = 3, \quad \operatorname{ord}(a^{25}) = 18.$$

Thus a^{25} is a generator of $\langle a \rangle$ despite its exponent being larger than the order of a. Exponents should always be interpreted modulo the order of the element; $a^{25} = a^7$ here, and $\gcd(7, 18) = 1$.

Example 4.5.20 (Finding all subgroups of a cyclic group)**.** Let $G = \langle a \rangle$ have order 36. Every subgroup corresponds to a positive divisor of 36, namely $1, 2, 3, 4, 6, 9, 12, 18, 36$. The subgroup of order d is $H_d = \langle a^{36/d} \rangle$. For instance, $H_9 = \langle a^4 \rangle$, $H_{12} = \langle a^3 \rangle$, and $H_6 = \langle a^6 \rangle$. The containment criterion is $H_d \subseteq H_e$ if and only if $d \mid e$. Consequently $H_6 \subseteq H_{12}$, but H_9 and H_{12} are incomparable. Their intersection is $H_{\gcd(9,12)} = H_3$, and the subgroup they generate is $H_{\operatorname{lcm}(9,12)} = H_{36} = G$. These relationships foreshadow the GCD and LCM identities for ideals of the integers.

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Example 4.5.21 (The order of a permutation after taking a power)**.** Let $\sigma = (1\,2\,3\,4\,5\,6)(7\,8\,9\,10)$, a permutation in S_{10}. Then $\operatorname{ord}(\sigma) = \operatorname{lcm}(6,4) = 12$. The order of σ^8 is $12/\gcd(12,8) = 3$. The parity of σ is $(-1)^{(6-1)+(4-1)} = (-1)^8 = 1$, so it is even; every power of σ is consequently even. To see the cycle structure of σ^8, note that on the six-cycle it acts as the second power, yielding two disjoint three-cycles, whereas on the four-cycle it acts as the identity. Thus $\sigma^8 = (1\,3\,5)(2\,4\,6)$. This supplies an independent check that its order is three.

Example 4.5.22 (Odd and even permutations via inversions)**.** A permutation is displayed in one-line notation as $\sigma = [4,1,5,2,3]$. Count inversions by fixing each entry and counting the smaller entries to its right. The entry 4 contributes three inversions; 1 contributes none; 5 contributes two; 2 contributes none. The total is five, so σ is odd. Tracing images gives $1 \mapsto 4 \mapsto 2 \mapsto 1$ and $3 \mapsto 5 \mapsto 3$, hence $\sigma = (1\,4\,2)(3\,5)$. A three-cycle contributes two transpositions and a two-cycle contributes one: again the total parity is odd. Calculating a quantity in two independent ways is an effective way to detect errors in permutation composition.

Example 4.5.23 (Why $\mathbb{Z}_2 \times \mathbb{Z}_6$ is not cyclic)**.** The group has 12 elements. Every element (x,y) has order equal to the least common multiple of the orders of x and y. The first coordinate has order at most two, and the second at most six, so every element has order dividing $\operatorname{lcm}(2,6) = 6$. Hence no element generates the group. Compare this with $\mathbb{Z}_3 \times \mathbb{Z}_4$, which also has 12 elements but is cyclic, since $\operatorname{lcm}(3,4) = 12$. The example makes clear why the cardinality of a finite group is not enough to determine its algebraic structure.

Example 4.5.24 (Finding the ideal generated by two polynomials)**.** Let $R = \mathbb{Q}[x]$, $f = x^2 - 1$, and $g = x^2 - 3x + 2$. Factor $f = (x-1)(x+1)$ and $g = (x-1)(x-2)$. The greatest common divisor is $x - 1$, up to multiplication by a nonzero rational number. Since $\mathbb{Q}[x]$ is a Euclidean domain, $(f) + (g) = (\gcd(f,g)) = (x-1)$. Similarly, their intersection is generated by the least common multiple, $(f) \cap (g) = ((x-1)(x+1)(x-2))$, while their product is $((x-1)^2(x+1)(x-2))$. Because the polynomials share the factor $x - 1$, the product is properly contained in the intersection. This is exactly analogous to the computation for ideals $4\mathbb{Z}$ and $6\mathbb{Z}$.

Example 4.5.25 (A ring of functions)**.** Let $X = \{p, q\}$ and let R be the set of all functions $X \to \mathbb{R}$, with pointwise operations. Then $R \cong \mathbb{R} \times \mathbb{R}$ through $f \mapsto (f(p), f(q))$. The function u with $u(p) = 1$, $u(q) = 0$ is nonzero, as is the function v with $v(p) = 0$, $v(q) = 1$, but $uv = 0$. Thus R has zero divisors although its coefficients are taken from a field. The subset $I = \{f : f(p) = 0\}$ is an ideal: it is closed under differences and multiplication by any function. The quotient $R/I \cong \mathbb{R}$ is obtained by evaluating a function at p. This example links the concepts of direct products, ideals, quotient rings, and evaluation maps in one setting.

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Example 4.5.26 (Distinguishing one-sided and two-sided ideals systematically)**.** Let $R = M_3(\mathbb{R})$. Let L be the set of matrices with third column zero. For $A \in R$ and $B \in L$, the third column of AB is A times the zero third column of B, so $AB \in L$. Thus L is a left ideal. It is not a right ideal: $E_{11} \in L$, but $E_{11}E_{13} = E_{13} \notin L$. Conversely, the set H of matrices with third row zero is a right ideal but not a left ideal: $E_{11} \in H$, whereas $E_{31}E_{11} = E_{31} \notin H$. The first test follows the columns under left multiplication, and the second follows the rows under right multiplication. For larger matrices, this viewpoint is far more efficient than multiplying arbitrary entries.

Example 4.5.27 (A quotient as an evaluation problem)**.** Describe $\mathbb{R}[x]/(x^2-1)$. Since $x^2-1 = (x-1)(x+1)$ and $(x-1)+(x+1) = \mathbb{R}[x]$ (their difference is 2, a unit), the Chinese remainder theorem gives

$$\mathbb{R}[x]/(x^2-1) \cong \mathbb{R}[x]/(x-1) \times \mathbb{R}[x]/(x+1) \cong \mathbb{R} \times \mathbb{R}.$$

Under the isomorphism, the class of $f(x)$ maps to $(f(1), f(-1))$. For example, the class of x maps to $(1,-1)$. Thus the quotient encodes simultaneous evaluation at the two roots 1 and -1. This also explains why the quotient is not a field: $(1,0)$ and $(0,1)$ are nonzero elements whose product is zero.

Example 4.5.28 (A nontrivial Chinese remainder computation)**.** Solve $x \equiv 4 \pmod 9$ and $x \equiv 5 \pmod{11}$. Write $x = 4+9t$. The second congruence becomes $9t \equiv 1 \pmod{11}$. Since $9 \cdot 5 = 45 \equiv 1 \pmod{11}$, it follows that $t \equiv 5 \pmod{11}$ and hence $x \equiv 49 \pmod{99}$. Direct verification gives $49 \equiv 4 \pmod 9$ and $49 \equiv 5 \pmod{11}$. Any other integer solution differs from 49 by a multiple of 99, because the difference must be divisible by both coprime moduli.

Example 4.5.29 (A failed Chinese remainder system)**.** Consider $x \equiv 1 \pmod 6$ and $x \equiv 2 \pmod 9$. If a solution existed, reduction modulo 3 would give both $x \equiv 1 \pmod 3$ and $x \equiv 2 \pmod 3$, a contradiction. More generally, $x \equiv a \pmod m$ and $x \equiv b \pmod n$ have a simultaneous solution if and only if $\gcd(m,n)$ divides $a-b$. Necessity follows by subtracting the congruences. For sufficiency, write $x = a+mt$ and solve $mt \equiv b-a \pmod n$, which is possible precisely under the stated divisibility condition. When a solution exists it is unique modulo $\operatorname{lcm}(m,n)$.

Example 4.5.30 (The modular law in concrete ideals)**.** Take $I = 12\mathbb{Z}$, $K = 4\mathbb{Z}$, and $J = 18\mathbb{Z}$. Because $I \subseteq K$, the modular law applies. Calculate

$$(I+J) \cap K = (6\mathbb{Z}) \cap (4\mathbb{Z}) = 12\mathbb{Z}$$

and

$$I + (J \cap K) = 12\mathbb{Z} + 36\mathbb{Z} = 12\mathbb{Z}.$$

Both methods give the same ideal. This calculation is simple, but its conceptual content is important: the modular law explains why the equality holds for every configuration of

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ideals with $I \subseteq K$, not merely for particular integers.

Example 4.5.31 (Nilpotents and zero divisors in residue rings)**.** Determine the nilpotent residues in $\mathbb{Z}_{72}$, where $72 = 2^3 \cdot 3^2$. If $[a]^k = [0]$, then $72 \mid a^k$, so the primes 2 and 3 both divide a. Conversely, if $6 \mid a$, then a^3 is divisible by $2^3 \cdot 3^3$ and hence by 72. Therefore the nilpotent residues are exactly $[0], [6], [12], \ldots, [66]$, the twelve multiples of six modulo 72. This illustrates the general fact that nilpotence modulo n is controlled by which prime divisors of n divide a representative, rather than by its absolute size.

Proposition 4.5.32. *Let $n \geq 2$ and let $p_1, \ldots, p_r$ be its distinct prime divisors. The residue $[a] \in \mathbb{Z}_n$ is nilpotent if and only if $p_i \mid a$ for every i.*

Proof. If $[a]^k = [0]$, then $n \mid a^k$, and hence each prime p_i divides a^k, so $p_i \mid a$. Conversely, write $n = \prod p_i^{e_i}$, and suppose every p_i divides a. If $k \geq \max_i e_i$, then $p_i^{e_i} \mid a^k$ for every i, so $n \mid a^k$. Thus $[a]$ is nilpotent. ■

Corollary 4.5.33. *The ring $\mathbb{Z}_n$ is reduced if and only if n is squarefree.*

Proof. If n is squarefree and $[a]^k = 0$, then every prime factor of n divides a, hence $n \mid a$ and $[a] = 0$. Conversely, if $p^2 \mid n$, write $n = \prod_i p_i^{e_i}$ and let a be the product of the distinct prime factors of n. Then $[a] \neq 0$, since $p^2 \nmid a$, but $[a]$ is nilpotent by the proposition. Thus $\mathbb{Z}_n$ is not reduced. ■

Example 4.5.34 (Why nilpotent matrices do not form an ideal)**.** In $M_2(\mathbb{R})$, both E_{12} and E_{21} are nilpotent because their squares are zero. However,

$$(E_{12} + E_{21})^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^2 = I_2,$$

so their sum is not nilpotent. Hence the set of nilpotent elements of a noncommutative ring need not even be an additive subgroup, much less a two-sided ideal. By contrast, in a commutative ring, the nilpotent elements do form an ideal: if $a^m = b^n = 0$, then every term in the binomial expansion of $(a + b)^{m+n-1}$ contains either at least m factors of a or at least n factors of b, so the entire sum is zero; also $(ra)^m = r^m a^m = 0$.

Proposition 4.5.35. *In a commutative ring R, the set $\mathcal{N}(R) = \{a \in R : a^m = 0 \text{ for some } m \geq 1\}$ is an ideal.*

Proof. The element 0 is nilpotent. If $a^m = 0$ and $b^n = 0$, the binomial theorem shows $(a - b)^{m+n-1} = 0$: in every term, either the exponent of a is at least m or that of b is at least n. If $r \in R$, then $(ra)^m = r^m a^m = 0$. Thus $\mathcal{N}(R)$ satisfies the ideal test. ■

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Example 4.5.36 (A compact proof using the group centre)**.** Suppose G is a group of order p^2, where p is prime. A standard result from the class equation says $Z(G)$ is nontrivial for a finite p-group, so $|Z(G)|$ is either p or p^2. If $|Z(G)| = p^2$, the group is abelian. If $|Z(G)| = p$, then $G/Z(G)$ has order p and is cyclic; by the theorem proved earlier, G is again abelian. Therefore every group of order p^2 is abelian. The class-equation result is additional background, but the last step illustrates the striking power of a simple statement about the centre.

Further questions for classroom discussion

The hypotheses in these results are significant: finiteness is essential to the cancellation criterion, commutativity is used in the binomial argument for nilpotent elements, and coprimality governs the simplest form of the Chinese remainder theorem. The following exercises explore these distinctions.

1. If a group has exactly one subgroup of each divisor order, must the group be cyclic? Test small groups before proposing a theorem.
2. Is every group of prime order cyclic? Prove the answer using Lagrange's theorem.
3. Can a nonabelian group have an abelian subgroup of index two? Give an example.
4. Is it possible for $IJ = I \cap J$ when $I + J \neq R$? Test $I = J$ for suitable idempotent ideals.
5. When do the sum and intersection of two ideals coincide? Derive a necessary and sufficient condition.
6. Can the units of a commutative ring form a noncyclic group? Examine $U(8)$.
7. Why is the set of all singular 2×2 real matrices not an ideal? Check closure under addition first.
8. Find two distinct nonzero polynomials $f, g \in \mathbb{R}[x, y]$ such that $(f) \cap (g) = (fg)$.
9. Give an example of a ring with a nonzero nilpotent ideal and describe the quotient by that ideal.
10. Explain how counting inversions and counting transpositions express the same permutation invariant.

A short collection of proof techniques

A substantial part of learning algebra is learning how to choose a proof rather than merely following one. The next results exhibit five recurring techniques: applying a homomorphism, using minimality, exploiting a finite counting argument, constructing a counterexample, and choosing a well-adapted representation.

Proposition 4.5.37. *Let $f : G \to H$ be a group homomorphism. If $a \in G$ has finite order n, then* $\operatorname{ord}(f(a))$ *divides n.*

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Proof. Since $a^n = e_G$, one has $f(a)^n = f(a^n) = f(e_G) = e_H$. If m is the order of $f(a)$, the order criterion for cyclic subgroups tells us that $m \mid n$. ■

Example 4.5.38 (Using orders to rule out an isomorphism)**.** There is no group isomorphism $\mathbb{Z}_8 \to \mathbb{Z}_4 \times \mathbb{Z}_2$. The group $\mathbb{Z}_8$ has an element of order eight, whereas an element of the product has order dividing $\operatorname{lcm}(4,2) = 4$. An isomorphism preserves element orders, so the two groups cannot be isomorphic. This argument is more informative than observing that their descriptions differ: both have eight elements, and the obstruction is structural.

Proposition 4.5.39. *If R is a finite integral domain, its characteristic is a prime number.*

Proof. Because R is finite and contains $1_R \neq 0$, there exist distinct positive integers $r > s$ such that $r1_R = s1_R$, so the characteristic is a positive integer n. If $n = ab$ with $1 < a, b < n$, then $(a1_R)(b1_R) = n1_R = 0$. By the minimality of n, both factors are nonzero, contradicting the absence of zero divisors. Thus n is prime. ■

Theorem 4.5.40. *Every finite commutative ring with identity and no nonzero zero divisors is a field.*

Proof. Fix $0 \neq a \in R$. Multiplication by a defines a map $L_a : R \to R$, $x \mapsto ax$. If $ax = ay$, then $a(x - y) = 0$. Since a is nonzero and there are no zero divisors, $x = y$. Thus L_a is injective. Finiteness makes it surjective, so $ab = 1$ for some $b \in R$. Every nonzero element is invertible, proving that R is a field. ■

The proof is worth comparing with the finite-cancellation criterion for monoids. In each argument an injective self-map of a finite set becomes surjective, producing a missing inverse. Recognizing this pattern makes it possible to transfer an idea from group theory to ring theory.

Example 4.5.41 (The hypothesis of finiteness)**.** The ring $\mathbb{Z}$ is an integral domain but not a field. Multiplication by 2 is injective on $\mathbb{Z}$, yet not surjective, because 1 has no integer preimage. Thus removing the finiteness assumption destroys the reasoning in the preceding theorem. This example pinpoints the gap that appears if one carelessly replaces "finite integral domain" by "integral domain".

Proposition 4.5.42. *Let R be a ring and $I \subseteq R$ a nonempty subset. Then I is a two-sided ideal if and only if $a - b \in I$ for all $a, b \in I$, and $ra, ar \in I$ for all $r \in R$ and $a \in I$.*

Proof. Necessity follows from the definition. Conversely, closure under subtraction makes I an additive subgroup by the subgroup test: taking $a = b$ gives $0 \in I$ and then $0 - a = -a \in I$, while $a + b = a - (-b) \in I$. The remaining conditions are precisely left and right absorption. Therefore I is an ideal. ■

Example 4.5.43 (An efficient negative test)**.** To show that a subset is not an ideal, it is enough to find one failed condition. For example, the collection S of invertible matrices in $M_2(\mathbb{R})$ is not an additive subgroup: I_2 and $-I_2$ are invertible but their sum is zero, which

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is not invertible. Hence S is not an ideal; there is no reason to begin checking absorption. Likewise the diagonal matrices form an additive subgroup and a subring, but fail the ideal test because $E_{12}E_{22} = E_{12}$ is not diagonal, although E_{22} is diagonal. The order in which one tests conditions can save considerable work.

Proposition 4.5.44. *If I and J are ideals of a ring R, then $I + J$ is characterized by the following property: it contains I and J, and every ideal containing both contains $I + J$.*

Proof. The set $I + J$ is an ideal: it is closed under subtraction, since $(i + j) - (i' + j') = (i - i') + (j - j')$, and it absorbs multiplication from either side because I and J do. For $i \in I$, $i = i + 0 \in I + J$, and similarly $J \subseteq I + J$. If an ideal K contains I and J, then $i + j \in K$ for every $i \in I$ and $j \in J$, because K is closed under addition. Hence $I + J \subseteq K$. ■

This universal property is often more useful than the formula for individual elements. For instance, if $I \subseteq J$, the smallest ideal containing both is simply J, so $I + J = J$ without any computation. Moreover, $I + J = I$ holds exactly when $J \subseteq I$.

Example 4.5.45 (A final synthesis involving $\mathbb{Z}_{24}$)**.** Consider the ring $R = \mathbb{Z}_{24}$. Its additive group is cyclic, generated by $[1]$, and has a unique additive subgroup of each order dividing 24. Its multiplicative monoid has identity $[1]$, but is not a group, since $[6]$ has no inverse. The group of units is $U(24) = \{[1], [5], [7], [11], [13], [17], [19], [23]\}$, and every unit satisfies $u^2 = [1]$; for instance $5^2 \equiv 1 \pmod{24}$, and the same can be verified for the other listed residues. Therefore the group of units is not cyclic, since it has eight elements but no element of order eight. As a ring, R is commutative and has zero divisors: $[6][4] = [0]$. The nonzero residue $[12]$ is nilpotent because $[12]^2 = [0]$. Finally, the ideals generated by $[6]$ and $[8]$ have sum $([2])$, intersection $([24]) = ([0])$, and product $([48]) = ([0])$: the intersection and product agree here, even though the ideals are not comaximal. Indeed, in $\mathbb{Z}_{24}$ both are zero because $\operatorname{lcm}(6, 8) = 24$ and $6 \cdot 8$ is divisible by 24. One structure has simultaneously illustrated cyclic groups, monoids, units, noncyclic groups, zero divisors, nilpotence, and ideal arithmetic.

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Concluding Note

Dear Students,

The completion of this course in *Abstract Algebra* marks an important stage in your study of mathematics. Through groups, subgroups, homomorphisms, permutations, rings and ideals, you have begun to see how apparently different mathematical objects are understood through common structures and principles.

These topics also form an essential foundation for higher study and for the algebra portions of competitive examinations, including **CSIR–UGC NET/JRF, GATE, NBHM, TIFR**, and various **M.Sc. and Ph.D. entrance examinations**. Careful attention to definitions, examples, counterexamples and proofs will serve you much better than memorizing isolated results.

As you continue your mathematical journey, develop the habit of asking not only *what* a theorem says, but also *why* it is true and *when* it can be applied. Work through problems independently, learn from unsuccessful attempts, and remain curious about the ideas that connect different areas of mathematics.

I hope these notes remain a useful companion in your future studies. May your interest in mathematics continue to grow beyond the classroom.

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